

UNIVERSIDADE DE LISBOA
FACULDADE DE CIÊNCIAS
DEPARTAMENTO DE INFORMÁTICA



Legal Judgment Prediction and Explanation in the Portuguese Context

Helton José Tavares Mendonça

Mestrado em Ciência de Dados

Dissertação orientada por:
Prof. Dr. Francisco Couto
Juíza Dra. Sofia Wengorovius

Acknowledgements

I would like to first thank God, for giving me strength, guidance and comfort, not only during this dissertation, but throughout my life.

Then I would like to thank my mother, not only for her unconditional love and support, but for sacrificing her life, her dreams and her comfort to save my life and to give me the chance to fight for my dreams.

Next, I would like to thank my girlfriend, again not just because of her unconditional love and support, but for being such a pillar of strength, hearing me and taking care of me, at the lowest points of my life.

Subsequently, I must thank my supervisor, Prof. Dr. Francisco Couto not only for his validation and supervision, but for the genuine inspiration and trust on myself and my work, without it this dissertation would have been much more difficult to complete.

After, I must also thank Judge Dr. Sofia Wengorovius, for her invaluable insights and explanations regarding our legal system and to have devoted time to validate my work, in a domain I had no background knowledge on.

Finally, I would like to thank all my family, friends and anyone that has helped me reach this far in my life.

For me, regardless of any goals, objectives or dreams, we are nothing without the people and the journey we take to reach them, so I express my utmost gratitude to all of you.

Abstract

Portugal's courts face a heavy backlog, with an official report counting 583,890 pending cases as of December 2024 and civil cases taking 24 months on average to resolve; the human cost is visible in cases like the GES/BES financial collapse, where alleged victims died while still waiting for a ruling.

Language models are increasingly proposed to help legal work move faster, but concerns about bias, transparency, and privacy remain, and evidence from the Portuguese legal context is scarce.

This thesis asks three questions: can LLMs outperform a fine-tuned LegalBERTimbau baseline at predicting outcomes in two types of Portuguese court cases; can these models explain their decisions in ways that align with legal reasoning; and do performance and explainability differ between a human-centered domain, domestic violence, and a documentary one, breach of contract?

To answer them, I built a dataset of 6,627 decisions scraped from two Portuguese court repositories, cleaned into 1,802 BoC and 1,126 DV cases, with labels extracted from verdict text and held-out gold test sets of 50 cases per task.

A fine-tuned LegalBERTimbau encoder was compared against a locally-run DeepSeek-R1-8B reasoning model, evaluated zero-shot and few-shot, using SHAP attributions and chain-of-thought traces for explainability and MCC, Macro-F1, and bootstrap confidence intervals for evaluation.

The encoder collapsed to majority-class prediction ($MCC = 0.000$) on both tasks, while the zero-shot LLM performed substantially better (DV: $MCC\ 0.592$, $Macro-F1\ 0.783$; BoC: $MCC\ 0.468$, $Macro-F1\ 0.663$).

Whether the model's explanations tracked legal reasoning, and whether DV and BoC truly differed in difficulty, remained inconclusive.

The results establish this direction as viable for the Portuguese legal context and offer a reference point for future work, while showing current models suit only low-risk, embedded workflow support, not autonomous decision-making.

Keywords: legal judgment prediction; large language models; explainable AI; domestic violence; breach of contract

Resumo

Os tribunais portugueses enfrentam um volume de processos pendentes, com 583.890 casos em dezembro de 2024 (dados oficiais) e os processos cíveis a demorar 24 meses; o custo humano é visível em casos como o colapso do GES/BES, em que alegadas vítimas morreram à espera de uma decisão.

Os modelos de linguagem têm sido propostos para agilizar o trabalho jurídico, mas persistem preocupações de enviesamento, transparência e privacidade, e a evidência no contexto português continua escassa.

Esta tese coloca três questões: se os LLMs superam uma linha de base LegalBERTimbau ajustada na previsão de resultados em dois tipos de processos judiciais portugueses; se estes modelos explicam as suas decisões de forma alinhada com o raciocínio jurídico; e se o desempenho e a explicabilidade diferem entre um domínio humano, a violência doméstica, e um domínio documental, o incumprimento contratual.

Para responder, construí um conjunto de dados com 6.627 decisões recolhidas de dois repositórios de tribunais portugueses, reduzidas a 1.802 casos de BoC e 1.126 de DV, com etiquetas extraídas do veredito e conjuntos de teste-ouro de 50 casos por tarefa.

Um codificador LegalBERTimbau ajustado foi comparado com um modelo de raciocínio DeepSeek-R1-8B local, avaliado em modo zero-shot e few-shot, usando SHAP e cadeias de pensamento para explicabilidade, e MCC, Macro-F1 e intervalos bootstrap para avaliação.

O codificador colapsou para a previsão da classe maioritária ($MCC = 0.000$) em ambas as tarefas, enquanto o LLM em modo zero-shot teve um desempenho superior (DV: $MCC\ 0.592$, $Macro-F1\ 0.783$; BoC: $MCC\ 0.468$, $Macro-F1\ 0.663$).

Se as explicações do modelo acompanhavam o raciocínio jurídico, e se DV e BoC diferiam em dificuldade, permaneceu inconclusivo.

Os resultados estabelecem esta linha de investigação como viável para o contexto jurídico português e oferecem um ponto de partida para trabalho futuro, mostrando que os modelos atuais servem apenas de apoio integrado de baixo risco, não de decisão autónoma.

Palavras-chave: previsão de decisões judiciais; grandes modelos de linguagem; inteligência artificial explicável; violência doméstica; incumprimento contratual

Resumo Alargado

Os tribunais portugueses enfrentam um problema de eficiência persistente e bem documentado. Um relatório oficial contabilizava 583.890 processos pendentes em dezembro de 2024, sendo os processos cíveis os mais demorados, com uma duração média próxima dos 24 meses. O Painel de Avaliação da Justiça na União Europeia coloca Portugal, de forma recorrente, a meio da tabela europeia nos indicadores de eficiência e de digitalização. O custo humano destes atrasos não é abstrato, e o colapso financeiro do GES/BES, em que foram noticiadas mortes de alegadas vítimas antes de qualquer decisão final, ilustra-o com clareza.

É neste contexto que a aplicação de grandes modelos de linguagem (LLMs) ao trabalho jurídico tem sido proposta como via para ganhos de eficiência. A proposta traz consigo preocupações legítimas de enviesamento, opacidade, privacidade e responsabilidade, agravadas pela sensibilidade do domínio. A literatura sobre previsão de decisões judiciais encontra-se relativamente consolidada noutras jurisdições, com trabalhos sobre o Tribunal Europeu dos Direitos Humanos, França, China, Índia, Brasil e Arábia Saudita, mas não encontrei qualquer estudo equivalente para o contexto português. Os dois casos mais próximos são imperfeitos, uma vez que o Brasil partilha a matriz jurídica mas não o sistema, e a Arábia Saudita apenas a condição de língua com poucos recursos.

Esta dissertação procura preencher essa lacuna e organiza-se em torno de três questões de investigação. A primeira pergunta se os LLMs conseguem superar uma linha de base de Deep Learning, um LegalBERTimbau ajustado, na previsão de resultados de processos judiciais portugueses. A segunda pergunta se estes modelos conseguem explicar as suas decisões de forma alinhada com o raciocínio jurídico. A terceira pergunta se o desempenho e a explicabilidade diferem entre um domínio centrado no ser humano, o da violência doméstica, e um domínio documental, o do incumprimento contratual. A estas questões correspondem três objetivos, o de estabelecer uma linha de base para esta tarefa em Portugal, o de avaliar técnicas de Deep Learning e de LLMs recomendando direções futuras, e o de explorar explicabilidade validada por peritos jurídicos.

Não existindo qualquer conjunto de dados público preparado para esta tarefa em português europeu, foi necessário construí-lo. Recolhi por scraping 6.627 decisões de dois repositórios públicos, o dgsi.pt e o jurisprudencia.csm.org.pt, geridos respetivamente pelo Ministério da Justiça e pelo Conselho Superior da Magistratura. Após limpeza, remoção de duplicados e de documentos sem texto integral, restaram 1.802 casos de incumprimento contratual e 1.126 de violência doméstica, menos de metade do total recolhido em ambos os domínios, o que por si só ilustra os problemas de qualidade destes repositórios. As etiquetas foram extraídas por expressões regulares aplicadas ao dispositivo das decisões, com o problema simplificado para duas classes na violência doméstica, decisão mantida ou alterada, e três no incumprimento contratual, favorável, desfavorável ou parcialmente favorável. Para cada tarefa reservei um conjunto de teste-ouro de 50 casos, definido por corte temporal, garantindo que nenhum caso de avaliação precedia os de treino.

A comparação central opõe dois paradigmas. Por um lado, um codificador LegalBERTimbau ajus-

tado, com uma arquitetura de janela deslizante para lidar com documentos longos e ponderação de classes para atenuar o desequilíbrio. Por outro, o DeepSeek-R1-0528-Qwen3-8B, um modelo de raciocínio de pesos abertos executado localmente, avaliado em modo zero-shot e few-shot. Como os documentos excedem largamente a janela de contexto do hardware utilizado, desenvolvi um pipeline de sumarização que seleciona as frases mais relevantes mantendo a ordem narrativa original do acórdão, uma vez que ordenar as frases por relevância destruiria a sequência lógica do raciocínio do tribunal. A avaliação recorre ao coeficiente de correlação de Matthews (MCC) e ao Macro-F1, com intervalos de confiança obtidos por bootstrap. A explicabilidade foi abordada com atribuições SHAP ao nível da frase para o BERT e com os registos de cadeia de pensamento nativos do LLM.

O resultado mais marcante é o do modelo ajustado. Dentro do protocolo de treino testado, o LegalBERTimbau colapsou para a previsão da classe maioritária em ambas as tarefas, com um MCC de 0,000. A avaliação sobre os próprios conjuntos de treino e de validação, e não apenas sobre o conjunto de teste, mostrou desempenho idêntico, o que aponta para subajustamento e não para sobreajustamento. O modelo não chegou a aprender qualquer sinal discriminativo, um resultado que deve ser lido como uma afirmação sobre o orçamento de treino e os dados disponíveis, e não sobre a arquitetura em si.

Em contraste, o LLM em modo zero-shot previu corretamente classes minoritárias e obteve um desempenho claramente superior, com um MCC de 0,592 e um Macro-F1 de 0,783 na violência doméstica, e um MCC de 0,468 e um Macro-F1 de 0,663 no incumprimento contratual. A primeira questão de investigação fica assim respondida dentro das condições testadas. Mais inesperado foi o efeito do few-shot. Acrescentar um exemplo por classe não melhorou o desempenho e, na violência doméstica, degradou a cobertura das classes minoritárias. No incumprimento contratual o padrão foi ambíguo, com o MCC a subir ligeiramente e o Macro-F1 a descer. Designo este efeito por paradoxo do few-shot, e a explicação mais plausível é a de que um único exemplo por classe não capta a variação interna dessa classe, acabando por ancorar o modelo à classe maioritária em vez de lhe servir de guia.

Quanto à explicabilidade, o balanço é menos favorável. As atribuições SHAP sobre o LegalBERTimbau revelaram-se dispersas e sem concentração em terminologia juridicamente relevante, o que é coerente com um modelo que não aprendeu a discriminar. Os registos de raciocínio do LLM são legíveis e frequentemente convincentes, mas identifiquei pelo menos um caso em que o modelo produziu uma justificação fluente e plausível para uma decisão errada. Legibilidade e correção são propriedades distintas, e nenhuma das duas abordagens foi validada por peritos em escala suficiente para sustentar conclusões estatísticas, pelo que a segunda questão de investigação permanece em aberto.

O exercício de validação por juízes, conduzido com um painel designado pelo Conselho Superior da Magistratura, tinha por objetivo confirmar a lógica de extração de etiquetas. A concordância foi elevada, 24 em 25 casos na violência doméstica e 21 em 25 no incumprimento contratual. Mais relevante do que os números foi a divergência entre os próprios juízes em alguns casos, e o facto de a terminologia das classes que construí não corresponder à que os tribunais efetivamente utilizam. Um dos magistrados assinalou ainda uma limitação estrutural, a de que, quando ambas as partes recorrem, o tribunal pode decidir parcialmente a favor de cada uma, produzindo resultados que não encaixam num esquema de três classes. Isto sugere que a previsão de decisões judiciais não é uma tarefa com uma verdade única para a qual um modelo suficientemente bom converge, com consequências diretas para a forma como estes sistemas devem ser treinados e avaliados.

A terceira questão não ficou resolvida. Os intervalos de confiança das duas tarefas sobrepõem-se e, com apenas 50 casos por tarefa, não existe poder estatístico para detetar diferenças que não sejam grandes. A hipótese correspondente é tratada como questão em aberto, ficando o teste formal de hipóteses, sobre um conjunto de teste maior, recomendado como trabalho futuro imediato.

Várias limitações condicionam a generalização destes resultados. O corpus é pequeno e de qualidade desigual, as etiquetas são heurísticas e imperfeitas, os conjuntos de teste-ouro não foram integralmente validados pelo painel, o pipeline de sumarização não foi ele próprio avaliado, e nem a geração aumentada por recuperação nem a ablação da cadeia de pensamento chegaram a ser implementadas. Não foi realizada qualquer auditoria de equidade aos dados ou às previsões, requisito prévio a qualquer utilização real. Acresce que a tarefa aqui estudada é de classificação retrospectiva a partir de acórdãos já decididos, e não de previsão a partir dos factos de processos ainda pendentes, que seria a aplicação verdadeiramente consequente desta tecnologia.

Face aos objetivos traçados, o primeiro e o segundo foram cumpridos, ainda que o segundo esteja limitado pelo protocolo de treino utilizado. O terceiro foi apenas parcialmente alcançado, uma vez que as explicações foram produzidas mas nunca validadas por peritos em escala. O contributo desta dissertação é sobretudo o de um ponto de partida, deixando um corpus, um sistema de avaliação e um conjunto delimitado de questões em aberto onde antes não existia base de trabalho para o português. O trabalho futuro deverá substituir a extração por expressões regulares por modelos treinados para o efeito, refinar o pipeline de sumarização, comparar vários modelos no mesmo sistema de avaliação, e desenvolver ferramentas de explicabilidade com interfaces adequadas a juristas, de modo a permitir avaliação em escala. No estado atual, estes modelos aconselham apenas aplicações integradas de baixo risco, e não apoio autónomo à decisão.

Table of Contents

List of Figures	xi
List of Tables	xii
1 Introduction	1
1.1 Context and Motivation	1
1.2 Research Questions, Hypotheses and Objectives	2
2 Background	4
2.1 Introduction	4
2.2 Portuguese Legal System	4
2.3 Overview of Large Language Models	6
2.4 LLMs in Legal Application	8
2.4.1 Main Use Cases	8
2.4.2 Current Barriers	9
2.4.3 Proposed Solutions and Future Directions	10
2.4.4 How are LLMs used currently in the Portuguese Judicial Context	10
2.5 Related Work	12
2.5.1 Classical Machine Learning Approaches	12
2.5.2 Neural Network Approaches	13
2.6 Ethical and Legal Considerations	15
2.6.1 Key Challenges	15
2.6.2 Key Regulations and Governance	16
2.6.3 Reflections on Ethical and Regulatory Considerations	16
2.7 Conclusion and Gaps in the Literature	17
3 Data	18
3.0.1 Preprocessing	18
3.0.2 Data Splitting Strategy	20
3.0.3 Example of document structures	20
3.0.4 Exploratory Data Analysis	22
4 Methodology	34
4.1 Overview	34
4.2 Materials and Computational Resources	35
4.2.1 Hardware	35
4.2.2 Software and Dependencies	35

4.2.3	Models and Pre-trained Weights	35
4.3	Fine-Tuned LegalBERTimbau	36
4.3.1	Motivation and Design Rationale	36
4.3.2	Sliding-Window Architecture	36
4.3.3	Window Aggregation via Mean Pooling	37
4.3.4	Training, Hyperparameters and Regularization	37
4.3.5	Limitations and Trade-Offs	39
4.4	Large Language Model Approach	39
4.4.1	Model Selection	39
4.5	Context Management & Smart Summarization Pipeline	40
4.5.1	Motivation	40
4.5.2	Summarization Architecture	41
4.5.3	Sentence Order Preservation	43
4.5.4	Limitations and Trade-Offs	43
4.5.5	Zero-Shot Prompting Strategy	44
4.5.6	Few-Shot Prompting Strategy	47
4.5.7	Limitations and Trade-Offs	48
4.6	Explainability Methods	49
4.6.1	SHAP Explanations for BERT	49
4.6.2	Thinking Logs and Chain-of-Thought for LLMs	50
4.6.3	Comparison Framework	50
4.7	Evaluation Methodology	51
4.7.1	Metrics	51
4.7.2	Bootstrap Confidence Intervals	51
4.7.3	Judge Validation Protocol	52
4.7.4	Limitations of Evaluation Approach	52
5	Results	53
5.1	Fine-Tuned LegalBERTimbau Phase	53
5.1.1	Training Dynamics	53
5.1.2	Train and Validation Set Performance	54
5.1.3	Gold Test Set Performance	56
5.1.4	Embedding Space Analysis after Fine-Tuning	58
5.2	LLM Phase	60
5.2.1	Zero-Shot with Smart Summarization	60
5.2.2	Few-Shot (k=1) with Smart Summarization	61
5.2.3	Summary Comparison: All Models	62
5.3	Judge Validation	62
5.3.1	Domestic Violence (DV)	62
5.3.2	Breach of Contract (BoC)	63
5.4	Summary Table: Judge Validation Overview	64
6	Discussion	65
6.1	Synthesis of Core Findings	65
6.1.1	Fine-Tuned BERT as Baseline	65

Table of Contents

6.1.2	DeepSeek Zero-Shot	66
6.1.3	Few-Shot Paradox	67
6.2	Addressing the Research Questions Directly	68
6.2.1	RQ1 — Can LLMs Outperform BERT?	68
6.2.2	RQ2 — Can Models Explain Their Decisions?	68
6.2.3	RQ3 — Does Performance Differ Between DV and BoC?	69
6.3	Judge Validation	69
6.4	Where AI Can and Cannot Help in Judicial Decision-Making	70
6.5	Limitations of This Study	71
6.6	Future Work	72
7	Conclusion	74
	Appendices	81
	Github Repository	82
	Data Files	83
	Class-Engineering Heuristics	84
.1	DV Binary Classification (KEPT vs. ALTERED)	84
.2	BoC Ternary Classification (Favorable vs. Unfavorable vs. Partial)	85
	SHAP Explanation Examples	88
	Example DeepSeek Reasoning Trace	91

List of Figures

3.1	Data Scraping & Preprocessing Pipeline	21
3.2	Document Structure: Structured vs Prose	23
3.3	BoC Class Distributions	24
3.4	DV Class Distributions	24
3.5	BoC Document Length Distributions	24
3.6	DV Document Length Distributions	25
3.7	BoC Most Common Bigrams	25
3.8	BoC TF-IDF Bigrams	26
3.9	DV Most Common Bigrams	26
3.10	DV TF-IDF Bigrams	26
3.11	BoC Cases per Year	27
3.12	DV Cases per Year	28
3.13	BoC Outcome Proportions Over Time	28
3.14	DV Outcome Proportions Over Time	28
3.15	BoC Outcomes by Judge Gender	29
3.16	DV Outcomes by Judge Gender	29
3.17	BoC Outcomes by Court Region	30
3.18	DV Outcomes by Court Region	31
3.19	UMAP Embeddings: DV and BoC Combined	32
3.20	UMAP Embeddings: BoC	32
3.21	UMAP Embeddings: DV	33
4.1	Smart Summarization Pipeline	41
4.2	Few-Shot Retrieval Mechanism	47
5.1	BERT Training Loss: DV	53
5.2	BERT Training Loss: BoC	54
5.3	UMAP After Fine-Tuning: DV	58
5.4	UMAP After Fine-Tuning: BoC	59
1	SHAP Attribution by Sentence Position	88
2	SHAP Top 10 Positive/Negative Sentences: DV Example	89
3	SHAP Top 10 Positive/Negative Sentences: BoC Example	90

List of Tables

2.1	Key Studies in LJP	15
5.1	BERT DV Performance — Train Set (n=904)	54
5.2	BERT DV Performance — Validation Set (n=160)	55
5.3	BERT BoC Performance — Train Set (n=1489)	55
5.4	BERT BoC Performance — Validation Set (n=263)	56
5.5	BERT DV Performance (n=50)	56
5.6	BERT BoC Performance (n=50)	57
5.7	DeepSeek Zero-Shot DV (n=50)	60
5.8	DeepSeek Zero-Shot BoC (n=50)	60
5.9	DeepSeek Few-Shot DV (n=50)	61
5.10	DeepSeek Few-Shot BoC (n=50)	61
5.11	All Models Summary Comparison	62
5.12	Judge Annotations: DV Cases	62
5.13	Judge Confidence: DV	62
5.14	Judge Annotations: BoC Cases	63
5.15	Judge Validation Summary	64

List of Abbreviations

AI Artificial Intelligence

AMP Automatic Mixed Precision

BERT Bidirectional Encoder Representations from Transformers

BoC Breach of Contract

CI Confidence Interval

CoT Chain of Thought

CSM Conselho Superior da Magistratura (Superior Council of the Judiciary)

DGSI Direção-Geral dos Serviços Informáticos (Portuguese IT Department)

DL Deep Learning

DV Domestic Violence

ECHR European Convention on Human Rights

F1 F1-score (harmonic mean of precision and recall)

GDPR General Data Protection Regulation

GPT Generative Pre-trained Transformer

GRU Gated Recurrent Unit

HAN Hierarchical Attention Network

JP Julgados de Paz (Portuguese Peace Courts)

LegalBERTimbau Legal-domain BERT model fine-tuned on Portuguese court decisions

LLM Large Language Model

LSTM Long Short-Term Memory

ML Machine Learning

MLM Masked Language Modeling

MCC Matthews Correlation Coefficient

List of Tables

NLP Natural Language Processing

RAG Retrieval Augmented Generation

ROUGE Recall-Oriented Understudy for Gisting Evaluation

SHAP SHapley Additive exPlanations

SVM Support Vector Machine

STJ Supremo Tribunal de Justiça (Portuguese Supreme Court of Justice)

TRC/TRP/TRL/TRX Tribunal da Relação (Portuguese Court of Appeal, by region)

TF-IDF Term Frequency–Inverse Document Frequency

TSS True Skill Statistic

UMAP Uniform Manifold Approximation and Projection

XAI Explainable Artificial Intelligence

Chapter 1

Introduction

1.1 Context and Motivation

According to the 2025 Rule of Law Report in Portugal¹ by the European Commission, the Portuguese judicial system must still overcome and improve in its efficiency, particularly regarding Administrative and Tax Courts. This metric is defined by the European Union(EU), with various factors, such as duration of procedures, clearance rates(i.e.,the ratio of resolved incoming cases in a given time period) and the number of pending cases. To understand the magnitude of this issue in Portugal, I reference the 2024 Statistical Justice Report², where it can be observed there were 583,890 pending cases as of 31st December 2024, with a clearance rate of 98,7%. Alongside this, the report highlights that the average duration of a Civil Law case is 24 months, while the other types of cases, have roughly 7-9 months average duration.

To put in perspective these allegations, I draw some comparisons of justice systems metrics across Europe, per the 2025 EU JUSTICE SCOREBOARD³, which highlights progression of various metrics in efficiency, quality and independence of justice systems across the EU from 2021 to 2023. Without going into much detail, it can be concluded that although Portugal has improved in the metrics the EU evaluates over the years, it still lags behind the EU, commonly being at the middle of the rank of countries, particularly at the metrics that define efficiency. Furthermore, when comparing Portugal, in terms of digitalization of the justice system, once again, at best it is at the middle of the group of countries. In this domain, further study of countries like Estonia, Spain or Denmark, that often rank high in terms of digitalization in their justice systems, is recommended. Further reading of this report by the European Commission is advised, particularly of Chapters 2.1 and 2.2.3, to understand better how Portugal compares to its EU peers.

Now that one of the main problems in the portuguese justice system is established, its lagging efficiency and constant backlog of cases, it is important to understand why faster justice matters.

The GES/BES case, that revolved around the financial collapse of *Banco Espírito Santo* and its group company in 2014, was and still is, one the biggest stress tests to the portuguese justice system. Based on official documents⁴ from the Public Ministry, summarizing the accusations, the case involved more than

¹https://commission.europa.eu/document/download/5a482f87-1f24-47bd-8595-d25f1ca29c6a_en?filename=2025%20Rule%20of%20Law%20Report%20-%20Country%20Chapter%20Portugal.pdf

²https://estatisticas.justica.gov.pt/sites/siej/pt-pt/Destaques/20251031_D103_IndicadoresEstatisticos_2007-2024.pdf

³https://commission.europa.eu/document/download/51b21eff-a4b0-4e73-b461-06bd23b43d4e_en?filename=2025%20EU%20Justice%20Scoreboard_template.pdf

⁴https://dciap.ministeriopublico.pt/sites/default/files/documentos/pdf/resumo_acusacoes_ges-bes_vf.pdf

300 crimes, ranging from criminal association, active and passive corruption, money laundering among others, spread across more than 20 defendants, of which 18 are individuals. According to this document there are various subprocesses occurring relative to this case, with thousands of proclaimed victims and hundreds of witnesses. From the initial investigation in 2014, it passed ten years, for the main trial to start in 2024, and there is no expected date for its conclusion. In that timeframe various crimes have prescribed and some alleged victims have passed away ⁵. Although this case is an extreme example, it highlights the importance of an efficient justice system, where justice must be served in a timely manner. The delay and backlog of cases, not only affects the involved parties, particularly the victims, but also affects the overall trust by the people in its justice system, so it's imperative to explore innovative solutions to mitigate this problem.

With the rise of Artificial Intelligence(AI) and particularly Large Language Models (LLMs) in the recent years, fields like Law, had a surge of interest in applying these tools to improve various legal tasks, with the promise they would increase efficiency, reduce costs and improve the access and quality of the justice system. (Remolina et al., 2024). However, despite the potential benefits, it's important to understand and mitigate the potential risks associated with the use of AI in law contexts, like bias, transparency issues and data privacy, all factors that undermine fundamental principles of justice like fairness, human dignity, privacy and equality before the law. (Pedroso et al., 2024; Remolina et al., 2024) Considering the aforementioned problems and the potential solutions, residing in AI, particularly LLMs, while acknowledging the ethical and legal concerns, this Master's dissertation in Data Science aims to explore the use of Deep Learning(DL), specifically LegalBERTimbau (Melo et al., 2023), and ultimately LLMs, to predict or classify judicial decisions in the portuguese legal context.

1.2 Research Questions, Hypotheses and Objectives

For this dissertation, I focus on finding answers to the following research questions:

- **RQ1:** Can LLMs outperform the LegalBERTimbau DL baseline in predicting outcomes for both type of cases?
- **RQ2:** Beyond prediction, can the aforementioned models explain how they reached their decisions, and if these explanations align with legal reasoning?
- **RQ3:** How does performance and explainability differ between a sensitive/human-centered domain like domestic violence and a documentary domain, like breach of contract?

To answer these questions, I test the following hypotheses:

- **H1a:** Open source LLMs, will achieve higher quantitative performance metrics (e.g., Macro F1-score, Matthews Correlation Coefficient, etc.) than the LegalBERTimbau DL baseline in both domestic violence and breach of contract cases.
- **H1b:** The implementation of advanced prompting techniques, such as Chain of Thought(CoT) prompting and Retrieval Augmented Generation(RAG), will further enhance the performance of LLMs in predicting judicial decisions in both legal domains.

⁵https://www.rtp.pt/noticias/pais/mais-de-100-lesados-ja-morreram-a-espera-de-ver-feita-justica_n1607773

- **H2:** With the implementation of Explainable AI(XAI) techniques (e.g, SHAP),LLMs will provide more insights into their decision-making process, and these will align with validation from an human legal expert.
- **H3:** Performance and explainability will vary significantly statistically (e.g, Mann-Whitney U test, Cohen’s d, etc.) between case types, with domestic violence having worse results due to its case subjectivity and human-centered evidence, when compared to breach of contract cases, that are more documental and objective in nature.

Finally, the proposed objectives for this dissertation are:

- **O1:** Establishing a research baseline for Legal Judgment Prediction in Portugal, due to the lack of prior work in this specific legal task and context.
- **O2:** Experiment and evaluate DL and LLM techniques, based on related work and the state of the art, recommending further research directions.
- **O3:** Explore the use of XAI techniques to improve explainability of LLMs in legal contexts, validating these explanations with a human legal expert, with the purpose of increasing trust in these models.

With these research questions and objectives framed, this dissertation proceeds as follows. Chapter 2 establishes the foundational context through an overview of the Portuguese legal system, the evolution of language models, and related work in legal judgment prediction. Chapters 3 and 4 detail the data collection and experimental methodology. Chapter 5 presents results, and Chapter 6 offers a comprehensive discussion of findings, limitations, and future research directions.

Chapter 2

Background

2.1 Introduction

Given the main objective of this dissertation, to explore the use of Large Language Models (LLMs) to predict judicial decisions, it's imperative to understand and contextualize the current landscape of both the Portuguese Justice System and the evolutions done in the field of Language Modelling. I first explain key concepts and terms in the Portuguese legal system, particularly those pertinent to my study, such as the structure of the sentences and *acórdãos* (court rulings) and what is and how jurisprudence is structured in Portugal. Then I explore the history and recent innovations in LLMs, to understand how these models evolved to their current state of the art. Furthermore, I analyze how LLMs have been applied in legal contexts globally, highlighting key use cases, current barriers and proposed solutions, with a special focus on the Portuguese legal context. Alongside the more technical aspects, due to the sensitive nature of the legal field, I must also explore ethical and legal concerns and discuss proposed solutions to the problems applications like these could bring to humanity. Finally, I explore related work, particularly those that have investigated the use of ML and DL techniques to predict judicial outcomes, highlighting key findings and inspiring methodologies for this study. At the end of this section, the reader should have a solid understanding of the context of the problem this thesis sets out to solve, a overview of current tools available and inspiration for potential methodologies to apply in this research and ultimately localize this work in the current landscape of LegalAI.

2.2 Portuguese Legal System

Portugal is a Democratic State of Law, where every citizen is assured the right of access to the law and to the courts of justice for defense of their rights and interests.

Tribunals are the sovereign entities to administer justice in the name of people and are organized as follows by the Portuguese Constitution and Law 62/2013 of August 26th ¹:

- **Constitutional Court:** assigned to administer justice in matters of constitutionality, enforcing the supremacy of the Portuguese Constitution.
- **Supreme Court of Justice:** highest court for judicial matters, with jurisdiction over all other judicial courts in all the Portuguese territory.

¹<https://dgpj.justica.gov.pt/Dossiers-tematicos/Sistema-de-Justica-Portugues>

- **Courts of Appeal:** these tribunals are the second instance, where appeals from the first instance courts are judged. They are divided into 5 regions: Lisbon, Porto, Guimarães, Coimbra and Évora.
- **Courts of First Instance:** these are the courts where cases initialize. They are divided into several competences, depending on the nature of the case:
 - **Civil Courts:** handle civil matters such as contracts, property disputes, family law, etc.
 - **Criminal Courts:** deal with criminal cases, including felonies and misdemeanors.
 - **Family and Minors Courts:** administer cases related to family law, child protection, custody, and adoption.
 - **Labor Courts:** specialize in labor disputes between employers and employees.
 - **Commercial Courts:** focus on commercial and business-related disputes.
 - **Execution Courts:** responsible for fiscalizing the execution of judicial decisions. This includes enforcing or extinguishing prison/expulsion sentences, emitting arrest or liberation warrants, among other powers.
 - **Specialized Courts:** such as Maritime Courts, Intellectual Property Courts, among other specialized jurisdictions.

Administrative and Fiscal Courts have a similar hierarchical structure to the Judicial Courts:

- **Supreme Administrative Court:** highest court for administrative and fiscal matters.
- **Central Administrative Courts:** these tribunals are the second instances for administrative and fiscal cases.
- **Administrative and Fiscal Courts of First Instance:** these are the courts where administrative and fiscal cases initialize.

Additionally, there are other tribunals with focus on specific matters like:

- **Courts of Auditors:** tasked with fiscalizing public expenditure.
- **Military Courts:** used only in times of war, however, in times of peace there are military judges within the judicial courts that handle military justice matters.
- **Arbitration Courts:** private entities that resolve litigations outside the public judicial system.
- **Peace Courts:** small extrajudicial entities, used to resolve minor disputes.

Moreover, there is the Public Ministry, an independent entity, representing the State, with the main objective of defending the law and the democratic values of the Portuguese Republic against criminal actors, whether individuals or corporations.

For this dissertation, I use data from the Appeal Courts, Supreme Court of Justice, that handle both domestic violence cases and more complex contract litigation cases. Alongside these, I also use data from the Peace Courts, that have authority over smaller contract dispute cases.

Sentences, Court Rulings and Jurisprudence in Portugal

With pertinence to this dissertation, I should also define and distinguish between sentences, *acordãos* (court rulings) and jurisprudence and how are these structured, so LLMs can be better applied to their predictions.

A sentence can be defined as a judicial decision made by a singular judge. These are majorly applied in first instance courts and simpler cases. On the other hand, *acordãos* or court rulings are decisions made by a collective of three judges, usually in courts of appeal or higher. Jurisprudence can therefore be defined as the set of past court rulings and sentences, on a specific legal matter, that can be used as supportive material for future rulings.

The structure of sentences and *acordãos* is set by the Article 374 of the Decree-Law 78/87 of the Penal Code ², which states that a judicial decision must have three main parts: The *Relatório* (Report), the *Fundamentação* (Foundation/Justification) and the *Dispositivo* (Decision).

- **Relatório:** This section should contain the identification of all parties (defendant, assistants and civil parts) and should serve as a brief contextualization of the case or the appeal if applied.
- **Fundamentação:** Here sits the main body of the case, where the judges must enumerate all facts, proven and not proven, and must clearly expose their motives and reasoning, based on contextual facts, relevant jurisprudence and doctrines and finally the applicable laws, to the given case.
- **Dispositivo:** Finally, they must end, by clearly stating the final decision they have reached, alongside all the legal implications of such decisions (e.g, sentences, fines, compensations, destinations of objects, etc.).

It's important to note, although this structure is mandatory, which in turn creates certain patterns that can be mined, there is still a great degree of freedom, judges have when writing their decisions, creating some variability and difficulties when mining these documents. This will be further explored and analyzed in the Methodology chapter.

2.3 Overview of Large Language Models

Over the past years, LLMs have gained substantial interest in various fields, particularly in Law. This inherent relevance derives from the central importance language has when considering human nature, enabling the expression of one's thoughts, emotions, and promoting the communication with others. However, it's important to keep in mind, these models cannot think like humans, rather, emulate language comprehension and generation behind mathematical and statistical frameworks. (Wang et al., 2024; Naveed et al., 2024).

Nevertheless, the chronological historical evolution of Language Models(LMs), date back to Statistical Language Models in the 1990s (Rosenfeld, 2000), where the problem of processing context and semantic dependent features of natural language, was approached from a statistical perspective.

At that time, major barriers were primarily related to the dimensionality curse(i.e., the exponential increase in computational complexity as context size grows) and the disparity between the over-simplistic models (e.g., n-gram model, that predicted a word's likelihood based on previous n-words), and the complex nature of human language. Rosenfeld (2000), recommended future work to focus on injecting prior human knowledge of language into the models, through statistical frameworks.

²<https://diariodarepublica.pt/dr/legislacao-consolidada/decreto-lei/1987-34570075>

Overtime, the resurgence of Neural Networks(NNs), enabled the application of these to the natural language problem. By vectorizing or tokenizing the words and symbols in a given sentence, it was possible to pass these numerical representations through NNs, that mapped similar words to proximate vectors that clustered efficiently semantically similar text, then outputting a probability over the density distribution of words/sentences in a given task.

Bengio et al. (2000) had noted that, utilizing a model, where each training phrase, exhibited systematic generalization to unseen word combinations, performing the objective in a more resource efficient manner. They proposed future research to study the implementation of Recurrent Neural Networks (RNNs), to permit taking advantage of the sequential nature of language, which basic NNs could not, and warrant the use of larger training samples (e.g., a whole paragraph, at the time) without increasing excessively the number of parameters of the network.

Subsequently, RNN's became the staple when it came to LMs, surpassing its precedents, in terms of accuracy, and the capacity of handling larger amounts of text (Mikolov et al., 2010). Even so RNNs, still struggled, due to the impossibility of parallelization of training examples, inherent to the sequential nature of the network and the “vanishing” gradient problem in longer context windows (i.e, extreme shrinkage of gradients during training) that reduced learning abilities.

Although architectural improvements were made, namely with the introduction of Long Short-Term Memory and Gated Recurrent Units(GRU), which utilized gate mechanisms to optimize information flow and reduce the aforementioned issues, these networks still struggled with massive context size inputs.

A breakthrough, happened when Vaswani et al. (2017) proposed using the Attention Mechanism (e.g. focusing on specific keywords in a sentence) as the main foundation for identifying patterns between the input and the output, and Self-Attention, capturing dependencies within the input. These architectural innovations, not enabled taking advantage of Graphics Processing Units (GPUs) technology, which are specialized in handling repetitive calculations in a parallel format, reducing computation time. Alongside, they also facilitated the detection of long-range dependencies. Self-Attention yielded inherently more interpretable models, by showing which input elements were prioritized in a given task. Both of these mechanisms are part of the Transformer Network, that permitted scaling LMs to today's standards.

With the previously mentioned innovation, the scale of training data increased substantially, particularly on unlabeled text (e.g., text found on websites like Reddit, Common Crawl, Wikipedia, etc.).(Wang et al., 2024). Thereby permitted LLMs to grasp fundamental language structure, such as vocabulary, semantics, syntax, and logic of various tongues and dialects. Consequently, LLMs, excelled in various Natural Language Processing(NLP) tasks such as text summarization, question-answering or even translation, all vital for legal applications.

More recently, with the emergence of LLMs like GPT-4.5 (OpenAI, 2025) or Gemini (Gemini et al., 2025) alongside open source models such as DeepSeek (DeepSeek-AI et al., 2025) or Llama 4 (Meta AI, 2025), and many others (Shukla, 2025), have pushed the capabilities of what and how these models can be applied. These state-of-the-art models pursue multi-modality, being capable of interconnecting text, images or videos in various tasks and show high reasoning capabilities.

Considering the history and recent innovations in the field of LLMs, their use in legal practice seems inevitable, and for the importance of this dissertation I explore how have they been applied in Law, particularly in the Portuguese context, and how can they be further explored in this area.

2.4 LLMs in Legal Application

AI, particularly LLMs have never been more popular in Law applications. Their use can be segmented into various legal tasks, ranging from document summarization or drafting to more complex tasks such as judicial outcome prediction. This section will explore in more detail recent key applications in the LegalAI field, what are the current limitations and lastly proposed solutions alongside future research directions.

2.4.1 Main Use Cases

The key applications of LLMs in Law practice can be segmented into 3 categories, such as Legal Prediction, Legal Document Review and Legal Research (Siino et al., 2025a). I further explore each main use and their possible branches of applications.

Legal Prediction

In this specific task, the literature highlights the natural evolution from using traditional ML methods (e.g. Tree Models, Statistical Models, etc.) to the more advanced use of LLMs in Judgment Prediction (e.g. BERT or GPT based models). This task goal is forecasting the outcome of a legal case, by analyzing a great amount of legal data, ranging from past court decisions, applicable laws and case context to predict a court verdict of current or future diligence's (Siino et al., 2025a; Zhong et al., 2020).

However, this common methodology approach has been criticized in the literature (Medvedeva et al., 2023), as they argue that using information available at or after the time of prediction, such as the final decisions, act as a form of data leakage, and therefore, the model is not truly predicting the outcome of a case, but rather learning to infer the final decision from retrospective narration and post hoc reasoning from magistrates.

Nonetheless, this is the chosen approach for this dissertation, and for simplicity purposes I still refer to the terminology as Legal Judgment Prediction, but the aforementioned criticism and limitations of this approach must be kept in mind.

Legal Document Review

This application incorporates a broad range of tasks, which the literature commonly categorizing into either classification, regression or generative problems:

- **Classification Tasks:**

- *Named Entity Recognition*: Identifying specific entities in text based on context, such as recognizing “Apple” as Apple Inc. rather than the fruit in a contract (Siino et al., 2025a; Shao et al., 2025).
- *Document Classification*: Categorizing legal documents based on their content, such as identifying contract types (e.g., distinguishing between permanent contract, fixed-term contract, indefinite-term contract, etc.) (Siino et al., 2025a; Shao et al., 2025).

- **Regression Tasks:**

- *Similarity Estimation*: The goal is predicting a continuous value (usually between 0 and 1), representing how similar given entities (e.g., documents, court cases, contracts, legal opinions, etc.) are (Siino et al., 2025a; Zhong et al., 2020).

- **Generative Tasks:**

- *Document Automation:* Generating or structuring legal documents, such as contracts or motions, improving lawyers and other justice system agents efficiency and scalability, by allowing them to center their attention on more high reasoning legal tasks, instead of mundane document preparation (Siino et al., 2025a; Shao et al., 2025).
- *Document Summarization:* By shortening the verbosity of a given legal document while retaining the main takeaways, a legal agent can in a more efficient manner analyze such documents (Siino et al., 2025a; Shao et al., 2025).

Legal Research

Finally, a justice agent, needs to research various legal documents to sustain his arguments. Trough traditional methods like simple Google searches or specific databases searches like dgsi.pt³, an experienced agent could get decent work done, but augmenting its research with LLMs in a responsible manner, the improvements in time efficiency could be beneficial.

The main branching tasks at this domain are:

- **Document Retrieval:** Searching for documents in a natural language format, instead of key word/sentences is pivotal in the scalability of an agent work. However this specific task has shown potential misuses in the past with models hallucinating made up court cases or law articles. It's important that document retrieval to be complemented with other techniques like (Retrieval Augmented Generation)RAG or Knowledge Graphs (Siino et al., 2025a; Shao et al., 2025).
- **Case Entailment:** Determining whether a document has verified facts, legal principles and arguments it's crucial to one's work. Case Entailment complements Document Retrieval by creating a system capable of identifying and associating paragraphs within a specific precedent case, that could be relevant in justifying a argument or decision in a case (Siino et al., 2025a; Shao et al., 2025).
- **Question-Answering:** Lawyers and other agents don't know everything about Law, so it's natural that they want a question answering system. This could be useful not only for a preparation for a case but also for consultancy purposes with clients (Siino et al., 2025a; Shao et al., 2025). Here chat bots like Chat-GPT have been implemented, particularly at firm levels. However, even with the perceived benefits of such system, there are major concerns in it's use like Hallucinations, Data Security and Privacy, Compliance Issues (i.e. can't give legal advice without professional supervision), among others further explained in Homoki et al. (2024)

2.4.2 Current Barriers

Even though LLMs possess many advantages in Law applications, there are both technical and non-technical barriers, currently, that need further research and mitigation.

Given the inherent static nature of LLMs, their outputs could generate occasionally false, yet plausible looking facts or references, such as non existing court cases or law articles (Armour et al., 2020; Surden, 2023), with even Portuguese news reporting on current investigations of this phenomenon in the

³<https://www.dgsi.pt/>

Portuguese court system⁴. Other issues such as limited, biased and novel data, can reduce performance since these models depend on their training data. Also LLMs might be difficult to interpret without certain explainable techniques, which could reduce the trust and accountability of the models outputs. (Armour et al., 2020; Surden, 2023). Moreover, there is the inherent computational cost of running these models, and in dealing with a field, where one could ingest a large volume of data (e.g., a large firm contract), this could result in a steep price to implement (Armour et al., 2020; Surden, 2023).

Alongside the technical difficulties presented above, there are other concerns regarding ethical, legal and regulatory domains of LegalAI, such as data privacy and security, bias and discrimination, liability or property rights. These will all be explored more, in a section further down.

2.4.3 Proposed Solutions and Future Directions

To mitigate some of the technical concerns, the literature unanimously suggests rigorous human supervision and validation of models outputs. At the same time, users must educate themselves on the strengths and limitations of these tools to avoid over reliance and malpractice. Moreover, to tackle the specific problems legal application brings, there must be reinforced research in specializing models in specific law contexts, taking advantage and improving on recent models multi modality and higher reasoning capabilities, complementing LLMs with other techniques and finally enhancing transparency and interpretability of these models. (Surden, 2023; Siino et al., 2025b)

Concerning the non-technical hurdles, there must be investment in secure data handling protocols, contractual safeguards and regulatory frameworks to avoid the pitfalls mentioned. (Surden, 2023; Siino et al., 2025b; Shao et al., 2025)

In my view, the interdisciplinary dialogue between AI experts and legal professionals, will be key, to advance this field.

2.4.4 How are LLMs used currently in the Portuguese Judicial Context

In Portugal's judiciary system, there is also a movement towards innovation, as portrayed in the "*Estratégia GovTech da Justiça*"⁵, a government initiative to digitally evolve the legal domain in Portugal. LLM's, due to their capabilities in natural language, as discussed above, become inevitably a vehicle towards that innovation, both in the public and private sector.

Public Sector

Recently, there have been a growing number of public entities actively or planning to implement AI solutions in various legal tasks.

Pedro (2024) discusses the most recent initiatives taken by public institutions regarding the application of LLM's into the justice system, and I present a few of them.

For example, in 2023, the *Supremo Tribunal de Justiça* (Supreme Court of Justice), presented "IRIS-Information, Rationalization, Integration and Summarization: Application of Artificial Intelligence Techniques at the Supreme Court of Justice"⁶, in collaboration with INESC-ID, with multiple objectives, such as assistance in case analysis, preparation, anonymization and finally transcribing of old transcripts to digital format. Alongside this, there was also the goal public dissemination of Supreme Court cases in their website. Per the researchers, this was based in techniques like optical character recognition to extract

⁴<https://www.publico.pt/2025/02/11/sociedade/noticia/alegado-uso-ia-acordao-relacao-alvo-processo-averiguacao>

⁵<https://govtech.justica.gov.pt/>

⁶<https://stjiris.github.io/>

the text from scanned documents, then complemented with pipelines of document search, retrieval and analysis.

Another case, is Guia Prático de Acesso da Justiça (GPJ)⁷, a chatbot, as per the public sources, specialized in Marriage and Divorce Litigation and Company Creation laws. It's stated that is based of the GPT-4 model of OpenAI and distributed through Microsoft Azure cloud platform.

Further reading of Pedro (2024) is recommend to learn about more recent applications of AI and LLMs in the Portuguese Public Legal System.

Private Sector

The portuguese private sector is rapidly evolving when it comes to the application of AI and LLMs, pushed mainly by big law firms, due to the necessity of improving staff efficiency and to address local and global competition in the sector.

One of the standout case studies, regards the portuguese startup Legau⁸, and their collaboration with firms such as CCA Law Firm⁹, RFF Lawyers¹⁰, Antas da Cunha ECIJA¹¹, among others. Based on information provided on their product, they leverage LLMs for legal documents research, planning, reviewing, translation and creation, with robust integration with common work tools such as Microsoft Word and Outlook. Antas da Cunha ECIJA and RFF Lawyers reported an increase of 20.7% and 19% average time gain per individual working with Legau's tools.

Another growing portuguese LegalAI startup is Affine Law by NeuralShift¹², with a specialized focus on deep research capabilities of both public legal information and firm specific document database, with claims of 60% faster research speeds. Some of the partnerships involve firms like Melo Alves, Proença de Carvalho, Vieira de Almeida and Gama Glória¹³, with the latter commenting an interesting observation regarding the use of Affine, in my opinion: *" Research is efficient, and the end result is equivalent to having an unlimited number of junior associates dedicated to legal research. This gives the team more freedom to focus on strategic, high-impact tasks, allowing us to achieve more with the same number of lawyers on a project."* Testimony's like this highlight the real world impact LLM based systems can have when done tailored to customer needs.

Meanwhile, Abreu Advogados and Morais Leitão, are firms, that adopt foreign tools, such a Microsoft 365 Copilot¹⁴ and Luminance¹⁵, that enable automated email drafting, contract processing and analysis, question-answering, among other features, based on the main use cases of LLMs in LegalAI.

A common thread between all, is the commitment to data privacy and security, with all giving guarantees that follow EU guidelines of the topic.

⁷<https://justica.gov.pt/Servicos/Guia-pratico-da-Justica>

⁸<https://legau.com/>

⁹<https://legau.com/customer-stories/cca-legau/>

¹⁰<https://legau.com/customer-stories/rff-legau/>

¹¹<https://legau.com/customer-stories/adcecija-legau/>

¹²<https://www.affine.law/pt>

¹³<https://www.affine.law/pt>

¹⁴<https://abreuadvogados.com/noticias/abreu/abreu-advogados-e-pioneira-em-portugal-na-adocao-transversal-da-p>

¹⁵<https://www.mlgtts.pt/pt/conhecimento/noticias/Morais-Leitao-desenvolve-projeto-inovador/>

2.5 Related Work

Regarding the theme of legal judgment prediction, there have been various studies through time, that have investigated the application of various ML and DL techniques. Unfortunately, I was not able to find any similar study, particularly in the portuguese context, which could signify the novelty of this dissertation in Portugal.

Nonetheless, based on my research, the most organized way of discussing these findings is based on the type of techniques used, from traditional ML methods, to more complex Neural Network architectures, that include LLMs.

2.5.1 Classical Machine Learning Approaches

Aletras et al. (2016) made a foundational study, that has been cited thousands of times, where they used primarily a linear kernel Support Vector Machine(SVM) to predict whether there was a violation or not of Article 3 (prohibition of torture), Article 6 (right to a fair trial) and Article 8 (respect for one's privacy and family life) of the European Convention on Human Rights (ECHR). A N-gram algorithm was used to extract the textual features from the 250 cases regarding Article 3, 80 cases regarding Article 6 and 254 regarding Article 8. Finally, they achieved an average accuracy of 79% across all articles. An interesting pattern noted was that the most decisive predictive factors, were based upon a case circumstances, rather than raw legal principles. They conclude that this can be an indicative on how judges make decisions, particularly in ethically charged cases, like the one's treated at ECHR.

Another study, made in Europe, was Sulea et al. (2017), where they also used linear SVMs to predict a case outcome, but this time on French Supreme Court rulings. The dataset was made of 126,865 cases, spanning from the 1880s until the 2010s, from various doctrines, such as civil, criminal, social and commercial. The textual features were extracted using common NLP techniques, such as Bag-of-Words(unigrams) and bi-grams, and the results obtained were an accuracy of 96.9% and a f1-score of 97%. When the researchers compared their results with previous work, they found that their model outperformed various studies, but highlighted that true comparisons were difficult, due to language and most importantly jurisdiction differences, where they conclude french justice leaves less room for interpretation for judges, when compared to the previous studies cited in their paper. This highlights an key aspect, where the legal context might difficult models generalization, underlining the importance of specialized studies based upon local jurisdiction, like the one proposed in this thesis.

More recently, a study made in Thailand by Anantathanavit et al. (2024), also employed various classical ML methods like Logistic Regression, Decision Trees, Random Forests, SVMs and Naive Bayes, to predict the outcome of Supreme Court Cases. This time the textual features were extracted in various ways, such as Count Vectorizer, TF-IDF and Word2Vec and Doc2Vec embeddings. These featurizers + ML models were compared to a thai fine-tuned BERT model, and in average the BERT model outperformed all previous combinations, with an accuracy of 74.6%. Due to these findings, the researchers recommended the further exploration of DL methods, to handle the complexity of legal documents.

These methods were the first to be applied and remain important to contextualize the evolution of legal judgment prediction. However, after revisiting the proposal scope and expected performance trade-offs, classical ML methods will not be implemented in this dissertation, as the experimental focus is on DL and LLM approaches.

2.5.2 Neural Network Approaches

An early study, that applied DL methods to this problem, was Long et al. (2018), used a Bi-directional GRU to encode the textual features, then a pair-wise attention mechanism to model the complex interactions of the law documents, while increasing its interpretability. Finally, final convolutional layer is employed to predict the final outcome of the case. Their model was trained and evaluated on 100,000 from the Supreme People’s Court of China, and achieved an accuracy of 82.2% and a f1-score of 83.4%, surpassing all ML or simpler DL baselines. An interesting aspect of this study, and many other chinese ones, is their large datasets, commonly taken from the China Judgments Online website¹⁶, which contains hundreds of millions of carefully structured legal documents. Other examples of massive legal databases are the USA’s Case Law¹⁷ or the EU’s EUR-LEX¹⁸. When considering this dissertation data source, dgsi.pt¹⁹, one must be aware of the smaller size of the database. Another interesting field of approach in the portuguese context, could be the enhancement of this public source of legal documents, that could be further explored in future research.

Continuing with Neural Network approaches, I highlight a brazilian study, Bertalan et al. (2024), where they used various classifiers, including a Hierarchical Attention Network(HAN), composed by a mix of GRU and Attention mechanism layers, to predict the outcome of 2000+ homicide and corruption cases in the state of São Paulo. The HAN model achieved an accuracy of roughly 99% in both types of cases, but did not significantly outperform other simpler baseline models, such as a simple Regression Tree, a SVM or a simple GRU network, which achieved accuracies between 95-99%. One could argue , the deployment of more complex models, like HAN, particularly in a small dataset like this, may not be as cost effective, but more research analysis for my hypothesis is needed. Another brazilian investigation, Menezes-Neto et al. (2022), predicted the reversal or confirmation of appeals in the 5th Regional Federal Court in Brazil. This study, had to its disposal a much larger dataset, than the previous one, with over 700 000 cases. They used various DL architectures, such as AWD-LSTM, BERT+LSTM and BigBird, and all surpassed legal experts in terms of Matthews Correlation Coefficient(MCC), with the best model, AWD-LSTM achieving a MCC of 0.3688 against the experts 0.3531 average MCC score. Although these values are not distant, by quoting the researchers, a major justification for these ML/DL models can be observed: *“It would be nearly impossible for our expert panel to evaluate all entries within our validation and test datasets.”* Basically, instead of dealing with roughly 76 000 test cases, these legal experts, only dealt with 690. Ultimately, the relevance of brazilian studies, is due to the lack of similar research in Portugal and also due to the similarities between the portuguese and brazilian legal systems, due to the inheritance of the judicial matrix during the colonization period.

Finally I mention and analyze a few studies regarding the use of LLMs in this problem, since it’s the main focus of this thesis.

Kmainasi et al. (2025) is a recent study, with very similar objectives and methodology as the proposed in this thesis. They utilized public data from the Saudi Ministry of Justice, with 3752 training samples and 538 testing samples. One of the limitations of this investigation is the smaller size dataset, but most importantly the niche language they are in, Arabic, which has less research associated with it, similarly to European portuguese, in this specific problem. To achieve their goal of judicial sentence prediction, they compared baseline Llama-3.2-3B and 8B models, only with prompt engineering techniques, like few-shot learning for example, versus the same model with the same instructions, but fine-tuned to the specific

¹⁶<https://wenshu.court.gov.cn/>

¹⁷<https://case.law/>

¹⁸<https://eur-lex.europa.eu/>

¹⁹<https://www.dgsi.pt/>

legal context. They found an increase of an BERT-score-F1-score based on contextual embeddings to measure semantic similarity) of 54% and 58% for the 3B and 8B non fine-tuned models respectively, versus the 74% and 76% of the fine-tuned counterparts, concluding that fine-tuning was a key aspect in the performance improvement. Alongside these quantitative results, they also analyzed the qualitative outputs of the models, and found that the fine-tuned models, were able to justify their predictions better in aspects like legal language, coherence and clarity to name a few, when compared to the non fine-tuned models. This research, will be a key reference for my dissertation, due to the similarities with my proposed methodology in the LLM modelling.

Finally, I highlight studies that extrapolate the simple use of LLMs, but enhance them with other techniques, such as RAG, to improve models performance and interpretability in case prediction. Analyzing Nigam, Patnaik, et al. (2025) and Peng et al. (2024), both utilized RAG techniques, to improve the use of LLMs, open source and proprietary, respectively. In the first study, their case text + RAG system with combination of statutes, precedents outperformed the baseline predictions, using only the integral text of the case, with an accuracy of 65.86% and a f1-score of 63.96%. On the second study, had between 85% and 91% accuracy for the proprietary OpenAI models they used. In other lexicial and semantical metrics, such as ROUGE-L or BERT-score, the RAG system also outperformed the baseline. The second study, had a similar approach, and also concluded that their RAG system "Athena", exceeded the baseline LLM model in all evaluation metrics. These studies, serve as key references, by showing the potential external database systems can have in the task of legal judgement prediction. If possible, I will try to implement this technique and compare its predictive but also interpretable power, versus the use of just LLMs.

Across these studies a few differences from the present work stand out. None of them covers Portugal or European Portuguese, and the two closest proxies are imperfect, with Brazil sharing a judicial matrix but not a legal system, and Saudi Arabia sharing only the condition of a lower-resource language. The comparisons they draw tend to stay within a single family of methods, classical ML against classical ML, ML against a fine-tuned BERT, or a prompted LLM against its fine-tuned counterpart, while this thesis places a fine-tuned domain-specific encoder and a locally-run open-weight reasoning LLM on the same held-out test set and the same hardware. Each of them also works within one legal domain, whereas the same pipeline is applied here to two deliberately contrasting ones. Lastly, explainability appears in this literature mostly as an architectural property, through attention weights in Long et al. (2018) or retrieved statutes in the RAG systems, and is rarely put before legal practitioners. This dissertation does not close that gap either, but it does submit the heuristic label extraction to a panel of practising judges (section 6.3) rather than treating the labels as given.

To conclude, with this small review of what's been done in the field of legal outcome prediction, I hope to have shown the main patterns and methodologies uses, as well as their most notable differences to my own methodology. They will serve as foundational work to build and adapt to the portuguese legal context.

Table 2.1: Summary of Key Studies in Legal Judgment Prediction.

Legend: ML-only ML vs DL DL focused LLMs focused					
Author/Year	Jurisdiction	Data	Method	Results	Trade-offs
Aletras et al. (2016)	ECHR	250 (Art. 3), 80 (Art. 6), 254 (Art. 8) cases	SVM, N-gram	79% accuracy; case circumstances most predictive	Pioneering work; Small dataset limited to 3 articles
Sulea et al. (2017)	France	126,000+ Supreme Court cases	SVM, BoW, bi-grams	97% accuracy, 97% F1	Large dataset; Warning to jurisdiction/language differences; Recommend specialized studies
Anantathanavit et al. (2024)	Thailand	1,200+ Supreme Court cases	ML (LR, DT, RF, SVM, NB), BERT	BERT: 74.6% accuracy; outperformed classical ML	Classical ML underperformed, DL recommended; small dataset
Long et al. (2018)	China	100,000 Supreme Court cases	Bi-GRU, attention, CNN	82.2% accuracy, 83.4% F1	More complex architecture ; Large dataset
Bertalan et al. (2024)	Brazil	2,000+ cases of homicide and corruption of a State Court	HAN (Bi-GRU+Att), RT, SVM, GRU	HAN: ~99% accuracy; baselines: 95-99%	Small dataset; HAN more complex architecture but not much better than baselines
Menezes-Neto et al. (2022)	Brazil	700,000+ Supreme Court appeals	AWD-LSTM, BERT+LSTM, Big-Bird	AWD-LSTM: Models MCC 0.37 vs experts MCC 0.35	Models scale much higher than experts but only perform slightly better
Kmainasi et al. (2025)	Saudi Arabia	4,000+ Commercial Law cases	Llama-3.2-3B/8B, prompt, fine-tune	Fine-tuned: 74-76% BERT-score-F; non-tuned: 54-58%	Small dataset; niche language; fine-tuning crucial
Nigam, Patnaik, et al. (2025)	India	56,000+ Supreme Court case	LLM + RAG	66% accuracy, 64% F1; RAG outperforms baseline	RAG improves interpretability; Large Dataset; Open-Source models only;
Peng et al. (2024)	China	256 Supreme Court cases	LLM + RAG	RAG system outperforms baseline LLMs in Accuracy (91%)	RAG improves interpretability; Proprietary models only; Limited data

2.6 Ethical and Legal Considerations

2.6.1 Key Challenges

It's impossible to discuss applications of AI, particularly LLMs in Law, without addressing in a thorough manner, the ethical and legal implications that arise from it.

Studies like Remolina et al. (2024) or Jialin (2026), highlight that one of the major risks of using LLMs in law practice, is the perpetuation and amplification of existing biases, inherent in the training data. This problem, leads to an increased unfairness in legal outcomes, by enhancing existing prejudices that may exist regarding gender, race, socioeconomic status, etc.

Other common concerns, among peers is the lack of transparency and explicability of generative models, due to their complex architectures and processes of decision making, known as the "black box" problem. Alongside this, the natural risk of hallucinations and non-deterministic model outputs,

exacerbates the trust issues. For example, there have been many reports of misuses of LLMs, particularly popular chatbots in the generation of legal analysis, outputting inexistent laws, jurisprudence or doctrine. Even in Portugal, there have been news reporting such cases like ²⁰. In my view, this problem goes beyond model technical issues, and bleeds into user-side malpractice, usually associated with lack of literacy regarding these tools. In this study, I try to mitigate these issues, by using explainability techniques and if possible validation by legal experts, to evaluate if the outputs are coherent, justifiable and trustworthy.

The last cited concern above, leads to the next major ethical question: who is held accountable for artificially generated content, particularly in the legal practice? For example, if a judge uses an LLM or any AI tool, to analyze, draft or even decide over a given case, is the judge liable? Could it be the developers or the suppliers of the tool? When one pays for the services of a legal professional, the cost goes beyond the output of a document output or analysis. The biggest value, lies in the human aspect of understanding and dealing with complex cases.

In my view, the ethical and legal implications of using these AI tools in the justice system, must be carefully evaluated, due to their potential harmful and complex outcomes. And to mitigate these concerns, it must be clear what they can or cannot be used for, and when human supervision should be mandatory. Fortunately, there is an increase in literature, regarding on how to combat some of these issues, aforementioned. Let's explore them in the next section.

2.6.2 Key Regulations and Governance

To mitigate concerns like bias, transparency or liability, there have been introduced some regulatory frameworks, particularly in the EU, with the EU AI Act ²¹ and the GDPR ²².

The GDPR, focuses on users data privacy and security, mandating strict guidelines on how business collect, process and manage personal data. In an age where LLMs are trained on a voluminous amount of data, it's crucial that users personal data does not get or at the least minimize their utilization, for the improvement of these models. If not, in case of model adversarial attacks, personal data could be hurtfully extracted, causing huge damages to users. Also, with less personal data, risks of biases and discriminations minimize, since these models won't learn from sensitive features.

The EU AI Act establishes a comprehensive set of guidelines on how AI models must be operated within European territory. Briefly, it clearly segments AI systems into 4 risk categories: unacceptable risk (e.g. manipulative systems, discriminative systems), high risk (e.g. systems that if fail or are misused, could cause immense harm to people), limited risk (e.g. chatbot systems or deepfakes) and minimal risk (e.g. most AI applications, like classical ML and DL approaches). Moving forward, the AI Act also mandates clear transparency and accountability measures, from providers (developers) and professional users, to ensure AI systems are more understandable and trustworthy, to the end users.

Both of these regulatory frameworks, serve as major guidelines in Europe, particularly in Portugal, with literature such as Wengorovius (2025), recommending their strict adherence, while discussing the current applications both operational and planned, some of them aforementioned in Chapter 2.4.4.

2.6.3 Reflections on Ethical and Regulatory Considerations

The domain I'm researching would fall into the high risk category, even if done with human supervision and liability. Such systems, particularly those dealing with major human rights like fair trial, privacy,

²⁰<https://www.publico.pt/2025/02/11/sociedade/noticia/alegado-uso-ia-acordao-relacao-alvo-processo-averiguacao>

²¹<https://artificialintelligenceact.eu/the-act/>

²²<https://gdpr-info.eu/>

freedom, to name a few, cannot be used without strict rules and oversight, in my perspective. Although some of the key recommendations and guidelines cited here are a great start, the most important aspect is the human-in-the-loop, whether it's a judge, a lawyer or another legal professional, they are the most important factor to consider. They must be responsible for the use of these tools, because these are not made to replace them, rather to augment their capabilities and efficiency, improving the justice system as a whole. For this to be effective, training and education of these professionals is fundamental, to avoid malpractices.

To conclude, I believe LLMs and other AI tools, can have a huge positive impact in the Portuguese judicial system, but only if done and used in a responsible and ethical manner.

2.7 Conclusion and Gaps in the Literature

In this stage of the dissertation, I've presented a comprehensive review of existing literature on the applications of LLMs in the legal domain, particularly in the task of legal outcome prediction. I initiated with a brief review of the Portuguese Justice System structure and terms, to contextualize the problem this thesis tackles. Then I explored the evolution of Language Models, from statistical methods to the current state of the art LLMs, highlighting at each stage the main advantages and limitations of such models. Following this, I analyzed main use cases of LLMs in Law practice, from Legal Prediction, Document Analysis and Research, underscoring specific tasks within each category. Furthermore, I presented current limitations and challenges in the field, both technically and ethically, spanning from model costs, interpretability, data privacy to bias and liability concerns. However, I also proposed potential solutions and future indications, based upon recent literature and guidelines, to mitigate these issues such as strict human supervision and responsibility, model specializations in law contexts and adherence to regulatory frameworks like GDPR and the AI Act. Finally, I reviewed some related work regarding legal judgement predictions, from classical ML methods to more complex DL and LLM methodologies, spotlighting key studies and their findings.

Given this review a key gap can be identified in the literature, such as the lack of studies in the portuguese legal context, mainly regarding legal outcome prediction, let alone with LLMs. Due to the novelty of this task in the portuguese domain, I adopt and adapt various existing methodologies from abroad, such as the ones presented previously, to the specificities of the portuguese legal system. This study presents itself as a baseline for future research in this field locally, and I hope to contribute to the advancement of LegalAI in Portugal.

Chapter 3

Data

As aforementioned in the Background Chapter, I used publicly available pseudonymized data of judicial decisions made in portuguese courts, meaning sensitive and confidential information was removed to protect the involved parties privacy.

To extract it, since no API exists, I had to scrape the webpages directly. For that I used Python libraries like BeautifulSoup¹ for HTML parsing and manipulations, Crawl4AI² to asynchronously scrape the documents in the webpage, which becomes useful when scraping a large corpus, which was not the case, so for consistency reasons I scaled down to an ordinary sequential scraping.³

Now describing the data more precisely, I scraped decisions from Domestic Violence(DV) and Breach of Contract(BoC) related decisions, from several portuguese courts like Peace Courts, second instance courts and the Supreme Court of Justice. Behind this choice, were previous conversations with lawyers and law professors, which pointed conflicting reasons in which type of cases to tackle for this novel challenge in Portugal. Some argued that Penal cases like Domestic Violence, would be more interesting, due to their social impact and more simple law's application, but noted that dealing with these more subjective and ethical cases, could lead to less trust by experts in the models outputs. Others argued that Civil cases like Breach of Contract, since structured in a more objective manner, would be more suitable for this study, but warned that it deals with a broader and nuanced law code that could affect the final outcome, making it harder for the models to learn. Given both perspectives, and the novelty of this research locally, I decided to use both domains, compare results, and conclude which type of cases would be more suitable in future research or if there is no significant difference.

At the end of the scraping phase, I obtained a total of 6,627 documents, which 4,368 were BoC cases and the remaining 2,259 were DV, extracted from dgsi.pt⁴ and jurisprudencia.csm.org.pt⁵ websites, which are official public repositories of judicial decisions in Portugal, managed by the Ministry of Justice and the Superior Council of the Judiciary respectively.

3.0.1 Preprocessing

It's important to note, that these documents had metadata associated with them, varying from court to court, but the ones most common were case number, case summary (which at times unreliable due to lack

¹<https://www.crummy.com/software/BeautifulSoup/>

²<https://docs.crawl4ai.com/>

³Scraping and preprocessing pipeline code available at: <https://github.com/Ecologykk/Master-Thesis---Msc-Data-Science/tree/main/src/scrapers>

⁴<https://www.dgsi.pt/>

⁵<https://jurisprudencia.csm.org.pt/>

of information) date of decision, keywords and final verdict. Some metadata was included during scraping to enrich the dataset and promote better analysis, like case type, court name, URL. Alongside these, I also used regular expression patterns developed after analyzing manually a large number of documents to extract the final verdict deliberations and separate it from the rest of the text. This was done based off the fact explored previously in the Background Chapter, that portuguese judicial decisions have a common structure, defined by law, where the final decisions are always at the tail of the document. Through this analysis, I found judges follow linguistic patterns when writing these deliberations, like first titling the section as *Dispositivo* or *Decisão*, and then using particular connectors like *Pelo exposto* or *Nestes termos* which signal the final decision is being presented. If a final decision could not be extracted, the algorithm would flag for my manual review of the document. This algorithmic extraction, although not perfect, proved effective in most cases with only 4.41% and 5.46% of total scraped documents of BoC and DV respectively, requiring manual review, due to not having a decision extracted while the full text contained the final decision.

After having all the data stored in structured format (i.e. JSON) , I proceeded to prepare the data for both predictive modelling but also for exploratory data analysis, as one of the sub-goals of this study is scrutinizing the publicly available data. The first was dropping the documents that did not have a final decision extracted, as they would not be useful for the predictive modelling task. Manually adding the decisions based off the full text was a possibility, but the manual labor required was not feasible given the small proportion such documents represented. Alongside this, I also removed documents with no full text. This is a problem of older cases (prior to 2002 roughly) present in both repositories, where only metadata was stored. Finally, I also removed duplicate documents, based of the case number identifier. At the end of this cleaning phase, I was left with a total of 1,938 BoC cases and 1,204 DV cases, which represents a reduction of 55.63% and 46.70% respectively from the initial scraped documents. This reduction illustrates data quality issues present in these public repositories. Other problems in these datasets, were the inconsistent summaries, where at times did not clearly reflect on the case, making hard understanding the context without reading the full text. Other issues were incorrect labelling of case types, where a case labelled as DV, was not necessarily about domestic violence, rather a different type of crime. Finally, the main concern with these public repositories is the lack of text standardization, were as of now it seems text is directly copied from the official PDF documents, leading to font and formatting errors, like special characters, line and spacing issues, that diffculted processing of the text. There are no public technical reports or documentation of how these repositories are executed and maintained, so these observations are not fully informed. Alongside this, I contacted the responsible entities, but at the time of writing this dissertation, I have not received any response, that could clarify these issues.

The final step of preprocessing was to manually engineer the classes to predict. Given the complexity of actual court decisions, it's unlikely that a model, particularly a BERT-based one, could predict the exact final decision, as judges must deliberate on multiple points on a case. Therefore, I simplified the problem to predict whether the appellate outcome was KEPT or ALTERED for DV cases, and FAVORABLE, UNFAVORABLE or PARTIAL for BoC cases. For that, I used regular expressions to identify key patterns in the final decision text, that would indicate the decision's direction. I mined for patterns in the final decision text or its shortened metadata form, for partial terms as *parcialmente provido* or *procedente em parte* to segment partial decisions. Then out of non partial decisions, I searched for favorable judicial expressions like *provido* or *procedente*. Contrarily, unfavorable decisions usually contained antonyms like *improcedente* or *desprovido*. I also flagged cases that contained the expression *revogado* and its inflexions as positive outcome, given this meant the appealed decision was overturned, most of the time. On the other hand, expressions like *mantido* or *confirmado* were flagged as negative outcomes, as they

meant the appealed decision was upheld. In hindsight, I noticed a small edge case in the pattern order used to engineer the BoC labels, where expressions like *parcialmente improcedente* are captured by the negation check (checking for unfavorable cues like *improcedente*) before the partial check runs, and are therefore classified as UNFAVORABLE rather than PARTIAL. This joins other reasons discussed in section 6.5 for why these simplified labels are an imperfect proxy for the underlying judicial outcome. Nonetheless, this approach produces two class schemas. DV uses KEPT and ALTERED, while BoC uses FAVORABLE, UNFAVORABLE and PARTIAL. Consider this a pragmatic approximation of appellate outcomes discussed during the proposal defense, and it should be read as a working simplification rather than a perfect reflection of every legal nuance.

After dropping documents where this engineering was not possible, I was left with 1,802 BoC cases and 1,126 DV cases. It's key to note, that these processing steps were validated by the co-supervising judge in this dissertation, to ensure as much legal rigor as possible, regardless of the inherent limitations these heuristics based approach has. Nonetheless, this dissertation also presented external judge validation relating to these engineered classes, as further discussed downstream in section 6.3.

The precise regular-expression patterns used for class engineering are provided in Appendix 7.

3.0.2 Data Splitting Strategy

The complete dataset (1,126 DV and 1,802 BoC cases after preprocessing) was split into three disjoint pools, using temporal order by decision date to prevent information leakage.

First, there is the Gold Test Set, composed by 50 cases per case type, reserved exclusively before any model development and stratified to approximate natural class distributions, with intention of evaluating model performance on genuinely unseen legal precedent. DV contained 29 KEPT and 21 ALTERED cases, while BoC contained 35 UNFAVORABLE, 9 PARTIAL, and 6 FAVORABLE cases. Cases were selected from on or after chronological cutoff dates (DV: 2025-05-01; BoC: 2023-10-01).

The remainder of cases before the gold cutoff date (1,064 DV and 1,752 BoC) were used for fine-tuning LegalBERTimbau and for few-shot example selection for the LLM experiments.⁶ For the training protocol of BERT, this pool was further subdivided temporally into training (85%) and validation (15%) splits (applied to whole documents and therefore floored to the nearest case, giving 904/160 for DV and 1,489/263 for BoC, that is 84.96%/15.04% and 84.99%/15.01% rather than exactly 85/15), ensuring that the validation set contained cases that were chronologically later than those in the training set, similarly to how the gold test set was selected, to ensure no future information leakage into training process.

This temporal split design ensures the gold test set represents predictions on genuinely unseen legal precedent, mirroring the temporal novelty a judge faces with new cases, though not the procedural stage at which a judge actually decides, since the model classifies already written appellate reasoning rather than predicting from unresolved case facts, a distinctive limitation of the field further discussed in section 6.5.

3.0.3 Example of document structures

Understanding these document structures is essential for prompting of these language models, moving beyond simple prediction to explainable legal outcomes. Figure 3.2 provides examples of segmented

⁶The cutoff dates were hand-picked rather than derived automatically, and the training pool was defined as every case dated strictly before the cutoff. For DV, 62 cases fall on or after 2025-05-01: the 50 most recent form the gold test set, and the remaining 12 end up in neither pool, even though they predate every gold test case and could therefore have been used for training without leakage. This was glossed over at the time and only noticed later; it amounts to 1.1% of the DV training data and does not affect the reported results. BoC has exactly 50 cases on or after its cutoff, so no such surplus arises there.

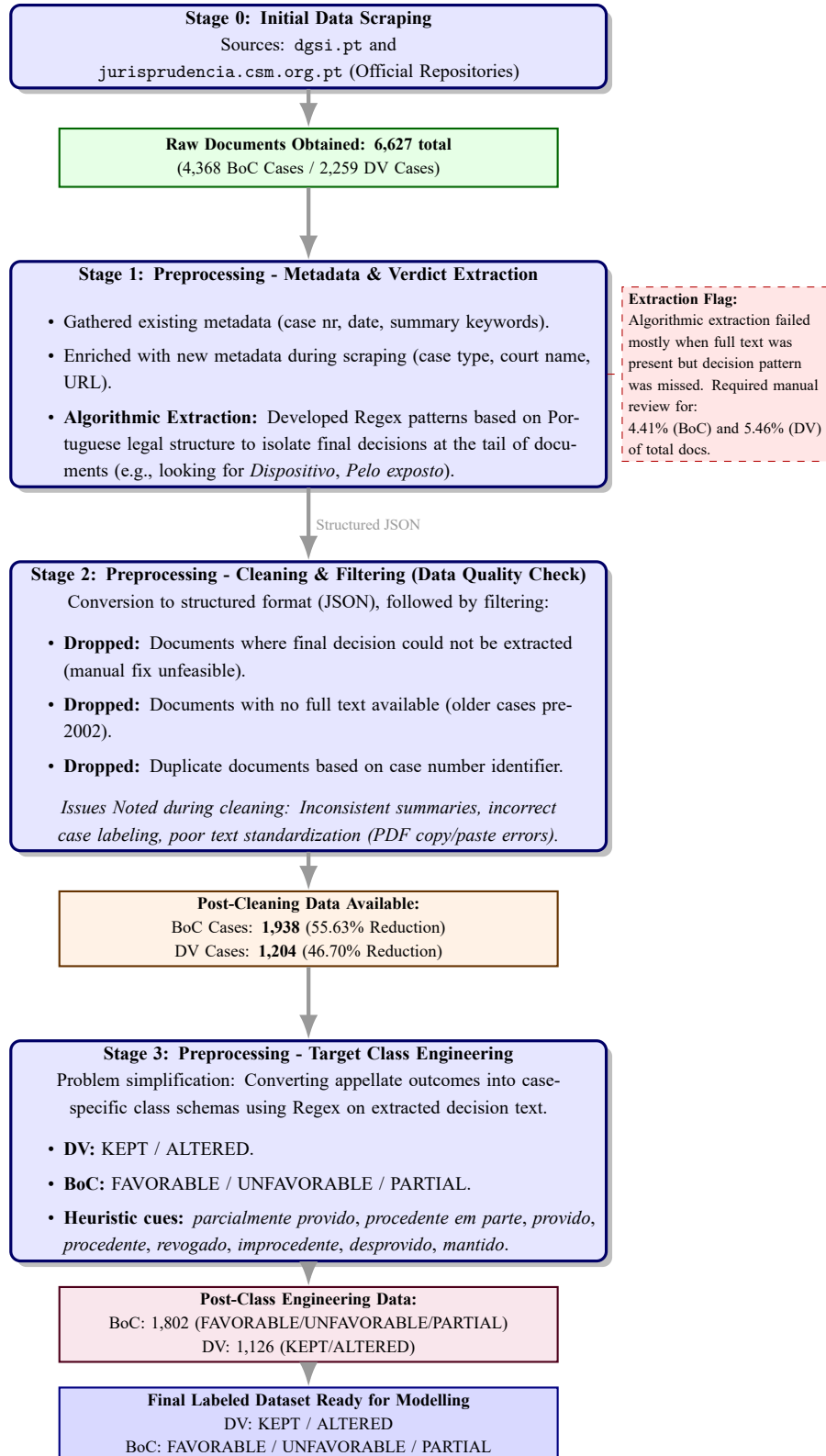


Figure 3.1: Summary flowchart of the data scraping, preprocessing pipeline, and resulting data reduction stages.

(green) and unstructured, prose-based (red) documents. In the left example, because each main section is titled and/or numbered, text pattern mining is easier. In contrast, the right example lacks this structure and adds complexity to decision extraction, relying only on the fact that final decisions are at the bottom of the document and that judges usually start with a formal connector like "*Pelo exposto...*". A few real document examples that are easier and harder to process can be found in the following footnotes.^{7 8}

3.0.4 Exploratory Data Analysis

Basic statistics

The first step of this exploration was analyzing the class distributions of the resulting dataset, because imbalanced classes may lead to biased predictive models, as seen in Figure 3.3 and Figure 3.4.

In the BoC dataset there was an imbalance, with 50.3% of cases being UNFAVORABLE, 28.3% being FAVORABLE and the remaining 21.4% PARTIAL. However, in the DV dataset, the class distributions are more balanced, with 52.2% of cases being KEPT and 47.8% being ALTERED. It's imprudent to speculate on the legal reasons behind these distributions, as it is not possible to know from the total cases, how many were appealed by public prosecutor or by private parties, which could affect how these results are interpreted. Nonetheless, from a data science perspective, it can be concluded that these imbalances, particularly in the BoC ternary dataset, must be addressed during model training, otherwise it could lead to biased classifications.

Another aspect I wanted to explore was document lengths, seen in Figure 3.5 and Figure 3.6. This would be important factor when choosing future LLM models, given their context window limitations. The first conclusion to formulate, based on the aforementioned figures, is that most DV and BoC cases seem to have similar length distribution, however, DV clearly has more lengthy outliers. One could argue the legal complexity of DV cases, that deal with a lot of witness statements and circumstantial evidence, but this would require more specialized legal knowledge to confirm, which is outside of the scope of this study. I took note in both case types of their most extreme length outliers, to further investigate them manually, and found that for BoC case, it was a simpler contract dispute at a Peace Court that ended with mutual agreement between parties. While in the DV case, the shortest document was a denial of further investigation by a Court, regarding an appeal made by the public prosecutor. As for their maximum lengths, both were complex cases, with BoC being a financial fraud case with multiple involved parties and the DV case being a murder trial with extensive context, evidence and analysis and reasoning of the case. All extremes seem to be what one could expected of these type of cases outliers.

Finally, the last statistics I wanted to learn, was term frequencies and most importantly their distinctiveness per class of documents, described by the TF-IDF metric(i.e, how frequent a term was in a document while being infrequent in all other documents in a given corpus). For this task, I had to remove portuguese stopwords like *o*, *a*, *de* or *para* to name a few. These were eliminated using NLTK⁹ Python library. Studying BoC most bigrams as seen in Figure 3.9, the obvious term of *contrato promessa* stands out as the most popular. When going down the list, the remaining words don't actually give much information regarding the class as many overlaps exist. More interestingly, most words seem to be com-

⁷Clean examples: <https://www.dgsi.pt/jtre.nsf/134973db04f39bf2802579bf005f080b/0411f0b2f6a1e84f80257fe7004d0111?OpenDocument>, <https://www.dgsi.pt/jtre.nsf/134973db04f39bf2802579bf005f080b/fa40d22e7c74973e8025849b0034dbf0?OpenDocument>.

⁸Messy examples: <https://www.dgsi.pt/jtre.nsf/134973db04f39bf2802579bf005f080b/2e257f8ff1747a0e80257de10056fb14?OpenDocument>, <https://www.dgsi.pt/jtre.nsf/134973db04f39bf2802579bf005f080b/7f38c6f270ba1442802586c0006a32af?OpenDocument>.

⁹https://www.nltk.org/howto/portuguese_en.html

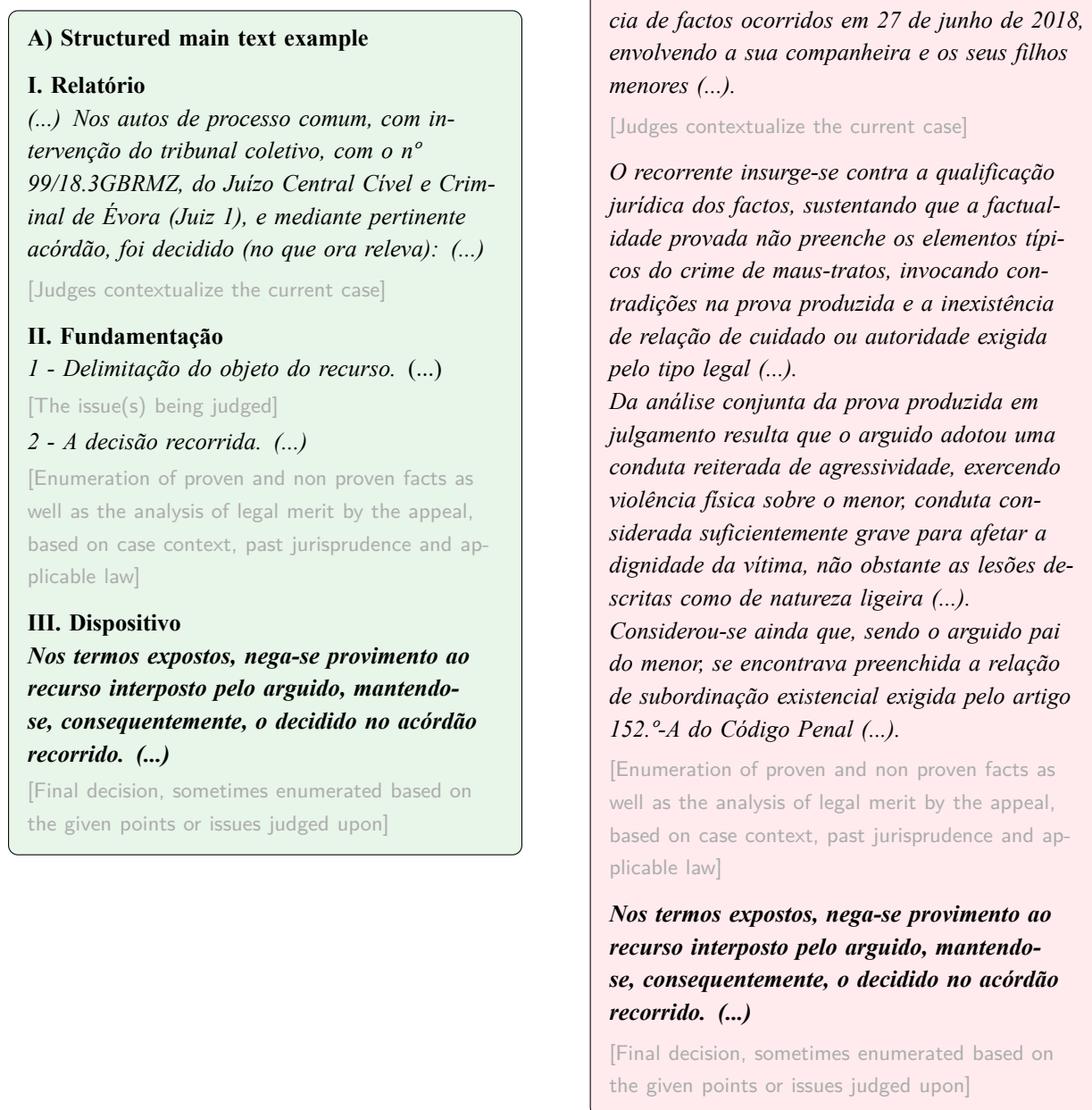


Figure 3.2: Structural contrast between section-segmented (green) and mostly prose (red) judicial decision documents. These documents are just toy representations of easier-to-process and extract final decisions in green and harder ones in red.

mon legal formalism, like *código civil* (civil code laws), *resolução de contrato* (contract resolution) or *compra venda* (buy-sell), that although important when considering their semantics, they act as noise for purely frequentist models. For that reason, I removed manually a few of these legal stopwords, to see if

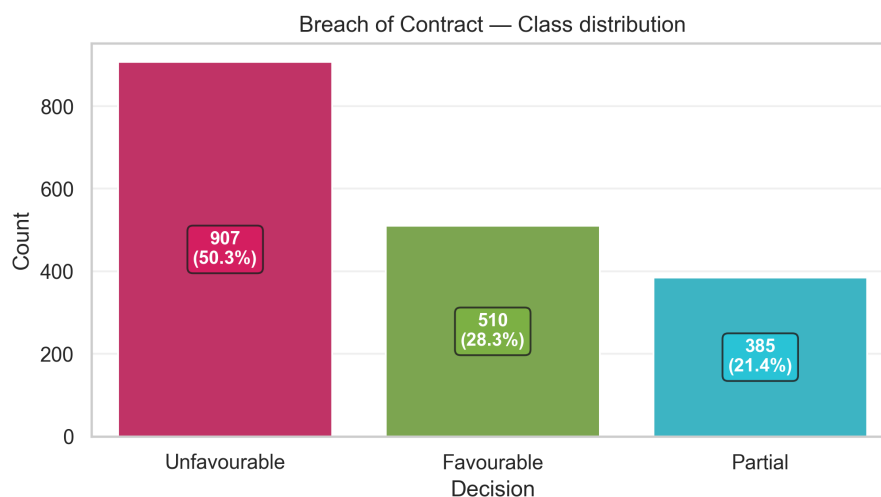


Figure 3.3: Breach of Contract Class Distributions

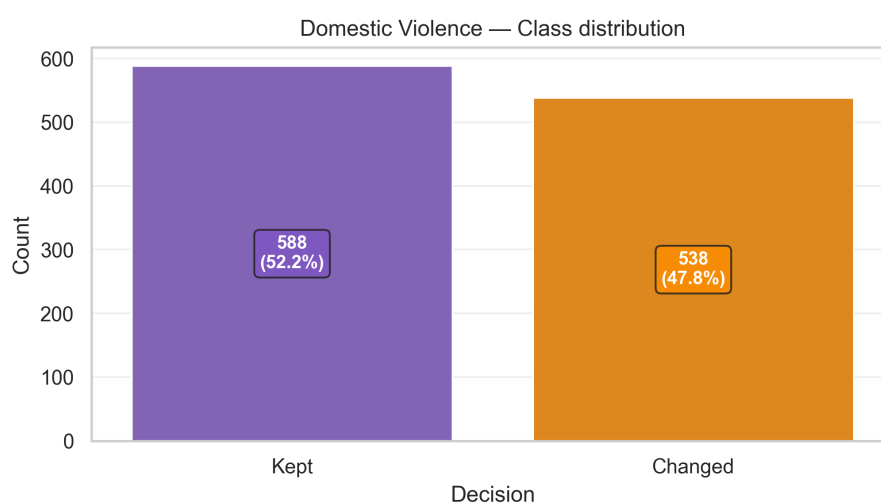


Figure 3.4: Domestic Violence Class Distributions

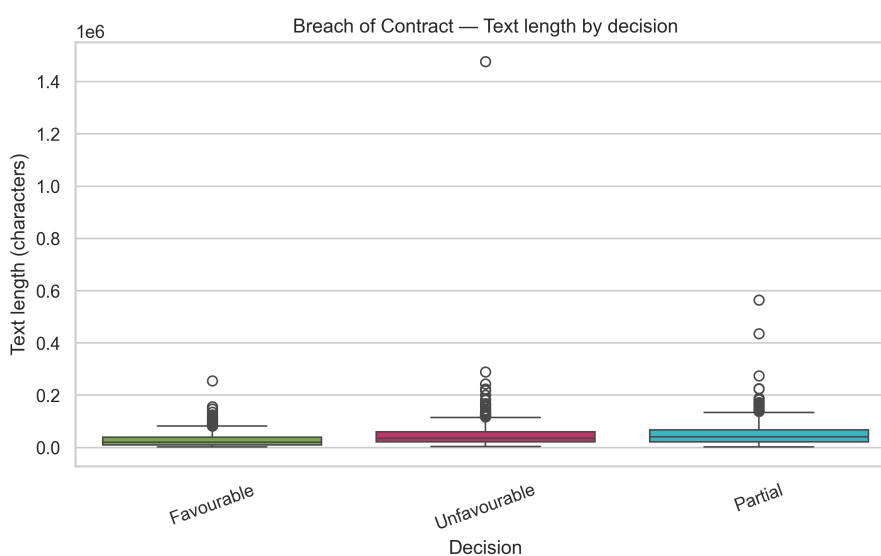


Figure 3.5: Document Length Distributions for Breach of Contract cases

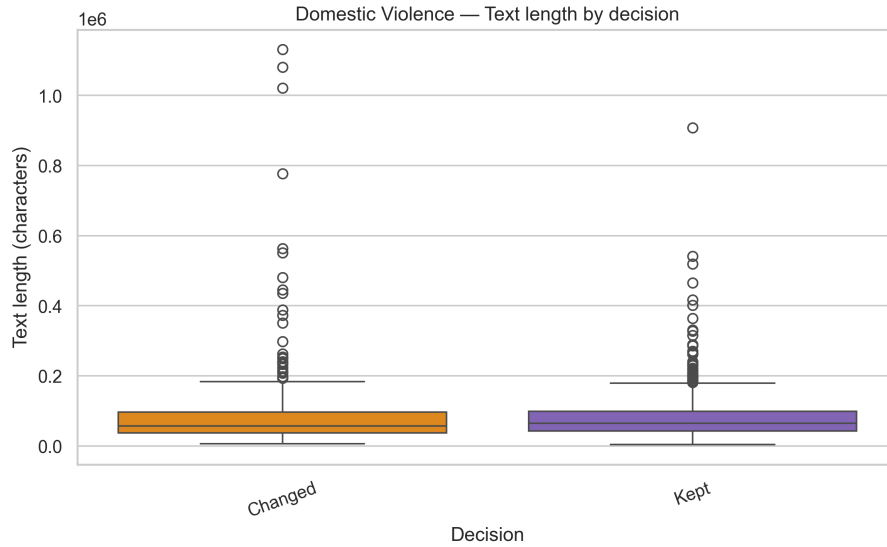


Figure 3.6: Document Length Distributions for Domestic Violence cases

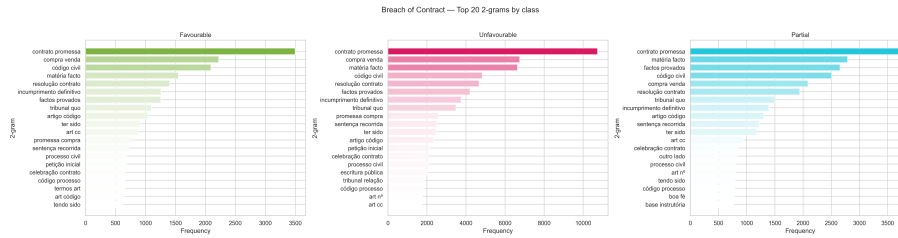


Figure 3.7: Top 20 Most common bigrams in the BoC dataset.

more distinctive terms appeared, and more case specific words start to appear, however, no clear pattern emerged, which is omitted from the figure for brevity purposes. When analyzing the most distinctive terms per class using TF-IDF, through Figure 3.10, a similar pattern of clear domination by legal formalism terms is visible. Similarly seen in DV figures in Figure 3.7 and Figure 3.8, a similar pattern of legal formalism domination appears, with terms like *código penal* (penal code), *violência doméstica* (domestic violence) being the most common and distinctive, adding no specific information regarding each class. The clear conclusion from this analysis, is that legal formalism is very present in the documents, and that if not filter, models that use frequency features, may struggle to learn classes. I searched for studies that explored Portuguese legal stopwords, and found the following technical work by Rodrigues et al. (2025), that used a large corpora of brazilian legal texts, to automatically extract legal stopwords while validated by brazilian legal experts. They composed a list of 3,602 terms, however in their study, they found no significant accuracy improvements in classification tasks, when removing these stopwords. Since in this case, the list of words was done manually and on a smaller scale, fair comparisons cannot be made. So, since their research found no significant improvements, I do not remove these legal stopwords when implementing predictive models. Nonetheless, this is an interesting avenue for future research, particularly in portuguese legal texts, where no such studies exist to the best of my knowledge.

Potential Biases analysis

As discussed in the Ethical Considerations section of the Background Chapter, it's important to analyze potential biases in the data, that could lead to unfair model predictions. Of course this is a complex

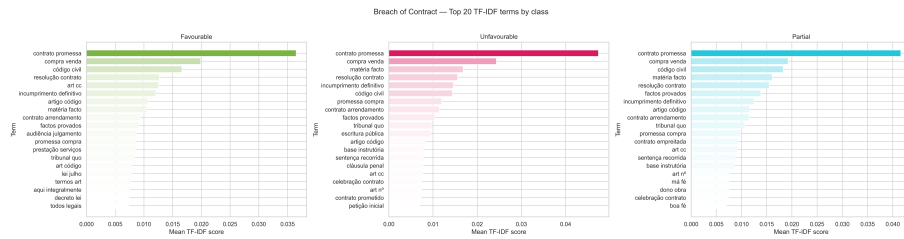


Figure 3.8: Top 20 TF-IDF score bigrams in the BoC dataset. Again at the top it' the unfiltered corpus and at the bottom some legal stopwords are filtered manually.

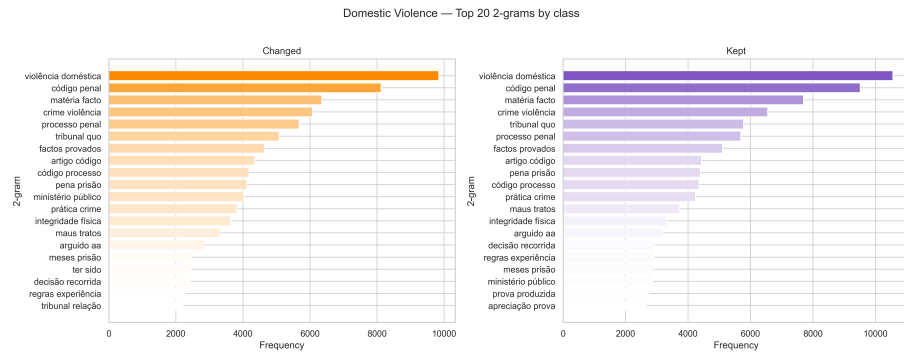


Figure 3.9: Top 20 Most common bigrams in the DV dataset.

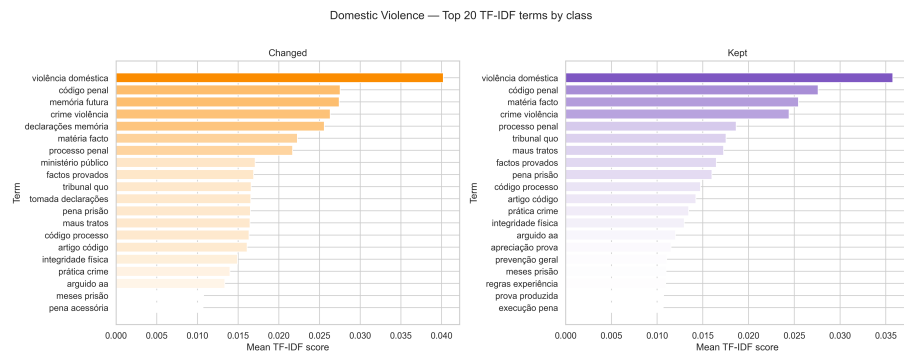


Figure 3.10: Top 20 TF-IDF score bigrams in the DV dataset. Again at the top it' the unfiltered corpus and at the bottom some legal stopwords are filtered manually.

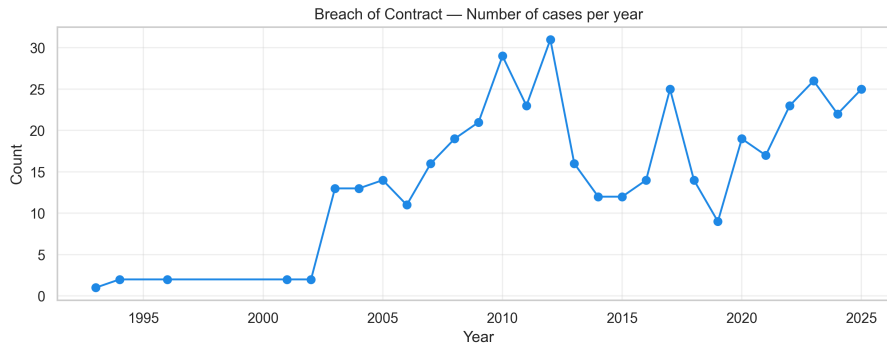


Figure 3.11: Number of cases per year over time for BoC datasets

and exhaustive task, and by no means did I manage to cover all potential biases. Initially it was my intention to further this analysis, using specialized tools like AI360 (Bellamy et al., 2018) or Microsoft Responsible AI Toolbox ¹⁰, however, time and problem complexity constraints, narrowed the analysis scope to the investigations described below.

Firstly, I investigated time-based patterns in the processed datasets. Figure 3.11 and Figure 3.12 show the number of cases per year for both BoC and DV datasets, respectively. It's clear that BoC has older cases, dating back to 1990. However, from the counts, it's only a few of cases per year, until roughly 2004-2005, where the count spikes upwards of dozens of cases per year, until maximum of over 30 cases in 2010 and 2012. In spite of this, the to recent years seems to be unreliable, with some years increasing and other decreasing, without a clear pattern. Comparing this to DV cases, the upwards trend is more noticeable, with roughly a linear increase in cases per year, peaking in 2025 with over 100 cases. This highlights the nature differences of both case types, where BoC cases, although more frequent, there is no cultural or social factor influencing their counts yearly. The same cannot be applied to DV cases, where social awareness and cultural changes over the years, may contributed to more cases being reported and consequently more judicial decisions being made.

Similar conclusions to Figure 3.3 and Figure 3.4 can be drawn from Figure 3.13 and Figure 3.14, respectively, where the class distributions per year, follow in general the same patterns as the overall distributions.

The temporal bias found is addressed through the data splitting strategy described in subsection 3.0.2, which reserves newer cases as the held-out gold test set and uses older cases for training and few-shot examples. This temporal ordering simulates real-world scenarios where models must generalize to genuinely future cases. It is important to note that although older cases may be less directly pertinent for predictive modeling under temporal conditions, they remain valuable for teaching models legal language and reasoning, since these fundamental aspects of legal practice remain relatively stable across periods even as specific laws evolve.

Secondly, I analyzed whether judges' gender had an impact on case outcomes. By scraping the reporting judge names and using a name entity recognition and classification library called gender-guesser¹¹, I compared the class distributions with judges' genders, resulting in Figure 3.15 and Figure 3.16 for BoC and DV datasets, respectively. From this analysis, the proportions of heuristical classes can be seen to be similar across genders, regardless of case type. Also another good indicator is the balanced representation of genders in the both case types, indicating at least in a superficial analysis, no significant gender bias

¹⁰<https://github.com/microsoft/responsible-ai-toolbox>

¹¹<https://pypi.org/project/gender-guesser/>

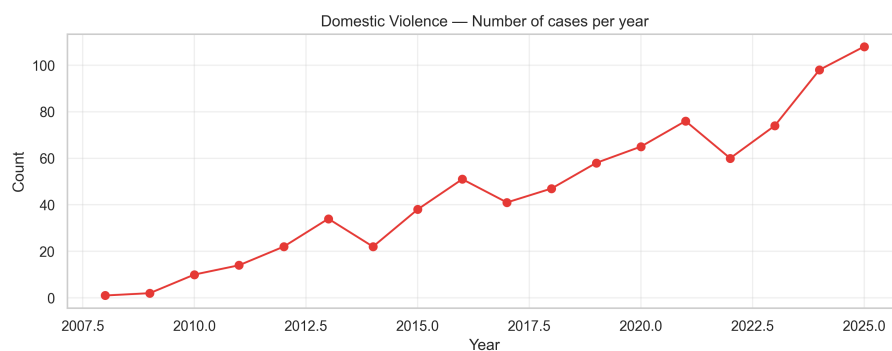


Figure 3.12: Number of cases per year over time for DV datasets

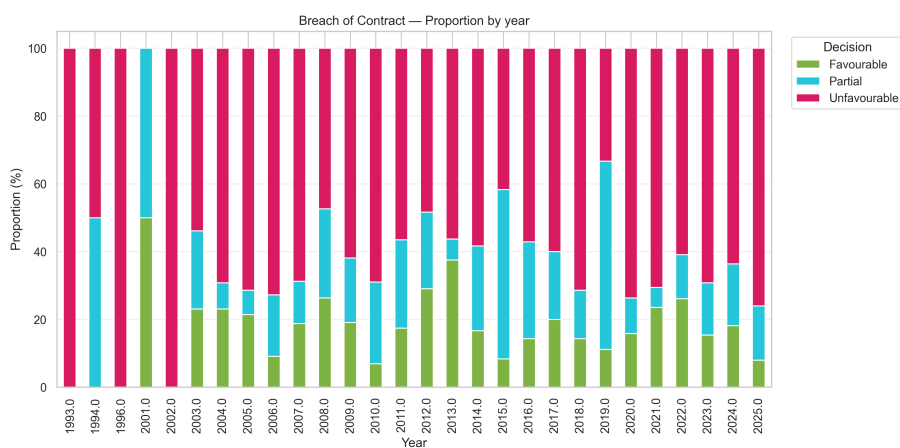


Figure 3.13: Proportions of case outcomes over time for BoC datasets

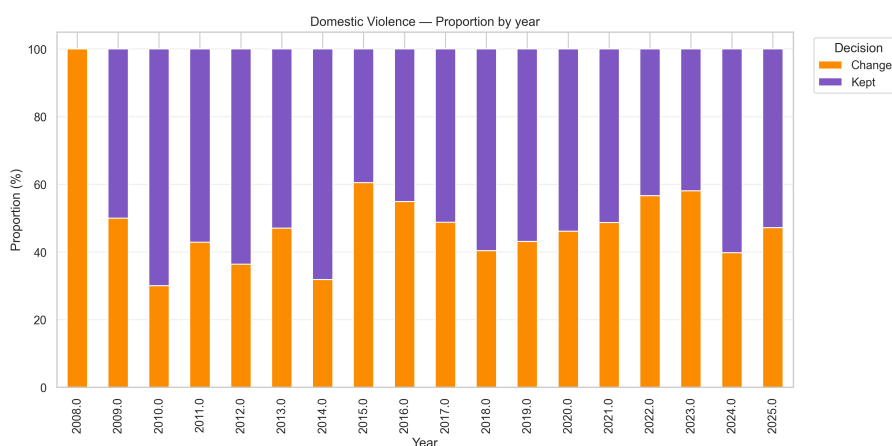


Figure 3.14: Proportions of case outcomes over time for DV datasets

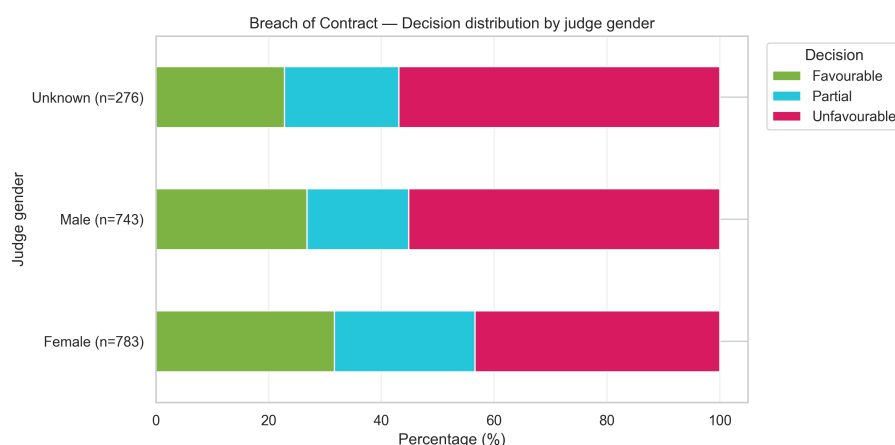


Figure 3.15: Class Distributions of BoC by gender using Python’s gender-guesser. In case of ambiguity, the judge was labelled as undetermined.

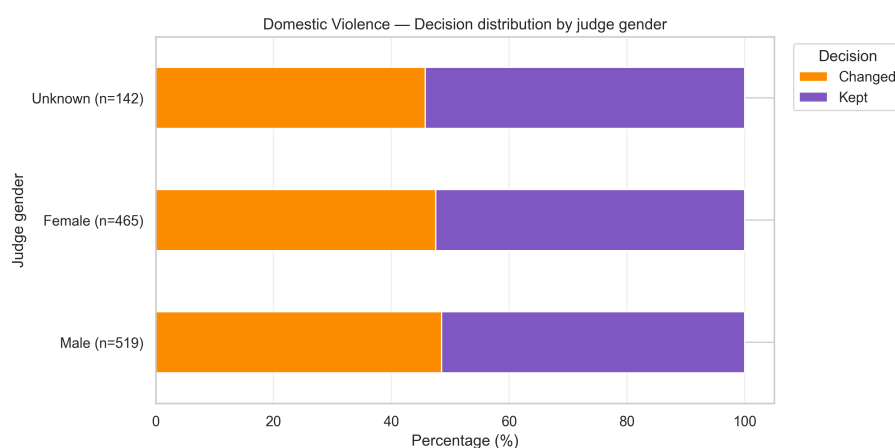


Figure 3.16: Class Distributions of DV by gender using Python’s gender-guesser. In case of ambiguity, the judge was labelled as undetermined.

present in the data.

Another pattern I wanted to explore was court decisions distributions by region. By cross-referencing court names with the respective proportions of classes, I obtained Figure 3.17 and Figure 3.18, which show concordant patterns with the overall class distributions, previously analyzed in Figure 3.3 and Figure 3.4 for BoC and DV, respectively. However, this analysis allows one to understand differences between court regions or types (e.g, comparing Peace Courts to second instance courts). Taking the total number of cases per court in consideration, Peace Courts have a higher proportion of favorable decisions, contradicting the trend of all the other judicial courts and the overall class distributions. Since Peace Courts deal with less complex civil cases, like contract disputes of lower monetary values, it’s often easier for appellant parties to have more favorable outcomes, due to the documental nature of evidences, that judges can analyze more objectively. For all the Appeal Courts and the Supreme Court of Justice, the same reasoning cannot be easily applied due to the greater complexity of cases, often associated with financial and bureaucratic crimes, that involve beyond simple contract analysis and require deeper investigations and reasoning from police and juridical authorities.

As for DV cases, since only dealt by judicial courts, the overall class distributions are observed in all courts, with exception of Coimbra and Guimarães Appeal Courts, where there is a inversion of pro-

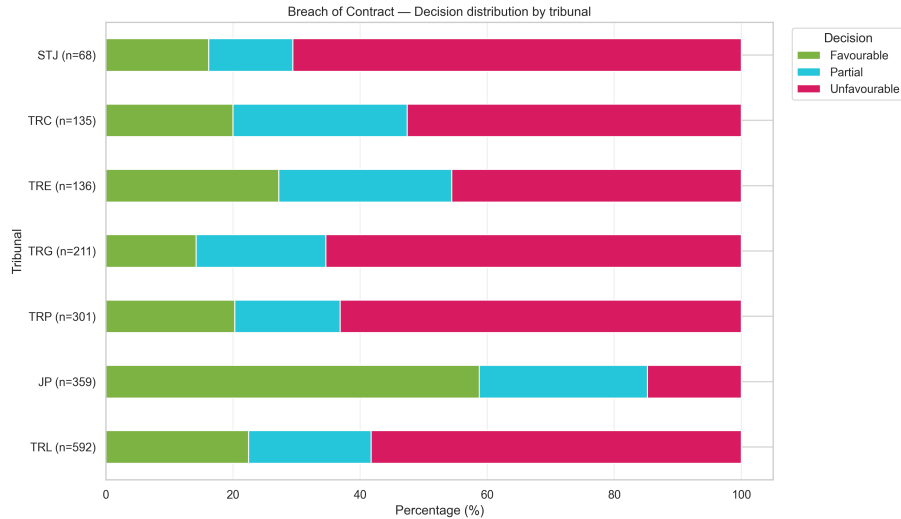


Figure 3.17: Class Distributions of BoC by court region. The y axis legend translated means: "STJ" stands for Supreme Court of Justice. "TRX" means an appeal court where X is the first letter of each city name(e.g, P for Porto, C for Coimbra and L for Lisbon). Finally "JP" stands for Peace Courts.

portions, with more unfavorable decisions, contradicting the overall trend. I cannot speculate on reasons behind these patterns, as it would require further legal and possibly even local sociological knowledge, that is outside the scope of this dissertation.

The last exploration I performed was analyzing words and phrases semantically associated with each class, given that frequency based analysis, was not sufficient to understand class separability. This analysis could prove valuable when choosing embeddings techniques for future predictive models.

Generic word-embedding experiments were explored initially prior to proposal defense but are omitted here in favor of the LegalBERTimbau document-embedding analysis presented below, which is more relevant to actual modeling approaches took in this dissertation.

Since fine-tuning these embedding models adds another layer of complexity and resource requirements, I searched for other pre-trained models that could be used off-the-shelf and that were trained on portuguese legal texts, reducing the need for fine-tuning. I found a sentence embedding model, namely LegalBERTimbau¹², developed during the Master's thesis of (Melo et al., 2023) during the "IRIS- Information, Rationalization, Integration and Summarization: Application of Artificial Intelligence Techniques at the Supreme Court of Justice"¹³ project. This model is fine-tuned from BERTimbau (Souza et al., 2020), a portuguese specialized BERT model. According to the author, the fine-tuning was performed using over 30,000 portuguese legal documents.

Using LegalBERTimbau, I generated sentence embeddings for both case types, and then aggregated them per document, to obtain document level embeddings, obtaining Figure 3.19. Analyzing the graph, a clear pattern emerges, where there is a distinct separation between DV and BoC document embeddings, indicating that LegalBERTimbau is able to capture semantic relationships that differentiate both legal domains. However, when observing Figure 3.20 and Figure 3.21, describing embeddings for BoC and DV classes, respectively, it's not clear the separations between outcome categories, with overlap between embeddings of the different classes. This results, indicate that although LegalBERTimbau is able to capture semantic relationships that differentiate legal domains, it struggles to capture class specific relationships

¹²<https://huggingface.co/stjiris/bert-large-portuguese-cased-legal-mlm-sts-v1.0>

¹³<https://stjiris.github.io/>

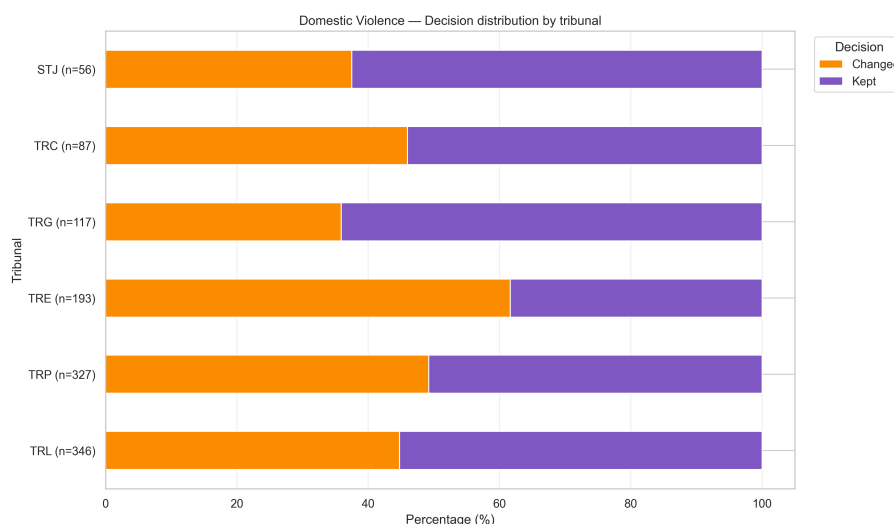


Figure 3.18: Class Distributions of DV by court region. The y axis legend translated means: "STJ" stands for Supreme Court of Justice. "TRX" means an appeal court where X is the first letter of each city name(e.g, P for Porto, C for Coimbra and L for Lisbon). Finally "JP" stands for Peace Courts.

within each domain. Nonetheless, given that the classes are engineered based on heuristics, rather than being inherent to the legal texts, it's understandable that embeddings may not reflect these separations clearly, out of the box. Although, UMAP representations need to be taken cautiously, due to its stochastic nature and required hyperparameter tuning needed, which I did not perform due to resource constraints, I would argue that these results are still promising for classification tasks, since the model seems to recognize and differentiate legal domains, which is a good indicator of its understanding of legal language and context.

For curiosity purposes, I investigated some outlier embeddings in the LegalBERTimbau embeddings, by manually checking the original documents, whose embeddings were distant from their document type clusters. I found that in the BoC case, the outlier case¹⁴ was a contract dispute that involved an insurance claim after a negligent dog attack, where the insurancy firm, refused to pay the claim. This case context dealt with personal injuries, emotional distress and medical reports, which are common factors in DV cases, explaining why its embedding was closer to DV documents. On the other hand the DV outlier case¹⁵ was a property dispute between conjugates, where the alleged aggressor prevented the victim from accessing their shared home. Similar to the previous case, the context of property disputes is a common factor in BoC cases, going against the typical DV case context, explaining its embedding proximity to BoC documents.

¹⁴<https://jurisprudencia.csm.org.pt/eccli/ECLI:PT:TRP:2023:282.17.9PCMTS.P1.BC/>

¹⁵<https://jurisprudencia.csm.org.pt/eccli/ECLI:PT:TRP:2022:1507.22.4T8MTS.P1.77/>

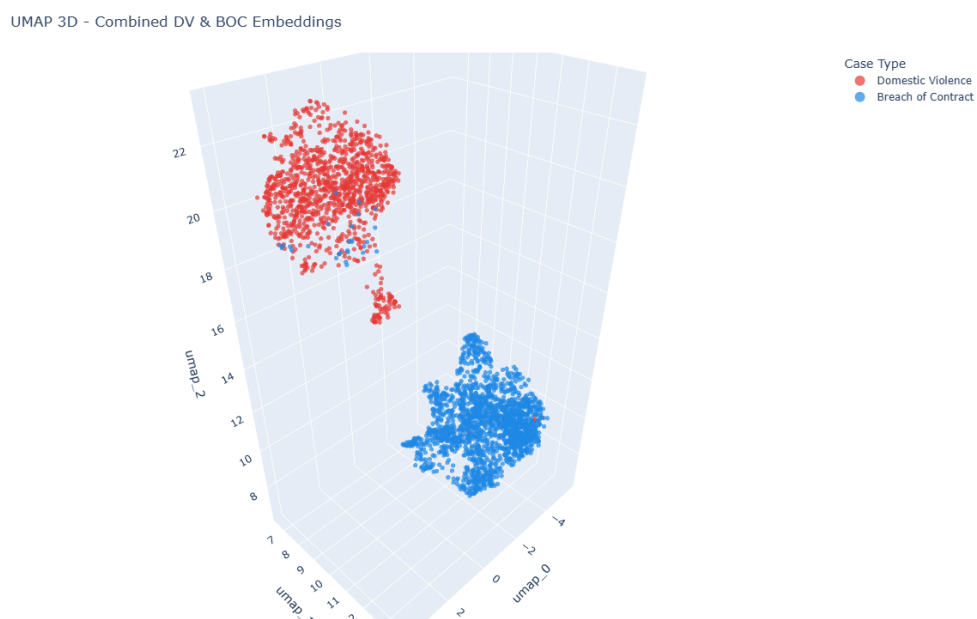


Figure 3.19: UMAP 3D visualization of LegalBERTimbau document embeddings for DV and BoC datasets.

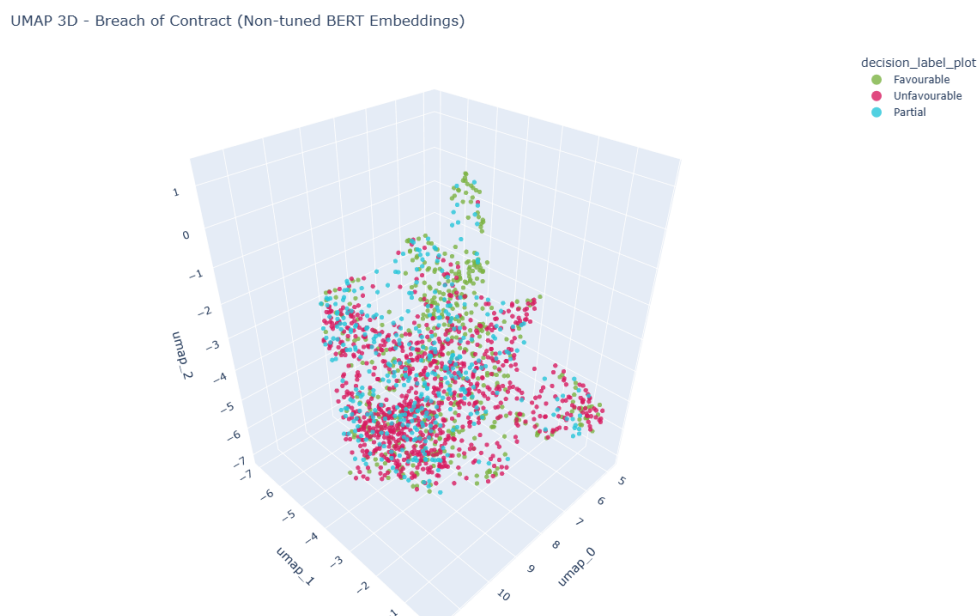


Figure 3.20: UMAP 3D visualization of LegalBERTimbau document embeddings for BoC datasets.

UMAP 3D - Domestic Violence (Non-tuned BERT Embeddings)

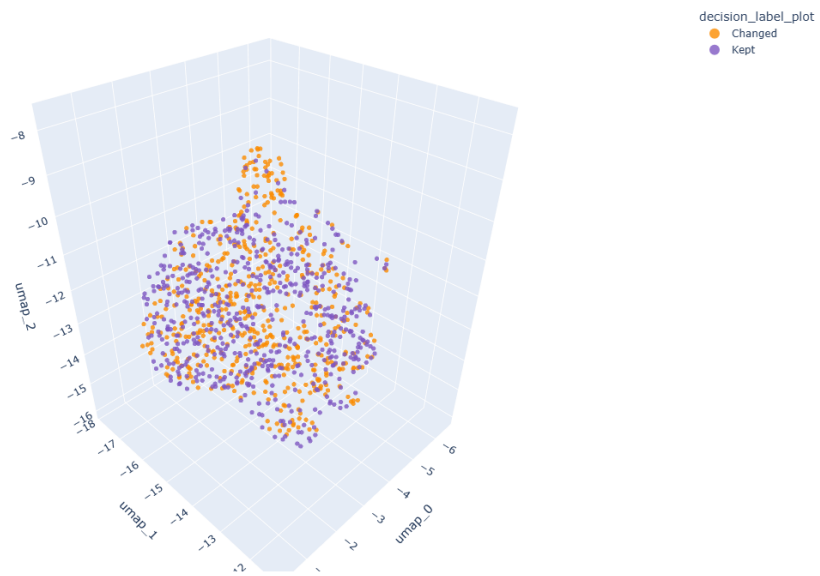


Figure 3.21: UMAP 3D visualization of LegalBERTImbau document embeddings for DV datasets.

Chapter 4

Methodology

4.1 Overview

This chapter describes the methodology for addressing the research questions (RQ1, RQ2, RQ3) and associated hypotheses (H1a, H1b, H2, H3) outlined in chapter 1. The study made a two-phase comparative approach:

1. Fine-tuning a domain-specific BERT model as a baseline.
2. Zero-shot and few-shot prompting of a smaller open-source LLM.

Phase 1 — Fine-tuned LegalBERTimbau: I fine-tune LegalBERTimbau (Melo et al., 2023), a portuguese legal-domain BERT variant, on the training set using a sliding-window architecture and use SHAP (Lundberg et al., 2017), to generate sentence-level attributions for explainability. This phase tests RQ1, RQ2 and RQ3, and associated hypotheses by evaluating predictive performance, analyzing SHAP attributions, between case types, and comparing them to LLM reasoning logs.

Phase 2 — LLM Prompting: I deploy DeepSeek-R1-0528-Qwen3-8B (DeepSeek-AI et al., 2025) via local inference with a custom smart summarization pipeline to manage context within hardware constraints. I explore zero-shot and few-shot prompting strategies, leveraging the model’s native CoT reasoning as explanations. Similarly to Phase 1, I address all research questions and associated hypotheses, with the exception of H1b, which is not empirically tested in this study, as the LLM’s native CoT is observed but not isolated or ablated, and RAG was not implemented.

Explainability comparison: I qualitatively compare BERT SHAP attributions with LLM reasoning logs, acknowledging they are fundamentally different objects. This analysis contextualizes RQ2 and H2, where it was not possible to infer on expert alignment as it was not possible to fairly compare the two methods, nor have them evaluated by legal experts due to time and resource constraints.

Evaluation: Both phases are evaluated on a held-out gold test set (n=50 per case type) using MCC, macro-F1, and TSS (for DV only), with supplementary judge validation (n=5 judges per type) to corroborate label reliability.

The following sections detail each component. Materials and computational resources are described first to ground the methodological choices in practical constraints.

4.2 Materials and Computational Resources

4.2.1 Hardware

Consider the following table, that summarizes the hardware specifications used for this dissertation. This information is relevant to understand the limitations discussed throughout the methodology but also to provide reference for future reproducibility of these experiments.

Component	Specification
Primary workstation	Lenovo LOQ 15IAX9
Processor	12th Gen Intel Core i5-12450HX
RAM	24 GB
GPU (primary)	NVIDIA RTX 3050, 6 GB VRAM
GPU (secondary)	Google Colab free tier (T4 clusters, intermittent availability)
GPU (tertiary)	NVIDIA P4/M10 cluster, 8 GB VRAM (faculty provided, unused due to massive latency)

4.2.2 Software and Dependencies

The following table only contains the main libraries and Python version used, from the data scraping, processing and modeling pipeline.

Tool / Library	Version
Python	3.12+
PyTorch	2.10.0+cu126
Transformers (Hugging Face)	4.57+
Pandas	2.3+
SHAP	0.51+
Ollama	0.6+
Selenium	4.35+
Crawl4Ai	0.7+
Streamlit	1.55+

4.2.3 Models and Pre-trained Weights

- **LegalBERTimbau:** stjiris/bert-large-portuguese-cased-legal-mlm-nli-sts-v1, via Hugging Face Transformers library (Melo et al., 2023) ¹.
- **LLM for inference:** DeepSeek-R1-0528-Qwen3-8B, via Ollama local deployment (DeepSeek-AI et al., 2025) ².

¹FinditonHuggingFace:<https://huggingface.co/stjiris/bert-large-portuguese-cased-legal-mlm-nli-sts-v1>

²FinditonOllama:<https://ollama.com/library/deepseek-r1:8b>

4.3 Fine-Tuned LegalBERTimbau

4.3.1 Motivation and Design Rationale

The first modeling approach employs LegalBERTimbau (Melo et al., 2023), a BERTimbau-Large variant pre-trained on over 30,000 Portuguese legal documents via MLM with the intent of clustering and semantic search. This model serves as a baseline against LLM approaches and was chosen over other transformer-based alternatives was motivated by three factors:

1. LegalBERTimbau is trained on Portuguese legal corpora, capturing domain-specific linguistic patterns out of the box;
2. BERT’s architecture, is lightweight compared to LLMs, permitting inference on personal hardware without quantization or distillation;
3. Extensive documentation and ablation studies exist for BERT fine-tuning on document classification tasks, providing methodological precedent (Sun et al., 2019).

However, BERT encounters a fundamental constraint, where the model accepts at most 512 tokens per forward pass. Given that documents in the dataset easily surpass this constraint (as seen in Figure 3.5 and Figure 3.6), direct fine-tuning on full documents is infeasible.

The sliding-window architecture described below, was done to accommodate this constraint, and it’s a well established method in the literature for handling long documents, while containing information loss (Alva Principe et al., 2025). It’s major drawback is the computational effort required for the multiple forward passes to encode the sentences, however, naively truncating these legal documents to fit the 512-token limit, would lead to substantial information loss.

Prior to tokenization, the *Dispositivo* (the operative part of the court decision) was excluded from the input text, leaving only the *Relatório* (narrative summary of facts and arguments) and *Fundamentação* (legal reasoning section) for both BERT training and inference, consistent with the retrospective classification framing where the outcome to be predicted has already been rendered and is contained within these sections.

4.3.2 Sliding-Window Architecture

To accommodate documents exceeding the 512-token limit, each document is tokenized into overlapping windows using the HuggingFace AutoTokenizer class. The tokenizer is configured with `max_length=512` and `stride=128`, which instructs it to create multiple 512-token windows with a stride of 128 tokens and return each window as a separate sequence.

Formally, for a document of length L tokens, this tokenization produces N overlapping windows:

$$W_i \in \mathbb{R}^{512}, \quad i \in \{1, 2, \dots, N\}$$

where consecutive windows overlap by 128 tokens, each window beginning 384 tokens after the previous one.

The tokenizer returns `input_ids` and `attention_mask` as 2D tensors of shape $(N, 512)$. For longer documents, N can exceed the computational budget, so to cap memory usage, if $N > 16$, I select 16 equally-spaced windows from the sequence:

$$\text{selected_indices} = \text{linspace}(0, N - 1, 16)$$

Notice that this was a personal choice, to balance document coverage against GPU memory constraints. With 16 windows and a stride of 128 tokens, this ensures coverage of approximately 6,272 unique tokens spaced across the document. Encoding all windows for documents with $N > 16$ required approximately 4–6 hours per training epoch on local hardware. With early stopping after approximately 7 epochs, this totaled roughly 28–42 hours of training before any inference could begin, rendering systematic experimentation with higher window counts infeasible. A personal decision was made to prioritize experimentation speed and fail-fast iteration over protocol optimization, setting the ceiling at 16 windows.

The 128-token stride creates relatively small overlap, ensuring that a fact or key legal phrase split across a window boundary would substantially appear in at least one of the windows. This is particularly important in legal documents, where critical reasoning often spans sentence or paragraph boundaries. However, information loss is inherent in this design, as selecting only 16 of potentially $N \gg 16$ windows discards entire sections of the document, and subsequent mean pooling of window representations weights all selected windows equally, losing any notion of positional importance or document structure.

4.3.3 Window Aggregation via Mean Pooling

Each selected window is independently encoded with frozen LegalBERTimbau (or fine-tuned top layers) to extract its [CLS] token representation. These window vectors are then aggregated via mean pooling into a single document representation, which is fed to the linear classification head. Mean pooling was selected over max pooling or attention-based aggregation to provide a simple and interpretable, document-level summary that weights all windows equally. However, this design approach may not capture the varying importance of different document sections, like the reasoning or the context of the case. Future iterations could explore attention-based pooling or positional approaches to address this limitation.

4.3.4 Training, Hyperparameters and Regularization

The training loop was designed to respect GPU memory constraints primarily. Furthermore, these hyperparameters are not the result of systematic hyperparameter tuning, which would require excessive computational resources. Instead, a trial and error approach was taken, by considering training time, GPU memory consumption and gold set performance. I cannot guarantee these are the optimal hyperparameters, and future work could explore a more systematic tuning approach, in case more computing resources become available.

- **Temporal validation split:** The training set was obtained after the exclusion of the gold set, described in chapter 3. This set was sorted chronologically by ruling date (*data_acórdão*) and partitioned into a training portion (first 85%) and a held-out validation portion (last 15%), with no shuffling. This preserves temporal ordering, where the model never observes future rulings during training, and ensures that the early-stopping signal is computed on cases that are strictly later in time than those used for weight updates, mirroring the realistic deployment scenario.
- **Batch Size:** 1 document per optimization step. Each document’s windows are processed in mini-batches of 16 windows as explained above. Higher batch sizes resulted in excessive training times and even out of memory errors.
- **Frozen Layers:** The first 20 layer’s of the 24-layer of LegalBertimbau were frozen. The premise behind this decision, was to leverage the pre-trained knowledge in portuguese legal language by Melo et al. (2023) and Souza et al. (2020), and to reduce the number of trainable parameters,

which would require more computational resources and increase the risk of overfitting on the small datasets used.

- **Optimizer:** AdamW with learning rate $lr = 5 \times 10^{-6}$ and weight decay = 0.05. The very low learning rate and strong L2 regularization were chosen after initial experiments showed severe overfitting on small datasets.
- **Learning Rate Schedule:** Linear warmup followed by linear decay, implemented using HuggingFace’s `get_linear_schedule_with_warmup`³. The schedule warms up over the first 10% of total training steps ($n_{\text{epochs}} \times n_{\text{documents}}$), then linearly decays to zero. The learning rate is updated after each document and not after each epoch, meaning the learning rate decays gradually throughout training. This industry-standard schedule stabilizes early training, mitigates the sharp loss oscillations observed in earlier experiments, and allows the model to approach convergence smoothly (Kalra et al., 2024).
- **Gradient Clipping:** Maximum gradient vector norm of 1.0 is applied to prevent exploding gradients and speed up training, again a standard practice in DL (Zhang et al., 2019).
- **Automatic Mixed Precision (AMP):** Trough PyTorch ⁴, fp16 mixed precision is automatically applied on certain operations during training, to reduce memory usage and accelerate computation without sacrificing stability(Lee, 2023).
- **Number of Epochs:** Training runs for up to 20 epochs at maximum. This was only chosen as a hard cap, to prevent excessive training times, while also being high enough to allow early stopping mechanism to trigger, and prevent overfitting.
- **Early Stopping:** I simply monitor the validation loss and save the best model at the minimum validation loss epoch. If the validation loss did not improve for 7 consecutive epochs, the training would halt and the best checkpoint would be restored. Although this approach is simple and standard in DL, the patience used is not empirically justified, rather chosen based on the prior intuition the model could overfit quickly, due to the small dataset size but that also could not learn any meaningful patterns as only the top 4 layers are fine-tuned.
- **Loss Functions:** All the windows of a single document are concatenated and mean-pooled into a single 1024-dimensional vector, which is then passed through a linear layer, outputting the model logits. These are then passed into the loss function, based on the case type:
 - **DV (binary):** BCEWithLogitsLoss ⁵ with $\text{pos_weight} = \frac{n_{\text{negative}}}{n_{\text{positive}}}$ to handle class imbalance, via inverse class frequency weighting. Although, as seen in Figure 3.4, the class imbalance is somewhat negligible, making the use of this weighting approach not strictly necessary.
 - **BoC (ternary):** CrossEntropyLoss ⁶ with balanced class weights and label smoothing coefficient of 0.1. This discourage overly confident predictions on the imbalanced BoC classes (Favorable and Partial), forcing the model to retain uncertainty and avoid degenerate solutions(Szegedy et al., 2016). DV did not require label smoothing due to better class balance.

³https://huggingface.co/docs/transformers/main_classes/optimizer_schedules

⁴<https://docs.pytorch.org/docs/2.12/amp.html>

⁵<https://docs.pytorch.org/docs/2.12/generated/torch.nn.BCEWithLogitsLoss.html>

⁶<https://docs.pytorch.org/docs/2.12/generated/torch.nn.CrossEntropyLoss.html>

- **Dropout:** Dropout rate of 0.3 is applied to the unfrozen layers to prevent co-adaptation of neurons and reduce variance during training.

4.3.5 Limitations and Trade-Offs

- **Computational Constraints on Window Coverage:** Encoding all windows for documents with $N > 16$ required approximately 4–6 hours per training epoch on local hardware, totaling roughly 28–42 hours across 7 epochs. In contrast, the summarization pipeline encoding required approximately 2–4 hours one-time for the entire dataset per case type on Colab free tier, making it substantially more efficient for downstream experimentation. This computational disparity forced a strategic choice between broader BERT window coverage and practical iteration speed, where the latter was prioritized to enable fail-fast iteration.
- **Sliding Window Information Loss:** The 16-window aggregation via mean pooling inevitably loses fine-grained positional information. For example, a critical fact appearing in window 8 contributes identically to windows 1 and 15. Furthermore, the overlapping windows may redundantly encode the same content, wasting computation without additional signal.
- **Mean Pooling Aggregation:** Mean pooling treats all windows equally. The training document’s tail (e.g., the *fundamentação* or legal reasoning and justification) may be disproportionately important but receives no additional weight. Alternative aggregation methods, that weight content and position, could retain more critical information.
- **Layer Freezing Rigidity:** Freezing 20 of 24 layers assumes pre-trained knowledge is uniformly valuable. However, the legal domain shift (from generic Portuguese text to legal decisions) and task shift (from legal text similarity to legal decision classification) may require unfreezing additional layers. Ablation studies varying the number of unfrozen layers were not performed, due to the computational cost and data scarcity, but could reveal more optimal configurations.
- **Limited Hyperparameter Tuning:** All training and regularization hyperparameters were set heuristically, via a few trial and errors, rather than via grid search or Bayesian optimization. Therefore, consider them as a starting point but are not claimed to be optimal.

4.4 Large Language Model Approach

4.4.1 Model Selection

The LLM component uses the DeepSeek-R1-0528-Qwen3-8B (DeepSeek-AI, 2025), served locally via Ollama⁷. The choice was motivated by three factors.

1. DeepSeek-R1-0528-Qwen3-8B is a base Qwen3-8B model optimised for reasoning capabilities, via distillation from the (at the time) flagship DeepSeek-R1:671B model. This means it was fine-tuned to output explicit CoT reasoning logs, which was one of the techniques I wanted to explore in this dissertation, to understand its impact on performance. Since this model came with this capability out of the box, it was a natural choice for this experimentation, as it would allow me to explore this technique without the need for specific prompting to elicit CoT reasoning. The experimentation with other open-source models, such as Llama or Mistral, was forsighted, but

⁷<https://ollama.com/library/deepseek-r1:8b>

ultimately not pursued due to time and resource constraints, and particularly for the lack of out-of-the-box reasoning capabilities, which would require additional work to implement and test;

2. The smaller 8B variant was selected over larger models due to the expected computational constraints of running inference locally. A smaller version could have been experimented with, but ultimately, due to the extensive intelligence required for legal decision prediction, 8B was chosen with the premise of a reasonable balance between performance and resource requirements.
3. Finally, being a open-source and locally served model, allowed for more control over its configuration and the reproducibility of results. Moreover, although the data was already pseudoanonymized, the fully local setup experimentation, ensures no data is sent through external servers, allowing full control over data privacy and security, which in a sensitive domain like the one in study, is a major concern.

Ultimately, all experiments were conducted at deterministic settings, with temperature = 0.0, top- k = 1, top- p = 1 and randomness seed = 42 to ensure token reproducibility across runs.

4.5 Context Management & Smart Summarization Pipeline

4.5.1 Motivation

Legal documents in the dataset substantially exceed available token budgets. This was expected based of document length analysis in Figure 3.5 and Figure 3.6. BERT’s sliding-window approach described above, was also employed to deal with this issue. On the other hand when experimenting in LLM inference on consumer or older architecture hardware such as my personal PC or the provided faculty GPU server, token management strategies were not as straightforward, as the only way to interact with the model was prompting. Initially, when first experimenting with zero-shot prompting approach, it was found the model was suffering from context dilution (Gu, 2026), specially due to a configuration error where the default token budget was still originally set to 2048 tokens. This was found through the thinking logs of Deepseek, where it showed meta-commentary on document structure, e.g., ”this is a long case”, rather than engagement with specific legal facts, articles, and reasoning, after low metric performance was observed. Naively, I tried to fix this by increasing the context window to 32,768 tokens, which was the maximum the hardware could handle. After a few trial and error experiments, I incrementally increased the context window until reaching the point of memory errors at this limit. However, dilution symptoms persisted. To manage prompt assembly within hardware constraints, I set an operational prompt budget of 28,000 tokens, reserving space for system instructions and case text.

Nonetheless, this made clear that a more sophisticated context management strategy was needed to ensure the model engaged with the most relevant legal information rather than being overwhelmed by the sheer volume of text.

Early experiments revealed that naive context management approaches like fixed-window extractions, random sentence sampling, TF-IDF filtering would lead to significant performance and reasoning failures, as it would often exclude critical legal reasoning and key facts or poison the input prompt with legal jargon, and there would be no control over the inputting information as document length varied by case.

The smart summarization pipeline described next addresses this by extracting and ranking sentences from the fundamentação using legal-domain embeddings, subject to linguistic and token constraints, while preserving document order to maintain narrative coherence of the court’s legal argument. Although

not a copy of a literature method, it follows the principles described in Mei et al. (2025), of Context Engineering, where it’s been proven that models need to be fed with only the most relevant information, to be able to perform well in downstream tasks, and that this information needs to be carefully selected and structured, to ensure the model can understand it and use it effectively. Thereby, consider this pipeline as a custom method for context management, and not a scientific contribution in itself, as it’s not been formally evaluated or compared to other methods, but rather an engineering solution to a practical problem encountered during experimentation.

4.5.2 Summarization Architecture

Figure 4.1 gives a high-level overview of the pipeline described below.

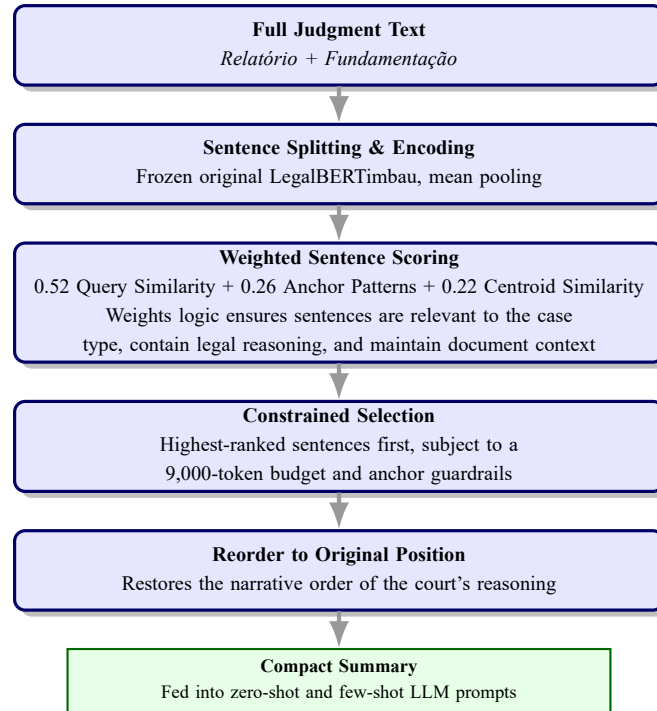


Figure 4.1: High-level overview of the smart summarization pipeline, from raw judgment text to the compacted summary used in LLM prompts.

1. **Sentence Splitting:** Portuguese legal text is split into sentences using a conservative regex-based splitter that respects Portuguese punctuation conventions. Fragments below 8 characters are discarded.
2. **Sentence Encoding:** Each sentence is encoded with frozen LegalBERTimbau (Melo et al., 2023), using mean pooling over token embeddings to produce a 1024-dimensional sentence vector.
3. **Case-Type Query Embedding:** A task-specific query embedding is constructed by averaging 5 templates per case type:
 - **DV templates:** e.g., “fundamentos jurídicos para manter ou alterar a decisão recorrida” or análise de “fundamentos rejeitados ou aceites que influenciam a manutenção ou alteração da decisão”

- **BoC templates:** e.g., “fundamentos jurídicos para procedência, improcedência ou procedência parcial da ação” ou “apreciação da prova e enquadramento legal para decisão total ou parcial da ação”

These query embeddings anchor the ranking toward decision-relevant legal reasoning rather than generic document importance.

4. **Linguistic Anchor Scoring:** Sentences were pattern matched by the presence of legal domain keywords grouped and weighted into six categories:

- Facts: *factos provados, matéria de facto, apurou-se que*
- Norms: *nos termos do artigo, de acordo com o disposto, regime jurídico aplicável*
- Proof evaluation: *apreciação da prova, valor probatório, não ficou demonstrado*
- Reasoning: *entende o tribunal, considera-se que, cumpre decidir*
- Jurisprudence: *jurisprudência, acórdão do, orientação jurisprudencial*
- Conflict/rejection: *não se verifica, carece de fundamento*

Each category was also weighted differently, with conflict/rejection, facts, proof evaluation and reasoning being slightly more weighted than norms and jurisprudence, to direct the extraction toward sentences that are more likely to contain decision directional content, rather than legal formalism and citations, which although important, are more important to provide explanatory context rather than direction to the model. The output of this step is a normalized score between 0 and 1 for each sentence, where higher values indicate stronger presence of the aforementioned legal anchors, and therefore a higher likelihood of being relevant to the court’s decision.

It’s important to be transparent that the above design decisions, are without a doubt, the least refined component of the pipeline, as they were manually engineered without any formal evaluation or optimization. They were designed based of the use of LegalBERTimbau, which is a model trained on portuguese legal documents, and therefore should be able to capture similar semantics of these templates, and also pattern matching with the linguistic anchors, to ensure the presence of relevant sentences in the final summarization output. However, future work, should explore more systematic and reproducible methods, such as fine tuning a model for this specific sentence extraction task, or validating these templates with legal experts, to ensure they are actually capturing the most relevant legal reasoning for each case type.

5. **Document Centroid Embedding:** The mean of all sentence embeddings, representing the document’s overall semantic center, to balance sentence relevance with document-level context.
6. **Sentence Scoring:** Each sentence receives a combined score:

$$\text{score} = w_{\text{query}} \cdot \cos(\text{sentence}, \text{query}) + w_{\text{centroid}} \cdot \cos(\text{sentence}, \text{centroid}) + w_{\text{anchor}} \cdot \text{anchor_score}$$

Each anchor group has a normalized weights. The default weights are 0.52 (query), 0.26 (anchor), 0.22 (centroid), so relevance to the case-type intent query was the dominant signal in sentence scoring, followed by the presence of linguistic anchors, with overall document-level similarity via the centroid contributing the least, to ensure the most relevant sentences were selected, even if they were not close to the document’s general semantic center. Once more, these were manually selected and without any formal evaluation, so they should only taken as a baseline configuration to solve this problem.

7. **Constrained Selection:** Sentences are selected greedily in ranked score order, then reordered back to their original document position for the final output, until token budget is exhausted, subject to guardrails:

- Summarization pipeline token budget cap of 9,000 tokens (character-to-token estimate⁸)
- Minimum 70% anchor ratio (at least 70% of selected sentences must contain legal anchors)
- Per-anchor-group caps (e.g., max 8 sentences per category)
- Max 6 anchorless sentences (fallback when anchor budget exhausted)

By applying a token budget constraint, I deterministically impose a hard limit on the amount of information fed to the model, which as a fallback ensures the model could at least perform inference on all cases, even the longest ones, which would otherwise be impossible on the hardware experiments were conducted on. The other guardrails described above, had the intent of forcing the selection of mostly anchor matched sentences and the reduction of sentences that did not contain any of these priors, again to ensure the richest information possible was input to the model. It's important to document that the anchor guardrails, managed to keep the token cap being reached, as no test case reached the 9,000 token limit, where both case types had an average of around 5,400 tokens after the summarization pipeline. Furthermore, the average number of selected sentences was 152 and 154, while the average reduction rate was 46.79% and 51.17% for DV and BoC cases, respectively. The output is a compacted *relatório e fundamentação* that preserves narrative structure while filtering procedural boilerplate and off-topic tangents, particularly in longer documents.

4.5.3 Sentence Order Preservation

A critical design decision was that inputting the selected sentences to the actual model had to respect their original positions in the original document, not in relevance score rank order. This stems from the problem that pure rank-based extractive summarization selects top- k sentences by score, which can fragment the document by jumping between high-scoring sections separated by context.

In legal documents, this fragmentation is particularly damaging because appellate reasoning is sequential, as the court presents facts, applies relevant law, addresses counterarguments, and reaches conclusions in a cumulative logical chain. Removing intermediate sentences breaks this chain and potentially omits critical context. Furthermore, had these sentences been input in ranked order, the narrative flow of the court's reasoning would have been lost, possibly leading to more confusion for the model, as it would for a human reader, as the logical connections between sentences would be disrupted.

This approach was based on the literature (Song et al., 2025) and ensures the summarized text remains a coherent sub-document.

4.5.4 Limitations and Trade-Offs

- **Heuristical Sentence Similarity and Pattern Matching:** The scoring relies on common text patterns found in legal documents. However, they may not capture all relevant sentences and may also introduce bias since they were manually designed. A clear example of this, is any legally relevant sentence that would not contain any legal linguistic anchor, would be heavily penalized

⁸This estimate is based on <https://help.openai.com/en/articles/4936856-what-are-tokens-and-how-to-count-them>. I am aware this is just an estimate and that the actual count for the model used may differ.

and possibly not selected. It's imperative for future work to explore more data-driven and expert validated methods for sentence scoring.

- **Information Loss:** I do not quantify relevant information loss per case, simply total token and sentence reduction. It's inherent that this approach can also omit critical legal information, even though guardrails were implemented to mitigate this, such as the anchor pattern matching.
- **Not Task-Fine-Tuned:** Sentence ranking uses linguistic patterns and embedding similarity, not task-specific importance learned from a labeled dataset. A sentence may be legally critical but linguistically uncommon and thus underranked. Task-specific fine-tuning could improve this, but requires labeled sentence-importance datasets, which would be costly to create and one already made was not found.
- **Sentence-Level Granularity:** The method cannot split long sentences. A single complex sentence may prevent inclusion of shorter, high-value sentences if the budget is tight. Furthermore, the contrary could also be true, where possible longer sentences or even paragraphs could be deemed more relevant than simply extracting sentences. Future work needs to explore different levels of document chunking, to understand its effects on summarization and downstream tasks.
- **Computational Cost:** Encoding every sentence in every document via LegalBERTimbau adds preprocessing latency. For 3,000 documents, this preprocessing step required approximately 2–4 hours one-time per case type on Colab free tier. This cost is amortized, as it's only performed once per dataset, in stark contrast to sliding-window BERT fine-tuning which required approximately 4–6 hours per training epoch across 7 epochs (roughly 28–42 hours total on local hardware, as seen in subsection 4.3.2). The one-time summarization cost proved substantially more practical for iterative experimentation, allowing multiple downstream inference strategies (zero-shot, few-shot) without re-encoding.

4.5.5 Zero-Shot Prompting Strategy

Each prompt consists of a *system message* and a *user message*. For this study, prompts were designed to be zero-shot, meaning the model would not receive any labeled examples of the task. Furthermore, my prompts incorporated several well-established prompt engineering practices such as acting as a domain expert persona (experienced appellate judge), explicit task instructions, clear constraints on model behavior (what to do and what not to do), definitions of valid output classes, and a strictly specified response format. These design choices follow best practices documented in both academic research and industry applications Schulhoff et al. (2024).

Domestic Violence Prompt**PT****Persona**

És um Juiz Conselheiro português com mais de 30 anos de experiência em tribunais de segunda instância, especializado em direito penal e, em particular, em recursos de decisões em casos de violência doméstica. Ao longo da tua carreira apreciaste centenas de recursos nesta matéria e conheces profundamente a doutrina, a jurisprudência dos tribunais superiores e os critérios que determinam a manutenção ou alteração de uma condenação em sede de recurso.

Tarefa

Classificação do resultado de recursos em casos de violência doméstica em tribunais portugueses de segunda instância.

Instruções

1. Lê o Relatório para identificar o contexto do caso: partes envolvidas, conduta alegada, decisão do tribunal de 1.^a instância e fundamentos do recurso interposto.
2. Lê a Fundamentação para compreender a análise jurídica do tribunal de recurso: factos provados e não provados, legislação aplicável e jurisprudência citada.
3. Determina se o tribunal de recurso manteve ou alterou a decisão de primeira instância com base na argumentação jurídica presente na Fundamentação.
4. Emite exclusivamente o JSON de classificação no formato indicado.

Deves

- Usar exatamente uma das labels válidas no campo "predicted_label"
- Basear a classificação apenas no texto fornecido
- Emitir JSON sintaticamente válido como única e exclusiva resposta

Não Deves

- Adicionar texto, comentários, explicações ou raciocínio fora do JSON
- Usar labels não listadas, nem variações ortográficas ou traduções
- Inferir o dispositivo, não está no texto e não te é dado acesso a ele
- Recusar classificar ou pedir mais informação, classifica sempre com base no texto disponível

Classes válidas

"decisao_mantida": o tribunal de recurso confirma e mantém a decisão do tribunal de 1.^a instância, o recorrente não obtém qualquer alteração favorável

"decisao_alterada": o tribunal de recurso revoga, modifica ou substitui a decisão do tribunal de 1.^a instância, parcial ou totalmente

Contexto

- Tribunal: segunda instância portuguesa (TRC, TRP, TRL ou STJ)
- Tipo de caso: violência doméstica (tipicamente art. 152.º CP)
- O texto contém o Relatório e a Fundamentação. O dispositivo foi removido, o teu papel é inferi-lo a partir da análise jurídica presente na Fundamentação.

Formato de resposta obrigatório

```
{"predicted_label": "<LABEL>"}
```

EN**Persona**

You are a Portuguese appellate judge with over 30 years of experience in second instance courts, specializing in criminal law and, in particular, in appeals of decisions in domestic violence cases. Throughout your career you have reviewed hundreds of appeals in this matter and you deeply understand the doctrine, the jurisprudence of the higher courts, and the criteria that determine the maintenance or alteration of a conviction on appeal.

Task

Classification of the outcome of appeals in domestic violence cases in Portuguese second instance courts.

Instructions

1. Read the *Relatório* to identify the case context: parties involved, alleged conduct, first instance court decision, and grounds of the appeal.
2. Read the *Fundamentação* to understand the legal analysis of the appellate court: proven and unproven facts, applicable legislation, and cited jurisprudence.
3. Determine whether the appellate court upheld or altered the first instance decision based on the legal argumentation present in the *Fundamentação*.
4. Emit exclusively the classification JSON in the indicated format.

You Must

- Use exactly one of the valid labels in the "predicted_label" field
- Base the classification solely on the provided text
- Emit syntactically valid JSON as the sole and exclusive response

You Must Not

- Add text, comments, explanations, or reasoning outside the JSON
- Use labels not listed, nor spelling variations or translations
- Infer the *dispositivo*, it is not in the text and you are not given access to it
- Refuse to classify or ask for more information, always classify based on the available text

Valid Classes

"decisao_mantida": the appellate court confirms and upholds the first instance court's decision, the appellant obtains no favorable alteration

"decisao_alterada": the appellate court revokes, modifies, or replaces the first instance court's decision, partially or totally

Context

- Court: Portuguese second instance (TRC, TRP, TRL, or STJ)
- Case type: domestic violence (typically art. 152 of the Penal Code)
- The text contains the *Relatório* and *Fundamentação*. The *dispositivo* was removed, your role is to infer it from the legal analysis present in the *Fundamentação*.

Required Response Format

```
{"predicted_label": "<LABEL>"}
```

Breach of Contract Prompt**PT****Persona**

És um Juiz Conselheiro português com mais de 30 anos de experiência em tribunais de segunda instância e julgados de paz, especializado em direito civil e, em particular, em recursos de decisões em litígios de incumprimento contratual. Ao longo da tua carreira apreciaste centenas de recursos nesta matéria e conheces profundamente a doutrina, o regime geral das obrigações e a jurisprudência aplicável aos contratos em direito português.

Tarefa

Classificação do resultado de recursos em casos de incumprimento contratual em tribunais portugueses de segunda instância.

Instruções

1. Lê o Relatório para identificar o contexto do caso: partes envolvidas, objeto do contrato, incumprimento alegado, decisão do tribunal de 1.^a instância e fundamentos do recurso interposto.
2. Lê a Fundamentação para compreender a análise jurídica do tribunal de recurso: factos provados e não provados, legislação aplicável e jurisprudência citada.
3. Determina o grau de procedência do recurso para o recorrente com base na argumentação jurídica presente na Fundamentação.
4. Emite exclusivamente o JSON de classificação no formato indicado.

Deves

- Usar exatamente uma das labels válidas no campo "predicted_label"
- Basear a classificação apenas no texto fornecido
- Emitir JSON sintaticamente válido como única e exclusiva resposta

Não Deves

- Adicionar texto, comentários, explicações ou raciocínio fora do JSON
- Usar labels não listadas, nem variações ortográficas ou traduções
- Inferir o dispositivo, não está no texto e não te é dado acesso a ele
- Recusar classificar ou pedir mais informação, classifica sempre com base no texto disponível

Classes válidas (do ponto de vista do recorrente)

"decisao_desfavoravel": o recurso é julgado totalmente improcedente, o tribunal de recurso não altera nenhum ponto da decisão recorrida em favor do recorrente

"decisao_parcial": o recurso é julgado parcialmente procedente, o tribunal altera alguns pontos da decisão recorrida, mas não acolhe todos os pedidos do recorrente

"decisao_favoravel": o recurso é julgado totalmente procedente, o tribunal acolhe integralmente os fundamentos do recorrente e altera a decisão recorrida

Contexto

- Tribunal: segunda instância portuguesa (TRC, TRP, TRL ou STJ)
- Tipo de caso: incumprimento contratual (breach of contract)
- O texto contém o Relatório e a Fundamentação. O dispositivo foi removido, o teu papel é inferi-lo a partir da análise jurídica presente na Fundamentação.

Formato de resposta obrigatório

```
{"predicted_label": "<LABEL>"}
```

EN**Persona**

You are a Portuguese appellate judge with over 30 years of experience in second instance courts and peace courts, specializing in civil law and, in particular, in appeals of decisions in breach of contract disputes. Throughout your career you have reviewed hundreds of appeals in this matter and you deeply understand the doctrine, the general regime of obligations, and the jurisprudence applicable to contracts in Portuguese law.

Task

Classification of the outcome of appeals in breach of contract cases in Portuguese second instance courts.

Instructions

1. Read the Relatório to identify the case context: parties involved, object of the contract, alleged breach, first instance court decision, and grounds of the appeal.
2. Read the *Fundamentação* to understand the legal analysis of the appellate court: proven and unproven facts, applicable legislation, and cited jurisprudence.
3. Determine the degree of success of the appeal for the appellant based on the legal argumentation present in the Fundamentação.
4. Emit exclusively the classification JSON in the indicated format.

You Must

- Use exactly one of the valid labels in the "predicted_label" field
- Base the classification solely on the provided text
- Emit syntactically valid JSON as the sole and exclusive response

You Must Not

- Add text, comments, explanations, or reasoning outside the JSON
- Use labels not listed, nor spelling variations or translations
- Infer the *dispositivo*, it is not in the text and you are not given access to it
- Refuse to classify or ask for more information, always classify based on the available text

Valid Classes (from the appellant's perspective)

"decisao_desfavoravel": the appeal is judged totally unsuccessful, the appellate court does not alter any point of the appealed decision in favor of the appellant

"decisao_parcial": the appeal is judged partially successful, the court alters some points of the appealed decision but does not grant all of the appellant's requests

"decisao_favoravel": the appeal is judged totally successful, the court fully accepts the appellant's grounds and alters the appealed decision

Context

- Court: Portuguese second instance (TRC, TRP, TRL, or STJ)
- Case type: breach of contract
- The text contains the *Relatório* and *Fundamentação*. The *dispositivo* was removed, your role is to infer it from the legal analysis present in the *Fundamentação*.

Required Response Format

```
{"predicted_label": "<LABEL>"}
```

The user message provides the case identifier and the judgment text from the gold set, which contains only the *Relatorio* and *Fundamentacao* sections. The *dispositivo* is structurally absent, eliminating the possibility of label leakage by construction. To manage token usage within the 28,000-token operational prompt budget, documents are summarized through the retrieval-based pipeline before prompting, subject to the 9,000-token summarization budget described in section 4.5.

4.5.6 Few-Shot Prompting Strategy

The few-shot prompting strategy was built upon the latter strategy and exact prompt design, with the addition of labeled examples of the task. Based on the literature on few-shot learning like Min et al. (2022) and Tang et al. (2025) the number and quality of examples given to the model is known to have a significant impact on performance. So, the question on how to select these examples arose. Initially, naive approaches like random sampling or deterministically selecting these examples was thought, however, after careful consideration, an embedding based retrieval strategy was chosen, as not only is backed by literature, as seen previously in chapter 2, with studies like Peng et al. (2024) and Nigam, Patnaik, et al. (2025), it also approximates to the way actual judges study and use past jurisprudence to make informed decisions, by looking for similar cases and legal reasoning, that could best be applied to the case at hand, rather than randomly or fixed choosing examples.

This strategy was built as a retrieval step over the LegalBERTimbau phase training corpus, by taking advantage of the already computed and stored sentence embeddings. It also takes advantage of the smart summarization pipeline, described downstream in section 4.5, where it compresses the document to a set of ranked relevant sentences, based on the similarity with legal linguistic anchors, document and class intent.

Figure 4.2 gives a high-level overview of the mechanism, detailed formally below.

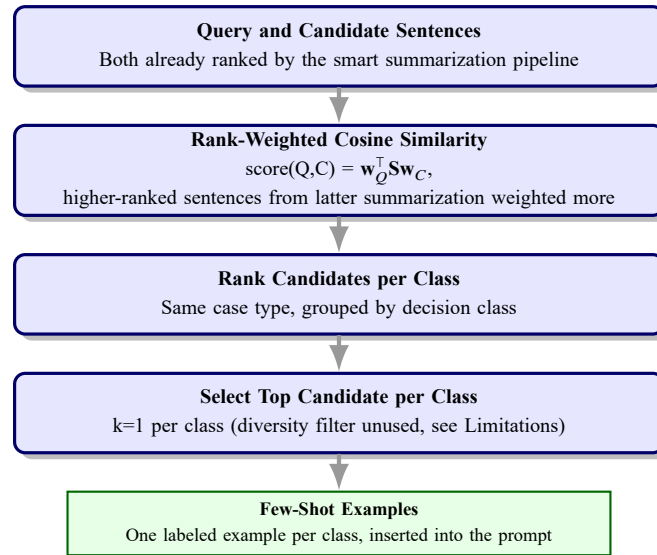


Figure 4.2: High-level overview of the embedding-based few-shot retrieval mechanism, from ranked query/candidate sentences to the final selected examples.

The similarity score between a test case Q and a candidate case C is computed from a cosine similarity matrix S . If Q keeps m selected sentences and C keeps n , then $S \in \mathbb{R}^{m \times n}$, and each entry S_{ij} is the cosine similarity between the i -th sentence embedding of Q and the j -th sentence embedding of C . The selected sentences are already stored in truncation order, so the first row in each embedding set corresponds to the most relevant sentence, the second row to the next one, and so on, according to the smart summarization pipeline's ranking. This ordered structure is converted into two normalized weight vectors, $w_Q \in \mathbb{R}^m$ for the query test case and $w_C \in \mathbb{R}^n$ for the candidate example case, giving more weight to earlier sentences than to later ones. The final score is a single scalar obtained by the matrix product:

$$\text{score}(Q, C) = w_Q^T S w_C \quad (4.1)$$

Expanding the expression, there is a double summation over all sentence pairs:

$$\text{score}(Q, C) = \sum_{i=1}^m \sum_{j=1}^n w_{Q,i} S_{ij} w_{C,j} \quad (4.2)$$

And since the weight vectors were normalized so their entries sum to 1:

$$\sum_{i=1}^m w_{Q,i} = 1, \quad \sum_{j=1}^n w_{C,j} = 1 \quad (4.3)$$

Then, they can be interpreted as probability distributions over the ranked sentences:

$$\begin{aligned} \sum_{i=1}^m w_{Q,i} &= 1, & \sum_{j=1}^n w_{C,j} &= 1, \\ p_{ij} &= w_{Q,i} w_{C,j}, & \sum_{i=1}^m \sum_{j=1}^n p_{ij} &= 1. \end{aligned} \quad (4.4)$$

So, the final score can be rewritten as the expected value of the cosine similarity under the weight vectors joint distribution p_{ij} :

$$\text{score}(Q, C) = \sum_{i=1}^m \sum_{j=1}^n p_{ij} S_{ij} = \mathbb{E}_p[S_{ij}] \quad (4.5)$$

In plain terms, the method compares every selected sentence from the test case with every selected sentence from the candidate case, and then computes the expected similarity value across all pairs. Furthermore, it gives more importance to the highest-ranked or earliest selected sentences, while lower-ranked progressively contribute less to the expected score. The design of this approach, was meant to keep the legal reasoning signal concentrated in the parts of the document that were previously identified as most relevant, while reducing the impact of boilerplate text, according to the pipeline presented in section 4.5.

After ranking the candidates, this approach keeps only the most relevant candidate case as the selected example for the few-shot prompt. I would have liked to have more than one example per class, to allow more diversity the model could learn from, however, once again, hardware constraints limited how much the context window could be filled, and with more than one example per class, Out of Memory errors would occur, breaking the experimentation.

4.5.7 Limitations and Trade-Offs

- **Heuristic truncation:** the smart truncation step preserves sentences judged to be most relevant, based on the factors mentioned, but it remains an approximation. In addition, the factors themselves may not capture all aspects of legal relevance, as the anchors were manually drafted and so were the weights assigned to each. Given that no domain expert defined the latter, the approach largely reflects developer choices.
- **Approximate few-shot retrieval:** the few-shot examples are selected through sentence-level cosine similarity and ranking order from the truncation step. In theory, this helps preserve legal reasoning signal, but it is still only a proxy for true judicial similarity and having so many heuristics in the pipeline may introduce noise and bias into the selection process.
- **Limited diversity in practice:** although the pipeline was designed to include a diversity filter, hardware constraints reduced the final configuration to one example per class in the few-shot setting. As a result, a more complete few-shot approach with multiple examples per class was not feasible, and it would be recommended to further explore this in future work.

- **Sensitivity to design choices:** despite the parts used in the pipeline being well known in the literature, all put together, especially with manual tuning, as not been replicated, to the best of my knowledge. Therefore, the results may vary with changes in the truncation thresholds, similarity thresholds, prompt wording, or the number of examples retrieved per class. These choices were made pragmatically to overcome the engineering challenges of working with a large corpus of legal data, with low hardware resources, and so they may not be optimal nor generalize well to other domains or tasks.
- **Interpretability versus optimality:** the pipeline is easier to inspect and explain than a more opaque learned retrieval system, but that interpretability comes at the cost of not fully optimizing every component jointly.

Overall, the approach is best understood as a controlled and practical approximation to legal reasoning retrieval, rather than as a theoretically complete solution. For this reason, I hope future work will thoroughly evaluate and improve this pipeline, making it less heuristic and more data-driven while preserving its interpretability as much as possible.

4.6 Explainability Methods

4.6.1 SHAP Explanations for BERT

I used SHAP to generate sentence-level attributions for the fine-tuned LegalBERTimbau model, where due to the computational cost of SHAP, this analysis was only done in the same 5 cases used as validation examples for the judges, in the streamlit annotation app. The method works by iteratively masking sentences in the input text and measuring the resulting change in model output probability, thereby computing each sentence’s contribution to the final prediction. It was implemented this way on purpose, instead of using the natural attention capabilities of the model, in case these would of been validated by judges to allow more human interpretability. However, this was not the case, as low model performance, time constraints, and uneven comparisons with LLMs explainability methods, made this validation unfeasible.

For each case, the implementation produced an HTML file with colour-coded text overlays, where red indicated sentences pushing toward the predicted class and blue indicated sentences pushing away, together with a bar chart of the highest and lowest contributing sentences. I additionally plotted the distribution of SHAP attributions across sentence position for every case, on a shared scale to allow comparison between documents and case types. All of these visualizations, with the exception of the HTML files, can be found in Appendix .2. Qualitatively, I inspected all of these visualizations, and they revealed a diluted pattern of sentence importance, where small spikes in SHAP values could visually be observed across the text, but did not necessarily concentrate on the legal keywords that I expected, nor any significant top sentences that could be used to explain the model’s predictions.

As such, it can be concluded that SHAP provided explanations for the BERT model phase, however, these were non informative and did not provide any specific patterns that could be humanly interpreted as decisive in the model decision making process. This is consistent with the model’s performance reported in subsection 6.1.1, where it collapsed to majority-class prediction on both case types and the loss curves showed it never learned class-discriminative features. With no input-dependent decision process to expose, the attributions most likely reflect secondary document variations rather than legal signal differences, which is a property of the collapsed model rather than a limitation of SHAP.

4.6.2 Thinking Logs and Chain-of-Thought for LLMs

For the LLM approach, the initial designed approach was to also use SHAP to generate sentence-level contributions, however, the computational constraints were only exacerbated by the larger model size and therefore was not pursued. Instead, I leveraged DeepSeek-R1-0528-Qwen3-8B native reasoning capability, which outputs structured thinking logs (enclosed in <think> tags) before producing a final answer. These logs capture the model’s step-by-step reasoning process identifying key facts, appellants, citing applicable legal articles, referencing past jurisprudence, and evaluating argument strength. As seen in section 4.5 I also used these logs to debug the model failure modes, where extreme low performances, were also accompanied by vague out of topic reasoning, which was a clear indicator of context dilution, and therefore the need for a more sophisticated context management strategy.

This approach however has the major limitation of being purely introspective, as it relies on the model’s own self-reporting of its reasoning process. It is known from the literature like Turpin et al. (2023), that reasoning logs can be misleading. In this exemplary study, the researchers prompted injected the model to misclassify the task, however the reasoning logs still produced convincing explanations for the wrong answers.

The same happened here, for example for case 236/23.6PBHRT.L1-9⁹, the model incorrectly predicted a kept decision by the appellate court, when the correct label was an altered decision. When inspecting the thinking logs, reproduced in full in Appendix .2, it is clear that the model correctly identifies the case context, the case facts, and the previous decision. However, it misinterprets the prosecution appeal’s arguments and the defendant’s response to it, as the second instance reasoning for the keeping the decision. By doing this, it confidently outputs a wrong label, while providing a somewhat plausible looking reasoning chain, that without human validation, could be interpreted as faithful reasoning, when in fact it is not.

In addition, examples like these, further highlight the limitation of any form of summarization or truncation, where information loss, can lead to the model to misinterpret case material and therefore output wrong predictions.

This limitation confirms literature findings, and highlights that reasoning outputted by the model, can only be used as a monitoring tool, rather than a faithful representation of the model’s actual reasoning process.

4.6.3 Comparison Framework

A quantitative and fair comparison between SHAP sentence-level attributions and LLM reasoning logs is not possible due to the fundamentally different nature of both. Although SHAP is a more established technique in literature and industry, it proved unfeasible for both, and ultimately uninformative for the BERT approach, while the reasoning logs of the models although interpretable, are not truly rigorous and are easily misleading as seen in the previous example.

Consequently, the explainability analysis remained qualitative and cannot be used for any scientific claims, nor to respond or corroborate the research questions and hypotheses involving explainability methods.

⁹<https://jurisprudencia.csm.org.pt/eccli/ECLI:PT:TRL:2025:236.23.6PBHRT.L1.9.58/>

4.7 Evaluation Methodology

4.7.1 Metrics

Model performance was assessed using three complementary metrics, based upon the confusion matrix of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN):

1. **Matthews Correlation Coefficient (MCC)** (Yule, 1912; Cramer, 1946; Matthews, 1975; Baldi et al., 2000) is a correlation-based metric designed for imbalanced classification, which considers all four quadrants of the confusion matrix. It's defined as:

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (4.6)$$

It ranges from -1 (perfect disagreement) to $+1$ (perfect agreement), and since it aggregates the full confusion matrix, makes it robust to class imbalance. MCC was the primary metric for both DV (binary) and BoC (ternary) tasks, as it avoids the pitfall of accuracy in imbalanced settings where a model predicting only the majority class appears deceptively competent.

2. **Macro-averaged F1-score** (Van Rijsbergen, 1979; Sasaki, 2007) computes F1 per class and averages unweighted. F1 is defined as the harmonic mean of precision and recall:

$$\text{F1} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (4.7)$$

It's another imbalance robust metric, that penalizes poor performance per classes.

3. **True Skill Statistic (TSS)** (Pierce, 1884; Youden, 1950; Woodcock, 1976) was applied to DV binary classification to directly measure the balance of sensitivity (true positive rate) and specificity (true negative rate), following recommendation from the thesis proposal defense. It's defined as:

$$\text{TSS} = \text{sensitivity} + \text{specificity} - 1 = \frac{TP \times TN - FP \times FN}{(TP + FN)(TN + FP)} \quad (4.8)$$

TSS is particularly informative for legal applications where both false positives (wrongly predicting changed decision) and false negatives (missing true changed decisions) carry significant responsibility costs.

Through forest plots, I was also able to assess per class performance, beyond the summarization metrics, detailed above, like precision, recall and F1-score, which allowed fine-grained inspection of model behaviour on each class.

4.7.2 Bootstrap Confidence Intervals

All reported metrics are accompanied by 95% confidence intervals (CIs) computed via stratified bootstrap resampling (Rao et al., 1984) with $n = 1000$ resamples. This procedure resamples from the test set with replacement within each class proportion, preserving the original class distribution in each resample. This stratified approach is especially important for small, imbalanced datasets, ensuring that rare minority classes are represented in every bootstrap sample.

A key artifact of stratified resampling on a model predicting only a single class, as occurred with LegalBERTimbau on both DV and BoC tasks, is that it produces identical metric values in every bootstrap iteration. Consequently, the confidence interval width is exactly zero (e.g., $[0.367-0.367]$), which is not a computation error but rather the correct output of the bootstrap procedure under deterministic predictions. See the BERT results tables (Table 5.5, Table 5.6) for examples of collapsed zero-width CIs and its interpretation in section 5.1.

4.7.3 Judge Validation Protocol

To assess reliability of the heuristic label extraction process, which as based on rule-based regex on dispositivos, as seen in subsection 3.0.1. a qualitative validation exercise was conducted with domain-expert judges. The protocol was as follows:

Judge Panel Composition

The panel was chosen by the *Conselho Superior da Magistratura* or Superior Council of the Judiciary, and I particularly thank Judge Dr. Ana de Azeredo Coelho for the logistic support and coordination with the judge panel. It originally consisted of 6 specialized judges for each case type. From these, 5 managed to finish the annotation task, but thanks is given to all, given the time constraints and workload of judges. Therefore, their efforts are particularly appreciated.

Annotation Interface and Task

An interactive web application was developed in Streamlit¹⁰ presenting judges with the full case text, as the goal was not to test their legal knowledge, rather purely to validate the label extraction process. Then the Judges were asked to:

1. classify the appellate outcome into the target scheme (KEPT / ALTERED for DV; UNFAVORABLE / PARTIAL / FAVORABLE for BoC)
2. indicate if they were confident or not confident on their classification, where non confident was said to be cases where judges felt the classes could not define the outcome, or they that justification did not allow them to clearly identify the outcome.
3. provide a written justification for their classification, which was optional if they were confident, but mandatory if they were not confident.

Cases annotated were 5 per case type, drawn randomly from the gold test set, again due to the time constraints of the panel.¹¹

4.7.4 Limitations of Evaluation Approach

The most significant limitation of the evaluation methodology is the small size of the gold test set and judge validation sample, which constrains statistical power and limits the ability to draw strong conclusions about model performance and even heuristic label extraction reliability.

As been said, time constraints and logistics played a major role in this limitation, were design decisions were made thinking judge availability could of been a bottleneck, and therefore the gold set was kept small to ensure judges could complete the validation task. But, even then, the size was reduced even more for the actual annotation task, as related to me, by Dr. Ana de Azeredo Coelho, the judges were extremely busy and had to prioritize their judicial duties over the annotation task, which is understandable.

Nonetheless, their feedback was invaluable, even beyond the annotation task, as seen further in the dissertation, where they're comments and even phone conversations, provided insights on the legal domain and the potential of AI in the Portuguese judicial system, which will be discussed in section 6.3.

¹⁰<https://streamlit.io/>

¹¹case IDs: DV: 65/24.0GEBRG.G1, 123/22.5PACTX.E1, 5052/21.7JAPRT-A.P1, 340/21.5TXLSB-E.L1-9, 87/25.3PGSXL-A.L1-5; BoC: 17126/26.0T8PRT.P1, 13951/22.2T8LSB-A.L1.S1, 1627/20.0T8PVZ.P1, 1980/19.8T8CTB.C1, 18203/23.8T8SNT.L1-6. Search them in <https://jurisprudencia.csm.org.pt/> to see the full text.

Chapter 5

Results

5.1 Fine-Tuned LegalBERTimbau Phase

5.1.1 Training Dynamics

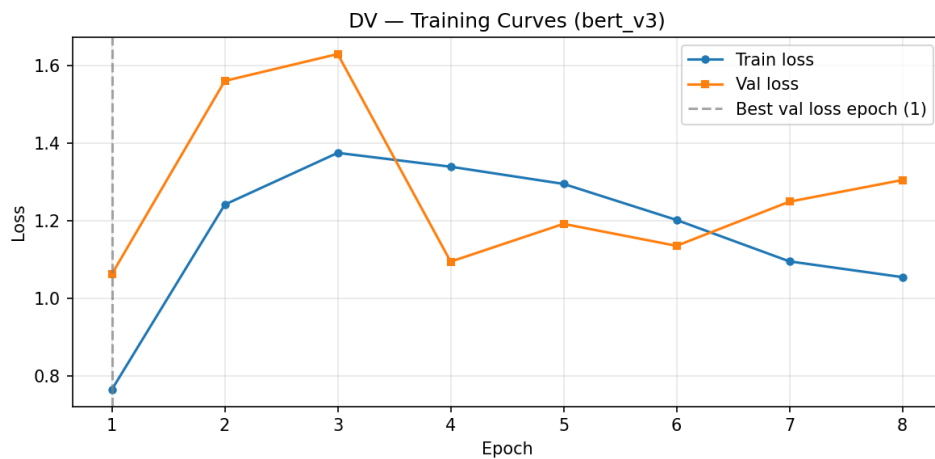


Figure 5.1: Training and validation loss curves for LegalBERTimbau fine-tuning on DV cases. The model was trained for up to 20 epochs with early stopping (patience=7) on raw validation loss.

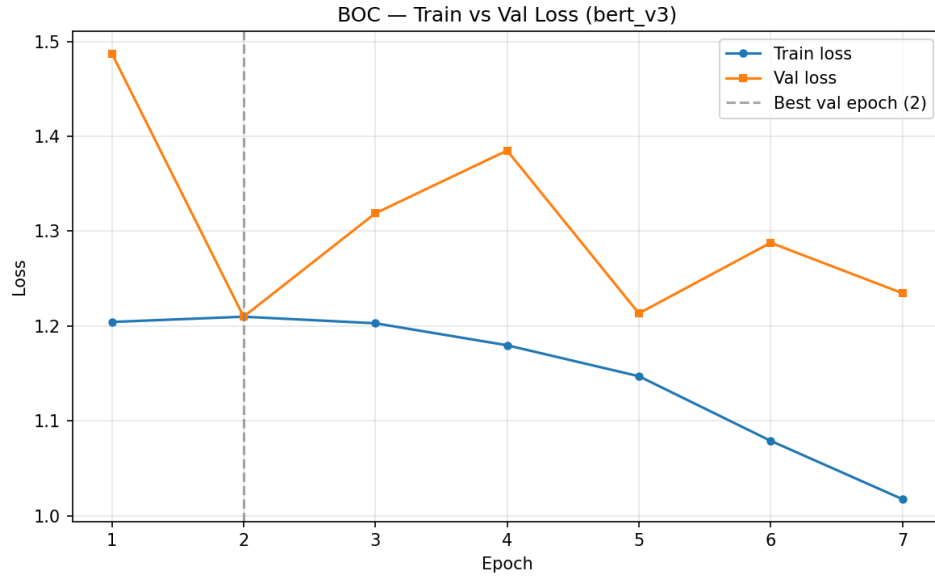


Figure 5.2: Training and validation loss curves for LegalBERTimbau fine-tuning on BoC cases. The model was trained for up to 20 epochs with early stopping (patience=7) on raw validation loss.

5.1.2 Train and Validation Set Performance

Domestic Violence (DV) — Binary Classification

Table 5.1: Fine-tuned LegalBERTimbau performance on the DV training set (n=904), the same data used to fit the model. The training set is near-balanced (469 KEPT / 435 ALTERED, 51.9%/48.1%), which rules out training-set class imbalance as an explanation for the collapse. The model predicted KEPT for all 904 documents, including 435 documents whose true label was ALTERED.

Metric	Value	95% CI	Class	Support
Precision	0.519	[0.519–0.519]	KEPT (0)	469
Recall	1.000	[1.000–1.000]	KEPT (0)	469
F1-Score	0.683	[0.683–0.683]	KEPT (0)	469
Precision	0.000	[0.000–0.000]	ALTERED (1)	435
Recall	0.000	[0.000–0.000]	ALTERED (1)	435
F1-Score	0.000	[0.000–0.000]	ALTERED (1)	435
Macro-F1	0.342	[0.342–0.342]		
MCC	0.000	[0.000–0.000]		
TSS	0.000	[0.000–0.000]		

Table 5.2: Fine-tuned LegalBERTimbau performance on the DV temporal validation set (n=160), used for early stopping during training. The model predicted KEPT for all 160 documents.

Metric	Value	95% CI	Class	Support
Precision	0.519	[0.519–0.519]	KEPT (0)	83
Recall	1.000	[1.000–1.000]	KEPT (0)	83
F1-Score	0.683	[0.683–0.683]	KEPT (0)	83
Precision	0.000	[0.000–0.000]	ALTERED (1)	77
Recall	0.000	[0.000–0.000]	ALTERED (1)	77
F1-Score	0.000	[0.000–0.000]	ALTERED (1)	77
Macro-F1	0.342	[0.342–0.342]		
MCC	0.000	[0.000–0.000]		
TSS	0.000	[0.000–0.000]		

Breach of Contract (BoC) — Ternary Classification

Table 5.3: Fine-tuned LegalBERTimbau performance on the BoC training set (n=1489), the same data used to fit the model. The model predicted UNFAVORABLE for 1144 documents and FAVORABLE for 345 documents, and never predicted PARTIAL despite 328 true PARTIAL examples in the training data.

Metric	Value	95% CI	Class	Support
Precision	0.584	[0.571–0.598]	UNFAVORABLE (0)	720
Recall	0.928	[0.908–0.946]	UNFAVORABLE (0)	720
F1-Score	0.717	[0.704–0.730]	UNFAVORABLE (0)	720
Precision	0.000	[0.000–0.000]	PARTIAL (1)	328
Recall	0.000	[0.000–0.000]	PARTIAL (1)	328
F1-Score	0.000	[0.000–0.000]	PARTIAL (1)	328
Precision	0.583	[0.544–0.629]	FAVORABLE (2)	441
Recall	0.456	[0.410–0.501]	FAVORABLE (2)	441
F1-Score	0.511	[0.473–0.551]	FAVORABLE (2)	441
Macro-F1	0.409	[0.394–0.426]		
MCC	0.303	[0.271–0.337]		

Table 5.4: Fine-tuned LegalBERTimbau performance on the BoC temporal validation set (n=263), used for early stopping during training. The model predicted UNFAVORABLE for 247 documents and FAVORABLE for 16 documents, and never predicted PARTIAL despite 48 true PARTIAL examples.

Metric	Value	95% CI	Class	Support
Precision	0.603	[0.587–0.621]	UNFAVORABLE (0)	152
Recall	0.980	[0.954–1.000]	UNFAVORABLE (0)	152
F1-Score	0.747	[0.730–0.764]	UNFAVORABLE (0)	152
Precision	0.000	[0.000–0.000]	PARTIAL (1)	48
Recall	0.000	[0.000–0.000]	PARTIAL (1)	48
F1-Score	0.000	[0.000–0.000]	PARTIAL (1)	48
Precision	0.625	[0.400–0.857]	FAVORABLE (2)	63
Recall	0.159	[0.079–0.254]	FAVORABLE (2)	63
F1-Score	0.253	[0.133–0.390]	FAVORABLE (2)	63
Macro-F1	0.333	[0.288–0.383]		
MCC	0.184	[0.079–0.278]		

5.1.3 Gold Test Set Performance

Domestic Violence (DV) — Binary Classification

Table 5.5: Fine-tuned LegalBERTimbau performance on DV gold test set (n=50). Metrics reported with 95% bootstrap confidence intervals (n=1000 resamples). The model collapsed to majority-class prediction (always predicting KEPT), resulting in zero MCC and F1 for the minority class (ALTERED).

Metric	Value	95% CI	Class	Support
Precision	0.580	[0.580–0.580]	KEPT (0)	29
Recall	1.000	[1.000–1.000]	KEPT (0)	29
F1-Score	0.734	[0.734–0.734]	KEPT (0)	29
Precision	0.000	[0.000–0.000]	ALTERED (1)	21
Recall	0.000	[0.000–0.000]	ALTERED (1)	21
F1-Score	0.000	[0.000–0.000]	ALTERED (1)	21
Macro-F1	0.367	[0.367–0.367]		
MCC	0.000	[0.000–0.000]		
TSS	0.000	[0.000–0.000]		

Breach of Contract (BoC) — Ternary Classification

Table 5.6: Fine-tuned LegalBERTimbau performance on BoC gold test set (n=50). Metrics reported with 95% bootstrap confidence intervals. The model collapsed to majority-class prediction (UNFAVORABLE), never predicting PARTIAL or FAVORABLE despite having 328 and 441 training examples of these classes, respectively.

Metric	Value	95% CI	Class	Support
Precision	0.700	[0.700–0.700]	UNFAVORABLE (0)	35
Recall	1.000	[1.000–1.000]	UNFAVORABLE (0)	35
F1-Score	0.824	[0.824–0.824]	UNFAVORABLE (0)	35
Precision	0.000	[0.000–0.000]	PARTIAL (1)	9
Recall	0.000	[0.000–0.000]	PARTIAL (1)	9
F1-Score	0.000	[0.000–0.000]	PARTIAL (1)	9
Precision	0.000	[0.000–0.000]	FAVORABLE (2)	6
Recall	0.000	[0.000–0.000]	FAVORABLE (2)	6
F1-Score	0.000	[0.000–0.000]	FAVORABLE (2)	6
Macro-F1	0.275	[0.275–0.275]		
MCC	0.000	[0.000–0.000]		

5.1.4 Embedding Space Analysis after Fine-Tuning

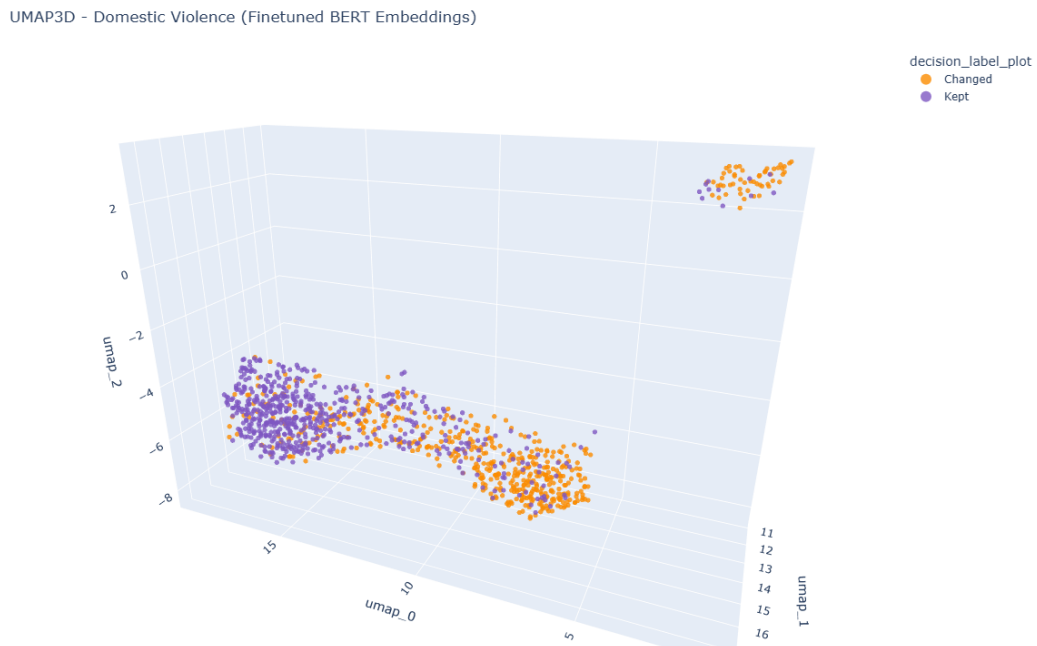


Figure 5.3: UMAP 3D projection of embedding space after fine-tuning. Consult Figure 3.20 for the pre-fine-tuning visualization. The figure shows one elongated and one smaller spherical cluster. This may indicate the model learned document-level structure. However, both DV classes (KEPT and ALTERED) remain somewhat intermixed within both clusters, suggesting the learned structure does not correspond to outcome classes but rather to linguistic or structural document features.

UMAP3D - Breach of Contract (Finetuned BERT Embeddings)

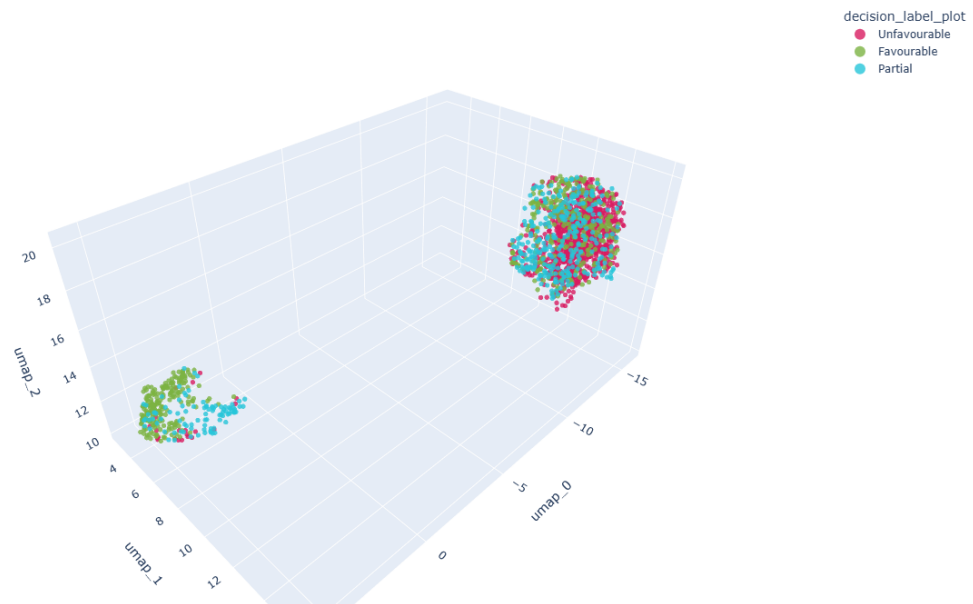


Figure 5.4: UMAP 3D projection of embedding space after fine-tuning. Consult Figure 3.20 for the pre-fine-tuning visualization. The figure shows approximately two spherical clusters. Similarly to DV, the BoC classes (UNFAVORABLE, PARTIAL, FAVORABLE) remain intermixed within both clusters, suggesting the learned structure does not correspond to outcome classes but rather to linguistic or structural document features.

5.2 LLM Phase

5.2.1 Zero-Shot with Smart Summarization

Domestic Violence (DV) — Binary Classification

Table 5.7: DeepSeek-R1-8B zero-shot performance on DV gold test set with smart summarization pipeline (context budget $\approx 9k$ tokens). Metrics with 95% bootstrap confidence intervals (n=1000 resamples). Substantially outperforms fine-tuned BERT (Table 5.5): MCC +0.592 vs 0.000, Macro-F1 +0.416 vs 0.367.

Metric	Value	95% CI	Class	Support
Precision	0.771	[0.684–0.875]	KEPT (0)	29
Recall	0.931	[0.828–1.000]	KEPT (0)	29
F1-Score	0.844	[0.762–0.918]	KEPT (0)	29
Precision	0.867	[0.700–1.000]	ALTERED (1)	21
Recall	0.619	[0.429–0.810]	ALTERED (1)	21
F1-Score	0.722	[0.545–0.872]	ALTERED (1)	21
Macro-F1	0.783	[0.661–0.895]		
MCC	0.592	[0.372–0.797]		
TSS	0.550	[0.325–0.775]		

Breach of Contract (BoC) — Ternary Classification

Table 5.8: DeepSeek-R1-8B zero-shot performance on BoC gold test set with smart summarization. The model correctly predicted all three classes, including FAVORABLE (recall = 0.833, F1 = 0.714) and PARTIAL (recall = 0.556), which fine-tuned BERT never predicted despite having 504 and 376 training examples. MCC improvement vs BERT: +0.468 vs 0.000.

Metric	Value	95% CI	Class	Support
Precision	0.867	[0.781–0.963]	UNFAVORABLE (0)	35
Recall	0.743	[0.600–0.857]	UNFAVORABLE (0)	35
F1-Score	0.800	[0.698–0.886]	UNFAVORABLE (0)	35
Precision	0.417	[0.214–0.643]	PARTIAL (1)	9
Recall	0.556	[0.222–0.889]	PARTIAL (1)	9
F1-Score	0.476	[0.222–0.700]	PARTIAL (1)	9
Precision	0.625	[0.400–1.000]	FAVORABLE (2)	6
Recall	0.833	[0.500–1.000]	FAVORABLE (2)	6
F1-Score	0.714	[0.462–0.923]	FAVORABLE (2)	6
Macro-F1	0.663	[0.521–0.805]		
MCC	0.468	[0.243–0.690]		

5.2.2 Few-Shot (k=1) with Smart Summarization

Domestic Violence (DV) — Binary Classification

Table 5.9: DeepSeek-R1-8B few-shot (k=1 per class) performance on DV gold test set. Few-shot degraded performance relative to zero-shot (Table 5.7): MCC 0.486 [0.257–0.681] vs 0.592 [0.372–0.797], Macro-F1 0.696 [0.554–0.822] vs 0.783 [0.661–0.895]. The performance pattern is consistent with few-shot examples anchoring the model toward the majority class, degrading minority-class recall. I refer to this counterintuitive degradation, where adding a labeled example reduced performance, as the few-shot paradox, analyzed in subsection 6.1.3.

Metric	Value	95% CI	Class	Support
Precision	0.700	[0.630–0.784]	KEPT (0)	29
Recall	0.966	[0.897–1.000]	KEPT (0)	29
F1-Score	0.812	[0.750–0.875]	KEPT (0)	29
Precision	0.900	[0.700–1.000]	ALTERED (1)	21
Recall	0.429	[0.238–0.619]	ALTERED (1)	21
F1-Score	0.581	[0.357–0.765]	ALTERED (1)	21
Macro-F1	0.696	[0.554–0.822]		
MCC	0.486	[0.257–0.681]		
TSS	0.394	[0.182–0.619]		

Breach of Contract (BoC) — Ternary Classification

Table 5.10: DeepSeek-R1-8B few-shot (k=1) performance on BoC gold test set. Few-shot slightly improved MCC over zero-shot (0.523 [0.314–0.736] vs 0.468 [0.243–0.690]) but decreased Macro-F1 (0.631 [0.449–0.792] vs 0.663 [0.521–0.805]). FAVORABLE recall dropped sharply (0.333 vs 0.833), consistent with few-shot examples anchoring the model away from the minority class.

Metric	Value	95% CI	Class	Support
Precision	0.906	[0.812–1.000]	UNFAVORABLE (0)	35
Recall	0.829	[0.686–0.943]	UNFAVORABLE (0)	35
F1-Score	0.866	[0.776–0.941]	UNFAVORABLE (0)	35
Precision	0.467	[0.312–0.667]	PARTIAL (1)	9
Recall	0.778	[0.444–1.000]	PARTIAL (1)	9
F1-Score	0.583	[0.385–0.783]	PARTIAL (1)	9
Precision	0.667	[0.000–1.000]	FAVORABLE (2)	6
Recall	0.333	[0.000–0.667]	FAVORABLE (2)	6
F1-Score	0.444	[0.000–0.800]	FAVORABLE (2)	6
Macro-F1	0.631	[0.449–0.792]		
MCC	0.523	[0.314–0.736]		

5.2.3 Summary Comparison: All Models

Table 5.11: Summary comparison of all models across both case types. Point estimates with 95% bootstrap confidence intervals (n=1000 resamples). Baseline Fine-tuned BERT (bert_v3) serves as the primary supervised baseline. DeepSeek-R1-8B zero-shot with smart summarization achieved the highest performance on both case types without any task-specific training.

Model	Case Type	Approach	Macro-F1 (95% CI)	MCC (95% CI)
Fine-tuned LegalBERT	DV	Supervised	0.367 [0.367–0.367]	0.000 [0.000–0.000]
DeepSeek-R1-8B	DV	Zero-shot	0.783 [0.661–0.895]	0.592 [0.372–0.797]
DeepSeek-R1-8B	DV	Few-shot	0.696 [0.554–0.822]	0.486 [0.257–0.681]
Fine-tuned LegalBERT	BoC	Supervised	0.275 [0.275–0.275]	0.000 [0.000–0.000]
DeepSeek-R1-8B	BoC	Zero-shot	0.663 [0.521–0.805]	0.468 [0.243–0.690]
DeepSeek-R1-8B	BoC	Few-shot	0.631 [0.449–0.792]	0.523 [0.314–0.736]

5.3 Judge Validation

5.3.1 Domestic Violence (DV)

Table 5.12: Judge annotations for DV cases (n=5 judges, 5 cases each). Each judge selected a decision label (KEPT or ALTERED) and confidence level (C = Confident, NC = Not Confident). Agreement with heuristic label = 24/25 (96%). K = KEPT, A = ALTERED, HL = Heuristic Label.

Case ID	J1	J2	J3	J4	J5	HL	Agree
65/24.0GEBRG.G1	K	K	K	K	K	K	5/5
123/22.5PACTX.E1	K	K	K	K	K	K	5/5
5052/21.7JAPRT-A.P1	A	A	A	K	A	A	4/5
340/21.5TXLSB-E.L1-9	K	K	K	K	K	K	5/5
87/25.3PGSXL-A.L1-5	K	K	K	K	K	K	5/5
<i>Overall DV agreement: 24/25 (96%)</i>							

Table 5.13: Judge confidence levels for DV annotations. C = Confident, NC = Not Confident. Total confidence: 20/25 (80%).

Case ID	Judge 1	Judge 2	Judge 3	Judge 4	Judge 5	Total C
65/24.0GEBRG.G1	C	C	C	C	C	5/5
123/22.5PACTX.E1	C	C	NC	C	C	4/5
5052/21.7JAPRT-A.P1	C	C	C	NC	C	4/5
340/21.5TXLSB-E.L1-9	C	C	C	NC	C	4/5
87/25.3PGSXL-A.L1-5	C	NC	NC	C	C	3/5
<i>Total confident: 20/25 (80%)</i>						

Selected Judge Commentary (DV)

Case 5052/21.7JAPRT-A.P1 — Scope Concern: Judge J4 (voted KEPT, not confident) noted that this case addresses a post-conviction procedural request for criminal record non-disclosure, not a domestic violence conviction appeal per se. The heuristic label ALTERED was extracted from the dispositivo of this procedural sub-decision, highlighting an annotation boundary condition addressed in the Discussion.

Case 123/22.5PACTX.E1 — Disagreement with Sentencing Practice: Judge J3 (voted KEPT, not confident) agreed with the appeal ruling on the compensation amount, but critiqued the first-instance sentence itself, describing it as “the paradigmatic example” of an excessive and systemic recourse to non-custodial measures that runs against treating domestic violence as a serious crime, and grounding this in GREVIO reports, the Istanbul Convention, ECHR jurisprudence and national plans against domestic violence. Although outside the strict scope of case classification, it shows that an outcome labelled KEPT can still be contested on its merits by a domain expert.

Case 87/25.3PGSXL-A.L1-5 — Disagreement with the Decision Itself: All five judges labelled this case KEPT, matching the heuristic label, yet two of them (J2 and J3) marked it as not confident for reasons unrelated to the label. J2 considered that the appellate court erred in its assessment of the established facts and in its application of the law, while J3 wrote that they would have granted the prosecutor’s appeal and revoked the decision. The label was correct, but the decision it summarises was contested by two of the five experts.

For complete judge comments and original Portuguese text, see the anonymised annotation dataset in the project repository (Appendix 7), under `judge_labeling_app/reports/anonymised/`.

5.3.2 Breach of Contract (BoC)

Table 5.14: Judge annotations for BoC cases (n=5 judges, 5 cases each). All annotations marked as Confident. Agreement with heuristic label = 21/25 (84%); excluding judge J2 (who produced annotations divergent from the other four judges on 4 cases), agreement among the remaining four judges = 20/20 (100%). U = UNFAVORABLE, F = FAVORABLE, P = PARTIAL, HL = Heuristic Label.

Case ID	J1	J2	J3	J4	J5	HL	Agree
17126/26.0T8PRT.P1	U	F	U	U	U	U	4/5
13951/22.2T8LSB-A.L1.S1	U	F	U	U	U	U	4/5
1627/20.0T8PVZ.P1	U	F	U	U	U	U	4/5
1980/19.8T8CTB.C1	P	F	P	P	P	P	4/5
18203/23.8T8SNT.L1-6	F	F	F	F	F	F	5/5
<i>Overall BoC agreement: 21/25 (84%); excluding outlier, 20/20 (100%)</i>							

Selected Judge Commentary (BoC)

Case 1627/20.0T8PVZ.P1 — Class Inadequacy Critique: Judge J1 proposed alternative terminology (“recurso procedente, recurso improcedente, recurso parcialmente procedente”) and identified a structural limitation: when both parties appeal, the appellate court can rule partially favorable to each, producing outcomes that do not fit cleanly into the three-class scheme. This expert feedback validates a known limitation discussed in the thesis evaluation section.

Case 13951/22.2T8LSB-A.L1.S1 — Multi-Court Hierarchy Navigation: Judge J1 provided an

explicit trace of the three-level court hierarchy: 1st instance declared competence → Tribunal da Relação revoked that decision (favorable to appellant) → STJ maintained the TRC ruling (unfavorable to the second appellant). This demonstrates the domain expertise required to disambiguate multi-instance confusion, a failure mode identified in model error analysis.

For complete judge comments and original Portuguese text, see the anonymised annotation dataset in the project repository (Appendix 7), under `judge_labeling_app/reports/anonymised/`.

5.4 Summary Table: Judge Validation Overview

Table 5.15: Summary of judge validation exercise across both case types. DV judges showed 80% confidence; BoC judges showed 100% confidence. High agreement on unambiguous cases validates the heuristic label extraction; disagreements point to genuine scope and boundary conditions in the classification task.

Case Type	Judges	Cases Annotated	Overall Agreement	Confidence
DV	5	25	24/25 (96%)	20/25 (80%)
BoC	5	25	21/25 (84%)	25/25 (100%)

BoC agreement excludes annotations from one judge whose pattern diverged from the other four. Consensus among remaining judges: 20/20 (100%)

Chapter 6

Discussion

6.1 Synthesis of Core Findings

6.1.1 Fine-Tuned BERT as Baseline

The fine-tuned LegalBERT_{Timbau} model collapsed to majority-class prediction on both case types. On DV, BERT achieved Macro-F1 of 0.367 and MCC of 0.000 (Table 5.5), predicting KEPT for all 50 test cases and never predicting ALTERED despite having 21 minority examples in the test set. In parallel, BoC performance was similarly poor with Macro-F1 of 0.275 and MCC of 0.000 (Table 5.6), with the model predicting only UNFAVORABLE and never PARTIAL or FAVORABLE, despite 328 and 441 training examples of these classes respectively.

This collapse is striking but warrants careful interpretation in the context of the study. BERT’s failure likely reflects the training protocol rather than fundamental architectural weakness. The model was subjected to heavy regularization, which was intentionally configured to prevent overfitting on the small training set. It had a dropout of 0.3, L2 weight decay of 0.05, and label smoothing of 0.1, applied to both case types to mitigate overconfidence on minority classes. However, these same aggressive regularization techniques, combined with a reduced learning rate (5e-6) and early stopping after 7 epochs of validation patience, appear to have restricted the model’s capacity to learn any class-discriminative features at all, causing it to default to majority-class prediction. From the loss curves (Figure 5.1 and Figure 5.2), training loss had a final downward trend but never approached convergence of 0.0.

To settle whether this reflects underfitting or overfitting, I scored both checkpoints on the training and validation splits themselves (subsection 5.1.2), not only on the gold test set. For DV, train, validation and gold test performance are identical to three decimal places (MCC 0.000 on all three, Table 5.1, Table 5.2, Table 5.5). The model predicts KEPT unconditionally, including for 435 ALTERED documents it was directly trained on, in a training set that is itself near-balanced (51.9%/48.1%). There is no train/held-out gap for overfitting to explain, so DV’s collapse is underfitting. As for BoC, a real but partial signal is visible on the training set (MCC 0.303), which degrades on the temporal validation split (MCC 0.184) and disappears entirely on the gold test set (MCC 0.000). PARTIAL, however, is never predicted on any of the three splits despite 328 training examples, so the dominant failure mode for BoC is also underfitting, with the FAVORABLE signal that was learned failing to transfer forward in time on unseen cases. Taken together, the loss curves, the train/validation results, and the gold test performance consistently show that both models had substantial difficulty learning patterns in the text sufficient to confidently and correctly classify cases under the heuristic labels used for training.

It was known beforehand that the task itself was genuinely difficult, where predicting outcomes from

intricate prose requires heavy reasoning and legal domain knowledge. The combination of demanding a domain shift (from legal text similarity to outcome classification), training on a small dataset without hyperparameter tuning, and applying heavy regularization, all likely contributed to this collapse rather than an architectural limitation of BERT.

Nonetheless, BERT proved extremely useful in practice. The smart summarization pipeline (section 4.5) relied on BERT for encoding capabilities to extract relevant facts from full judgment text before feeding summaries to the LLM, without altering the original words in the documents. This suggests that even if BERT struggles with fine-tuning for direct classification on this domain, it can still be valuable for other auxiliary tasks, to aid more powerful LLMs in downstream classification tasks.

For future work, this points to alternative roles, where BERT is fine-tuned for accessory tasks like information extraction and processing, rather than direct classification.

In summary, without deeper experimentation, this study cannot definitively rule out the used LegalBERTimbau as competitive, but rather conclude that this training protocol on this data proved insufficient to achieve strong performance.

6.1.2 DeepSeek Zero-Shot

DeepSeek-R1-8B achieved dramatically different results than LegalBERTimbau using zero-shot reasoning with the smart summarization pipeline. On DV, the model reached Macro-F1 of 0.783 [0.661–0.895] and MCC of 0.592 [0.372–0.797], correctly predicting both KEPT and ALTERED with reasonable precision and recall (KEPT precision 0.771, ALTERED recall 0.619). On BoC, it achieved Macro-F1 of 0.663 [0.521–0.805] and MCC of 0.468 [0.243–0.690], and critically, it correctly predicted all three classes (FAVORABLE with recall 0.833 and F1 0.714, and PARTIAL with recall 0.556). LegalBERTimbau never predicted any non-majority classes, for either case type.

This performance is meaningful because it required no task-specific training data, no fine-tuning, and no explicit few-shot examples. Given its inherent reasoning capabilities, the model applied reasoning directly.

This suggests that by leveraging the model’s pre-trained knowledge of legal logic and Portuguese language and carefully instructing it, the model outperforms a BERT model learned from limited labeled examples.

The core finding is that in this study’s setting, a larger reasoning-capable model generalized better than the small fine-tuned encoder under the protocol tested, and corroborates comparative studies in order jurisdictions, like Nigam, Deroy, et al. (2024), which found LLMs outperformed fine-tuned BERT on Indian Supreme Court judgements.

Furthermore, I must acknowledge extracting relevant information from the full judgment text was essential, as inputting the full documents presented multiple practical constraints like attention dilution (evidenced in the model’s reasoning logs), out-of-memory errors at hardware limits (32,768 tokens), and excessive latency. The summarization pipeline addressed all three constraints, as detailed in section 4.5.

Therefore, these results reflect the combined effect of the model’s reasoning capabilities, the prompting design, and the smart summarization pipeline, rather than the model in isolation.

I had hypothesized that advanced prompting techniques, namely CoT and RAG, would further improve performance, however, this study neither ablated CoT nor implemented RAG, leaving this hypothesis untested here.

Future studies should explore the individual and relative contributions components of the decision making system, mentioned above, to better understand the performance of LLMs on legal judgment

prediction tasks, in Portugal.

6.1.3 Few-Shot Paradox

Few-shot prompting, with one example per class, slightly degraded performance relative to zero-shot on both case types. On DV, MCC fell from 0.592 [0.372–0.797] to 0.486 [0.257–0.681], and Macro-F1 fell from 0.783 [0.661–0.895] to 0.696 [0.554–0.822]. The minority class (ALTERED) bore the cost with its recall dropping from 0.619 to 0.429. On BoC, the pattern was more complex. MCC slightly improved from 0.468 [0.243–0.690] to 0.523 [0.314–0.736], but Macro-F1 dropped from 0.663 [0.521–0.805] to 0.631 [0.449–0.792]. Most tellingly, FAVORABLE recall collapsed from 0.833 to 0.333, a pattern consistent with few-shot examples anchoring the model away from predicting rare classes.

Nonetheless, these results should be interpreted cautiously. Part of this outcome may also stem from my own methodology. As noted in subsection 4.5.6, examples were retrieved through previously extracted sentences cosine similarity, as seen in section 4.5, which is only a proxy for true judicial similarity, potentially losing important nuances, and the stacking of several heuristics in the methodology, could itself have added noise. In addition, the gold test set is small, and the confidence intervals overlap heavily. I therefore cannot conclude that few-shot prompting is definitively worse than zero-shot, only that it did not improve performance in this study, because it is not possible to cleanly separate the effect of few-shot prompting from the effect of these unvalidated design choices.

Even so, the difficulty appears to run deeper than this specific implementation. I found Alfieri et al. (2024), which implemented a similar approach to the one used here, that showed that dynamically selecting demonstrations by embedding similarity outperforms static selection curated in advance by experts or by LLMs across legal argument mining tasks. Yet a similarity driven approach still did not significantly improve few-shot here. I hypothesize that this points to a more fundamental limitation. With only one example per class, the demonstration is a thin, high variance anchor, and the model can latch onto the surface features of that single case rather than the underlying legal reasoning. Notice that recall for the anchored class fell in both case types. The model correctly recognized fewer true ALTERED cases on DV and fewer true FAVORABLE cases on BoC, misclassifying more of them into the other classes. This is what would be expected if a single example per class cannot capture the internal variation of that class, leaving the model less able to recognize instances that differ from the one anchor it was shown.

This connects to a broader, cross domain problem in large language models. Studies like, Hannan et al. (2026) document the same difficulty of selecting effective few-shot examples in software vulnerability detection, a domain far removed from law, which suggests that example selection fragility is a general property of in context learning rather than a quirk of Portuguese legal text. Earlier cited work in Min et al. (2022) and Tang et al. (2025) already noted that providing examples does not automatically improve results.

However, these findings do not invalidate few-shot prompting. They underscore that few-shot design is a non trivial discipline, and that solving it well in the Portuguese legal domain is itself a worthwhile research direction.

Knowing these limitations, future work could explore few-shot prompting with more examples per class once hardware allows, and could ablate which kinds of examples help or hurt, moving beyond raw sentence similarity toward selection criteria that better track the features determining a case outcome.

6.2 Addressing the Research Questions Directly

6.2.1 RQ1 — Can LLMs Outperform BERT?

RQ1 asked whether LLMs can outperform the fine-tuned LegalBERTimbau baseline on both case types, and H1a predicted this outperformance in quantitative terms. Under the specific conditions tested in this thesis, I conclude that both are confirmed. DeepSeek-R1-8B zero-shot reached MCC 0.592 (DV) and 0.468 (BoC), against 0.000 for LegalBERTimbau on both case types.

This answer, however, is constrained to this study’s protocol rather than a general claim about architectures. As discussed in subsection 6.1.1, LegalBERTimbau’s collapse may reflect training conditions (limited data, no hyperparameter tuning, domain shift) rather than an inherent ceiling for fine-tuned encoder models on this task. A more rigorous BERT protocol was outside the scope of this thesis, but I speculate it could narrow this gap.

It is also worth noting that neither model is close to production-ready for autonomous judicial classification. An MCC of 0.592 still means a substantial share of cases are misclassified, and the bootstrap confidence intervals are wide given the modest gold test set ($n=50$ per case type), so the estimated gap between models carries meaningful uncertainty.

In summary, RQ1 is fully answered, and H1a is supported, within the specific harness and training protocol evaluated here.

6.2.2 RQ2 — Can Models Explain Their Decisions?

RQ2 asked whether the models can explain their decisions, and whether these explanations align with legal reasoning, which H2 predicted would happen with the implementation of XAI techniques. However, this thesis compared two fundamentally different explanation artifacts. On one hand, I applied SHAP attributions over LegalBERTimbau’s sentence inputs, and on the other hand, I examined DeepSeek’s own chain-of-thought reasoning logs.

Both approaches produced explanations in the sense that they exposed something about the models’ internal decision process. SHAP attributions were dispersed across sentences rather than concentrated on a few clearly legally-relevant sentences, and their interpretation demands some technical familiarity with feature attribution methods. DeepSeek’s reasoning logs, by contrast, were human-readable prose, as the model cited case facts and applied what reads as legal logic, which is considerably more accessible to a legal professional without a machine learning background.

However, as discussed in section 5.3 and in section 6.3, neither explanation type was validated against expert legal judgment, due to time and resource constraints, and the two are not directly comparable as objects. This matters because interpretability and correctness are not the same thing, and this issue arose in this thesis. Based on the dispersed SHAP attributions from LegalBERTimbau, it is clear that SHAP attributions can be produced and read, but the sentences they highlight carry no legal significance, precisely because the model did not learn to discriminate outcomes in the first place. In parallel, I also observed that chain-of-thought log can be a fluent and plausibly sounding rationalization of an incorrect conclusion rather than a faithful reflection of the model’s decision process, of a wrongly predicted case. Without expert validation, I cannot confirm that either explanation aligns with actual legal reasoning, only that both are legible to different degrees.

Therefore, H2 remains unresolved rather than confirmed or refuted, and RQ2 is answered only partially, as it can be concluded that both models can produce explanations, but not that those explanations are legally sound or correct, with the methodology and resources available in this study.

6.2.3 RQ3 — Does Performance Differ Between DV and BoC?

RQ3 asked whether performance and explainability differ between DV, a sensitive and human-centered domain, and BoC, a more documentary and objective domain, and H3 predicted that DV would perform worse given its subjectivity.

The point estimates run counter to this prediction. It was found that DeepSeek zero-shot MCC was 0.592 [0.372–0.797] and Macro-F1 was 0.783 [0.661–0.895] on DV against 0.468 [0.243–0.690] and 0.663 [0.521–0.805] on BoC. Hence, one could say that DV performed marginally better, not worse. However, these confidence intervals overlap substantially, so this difference cannot be treated as statistically meaningful with the current sample.

Beyond the overlapping intervals, the gold test set is small, at 50 cases per task, which leaves little statistical power to detect anything short of a large difference between the two case types. H3 was therefore not formally tested, whether by Mann-Whitney U, Cohen’s d, or an equivalent procedure applied to the bootstrap distributions. Additionally, since no explainability analysis was specifically conducted to compare the models’ decision-making processes across case types, nor validated against expert judgment, the hypothesis that DV would be harder to predict than BoC, given its human-centered nature against BoC’s documentary one, cannot be corroborated here either.

Consequently, H3 is treated as an open question rather than a resolved finding, and RQ3 stays unanswered. Formal hypothesis testing on the bootstrap distributions, over a larger gold test set than the one available here, is recommended as immediate future work.

6.3 Judge Validation

The judge validation exercise (section 5.3) asked five legal professionals to independently label a small sample of cases per case type and compare their verdicts against the heuristic labels used for training and evaluation. Agreement in the study was high, when considering all cases, where 24 out of 25 (96%) for DV and 21 out of 25 (84%) for BoC, or 20 out of 20 (100%) when excluding the annotations from one judge whose pattern on four BoC cases diverged from the consensus of the other judges. However, with a panel of five judges annotating only five cases each per case type, no formal statistical claim can be drawn from these agreement rates. So one must treat the exercise as directional evidence, not a validated inter-rater reliability study.

What is more informative than the raw agreement numbers is that judges disagreed, or reported low confidence, even when independently reviewing the same short case. This matters beyond the specific disagreements. In my view, this suggests that legal judgment prediction is not a task with a single ground truth that a sufficiently good system converges on, but one where even expert human interpretation varies. A model trained on this kind of data does not just inherit the task’s genuine difficulty, it also inherits whatever interpretive tendencies, and potential biases, are latent in whichever labels it was trained on, heuristic or human. The written comments point the same way (section 5.3): on one case all five judges chose the same label, yet two marked it as not confident and disagreed with the appellate decision itself, and another judge contested the sentencing practice reflected in the decision being labelled. Agreement with a label, in other words, did not imply agreement with the decision it summarises.

A few judges also pointed out that the *acórdão* they were asked to annotate was not actually adjudicating the case type in question on its merits, but an accessory or procedural sub-decision, such as a request for future memory declarations or a conditional parole determination, that only touched the case type tangentially. Rather than treat this as a labeling defect to be patched with better regex rules, this points

to a more fundamental issue upstream, in the data extraction phase. The keywords (called *descriptores*) used to query the source database can be misleading, surfacing a document where domestic violence is mentioned, even when the decision being rendered is not actually about convicting or acquitting a defendant. Without domain expertise, this kind of mismatch can go unnoticed indefinitely, as in order to catch it required a legal professional reading the actual document. The problem resides that in order to make Data Science research on this domain, one must gather this data in streamlined and automated way, as manually reviewing each document is not scalable. This likely reflects that these public case databases are published primarily for legal professionals, and only secondarily, if at all, for computational reuse. Improving machine and non-expert human readability of these metadata sources, so that descriptive tags reliably reflect what a document actually decides, is a concrete direction for future work that would benefit legal AI research well beyond this thesis.

Finally, informal conversations held with several participating judges surfaced views that do not appear in the written annotations but are relevant context for this section. Judges expressed openness to AI assistance in principle, but clear reluctance toward its unsupervised, large-scale application, citing recent cases of legal professionals misusing AI tools, including submitting fabricated case citations and legal references, and general concern about misuse of this kind of technology in a sensitive field. Several also reiterated, consistent with the written commentary in section 5.3, that the heuristic labels are approximations of outcome categories, and that more established terminology, should be adopted instead. Given that this study’s results show LLM zero-shot prediction clearly outperforming the fine-tuned BERT baseline, revisiting the label scheme’s granularity, now with a stronger baseline established, is a natural next step for future work rather than being a limitation of this thesis.

6.4 Where AI Can and Cannot Help in Judicial Decision-Making

Neither model evaluated in this thesis is suitable for autonomous or direct decision-aiding in a judicial context. Even DeepSeek zero-shot, the best-performing configuration, demonstrated moderate correlation with true outcomes (MCC 0.592 on DV, 0.468 on BoC), well above chance but with wide confidence intervals and failure modes fundamentally incompatible with production use in a judicial context, and the misclassification analysis shows these errors are not simply random noise. Some of the recurrent errors clustered around specific, recurring failure modes, such as confusing which court level’s ruling is final in multi-instance appeals, misidentifying which party’s perspective the label should be read from, mistaking the Ministério Público’s non-binding recommendation for the actual ruling, or missing that a modification to a sentence’s ancillary conditions still counts as an altered decision. Of course, I acknowledge that the summarization pipeline applied to the document, could of also contributed to these mistakes in reasoning, due to loss of information, but the point remains that the model used not reliable enough to be trusted with consequential decisions. Surfacing an outcome prediction directly to a judge, or worse, to a party in a case, on the strength of this study’s models would risk introducing exactly the kind of confident, butwrong signal that erodes trust and could bias a decision rather than support it.

Therefore, I propose some examples of realistic, lower-risk applications:

- **Routing and Prioritization:** classifying incoming cases by predicted complexity or urgency to help distribute caseloads;
- **Process Automation:** extracting key facts, dates and procedural status to reduce administrative overhead;

- **Ranking:** surfacing cases that may be more likely to be appealed or that warrant closer review;

There are many other potential applications, but the common thread is the embedding of LLMs in the institutional workflow of the justice system, rather than a standalone piece in decision-making. In short, all of these examples, move the needle of Legal AI towards Agentic AI capabilities, as discussed thoroughly in studies like Liu et al. (2026).

However, realizing this in practice is itself a substantial task, beyond technical difficulties. It requires understanding the actual process steps, bottlenecks, and human patterns of the institution in question, which is beyond what this thesis, focused on establishing baseline predictive capability, set out to do.

6.5 Limitations of This Study

Before expanding on specific constraints of this study, it is important to note that the work frames the task as retrospective outcome classification from appellate court decisions, following standard framing in legal judgment prediction literature. The model reads the appellate court’s reasoning after the outcome has already been decided, and the task is to classify that outcome. This is not preliminary prediction from unresolved case facts, where a production system would instead receive proven facts and case context at an earlier procedural stage, with the model assisting in reasoning rather than post-hoc reading of finalized decisions. Therefore, this is considered a quirk of this research branch, rather than design limitation, as previously mentioned in section 2.4.1. However, future research could then expand this field by exploring predictions on unresolved cases, and then verifying the model’s predictions against the final court decision, which would be a more realistic and consequential application of this technology.

This dissertation had several limitations that constrain the generalizability of its findings, but that also point to concrete directions for future work.

The first constraint starts at the data level, where only a small subset of cases were able to be collected and processed for this study. Then, the quality itself of this data proved limited, as processing of this raw documents proved challenging and reduced even more the usable sample size, as seen in subsection 3.0.1.

Furthermore, the heuristical labels used for training and evaluation for this classification task, although simplified the study, were objectively imperfect, and legal experts pointed out that they do not reflect the most established terminology in the field and the actual complex decisions itself that the court is making. Adding to this imperfection, during implementation I identified a small edge case in the pattern order used to engineer the BoC labels. The function `extract_decision_ternary_from_summary_boc` (shown in section .2) checks negation and unfavorable cues (like *improcedente*) before checking partial cues (like *parcialmente*), so an expression such as *parcialmente improcedente* matches the negation pattern first and is classified as UNFAVORABLE rather than PARTIAL. This mismatch is symptomatic of a broader tradeoff in this heuristic labelling approach, where collapsing the many literal expressions judges actually use into a handful of engineered classes required an interpretive step that will always miss edge cases like this one. An alternative design for future work should keep the literal judicial expressions as labels directly, avoiding this interpretation step entirely. The resulting label diversity would not have been a problem for the LLM based approaches, since the relevant expression variants could simply be defined in context through the system prompt (as shown in the prompting examples). However, this was not a viable option for the fine tuned BERT baseline, which requires a small, fixed set of discrete output classes to train and evaluate against. Since a central goal of this study was to faithfully compare BERT and the LLMs under the same task definition, the simplified label scheme was adopted for both, at the cost of edge cases like this one.

Finally, The test sets used for evaluation were also extremelys mall, as they had the objective of being manually validated by legal experts, to at least directionally confirm the label extraction logic and the model classification alignment with judges decisions. Unfortunately, due to unexpected time constraints, even this small test set was not fully validated by judges, not fully corroborating the label extraction logic by reliable statistical measures. Even if the evaluation set was larger, limitations and possibly reducing of the test set would happened regardless, due to the low GPU compute budget available for this study, which affected the iterative experimentation a study of this nature requires.

The previous factors, set up the work discussed in section 4.5, where the smart summarization pipeline was implemented to extract relevant information from the full judgment text, to fit within the LLM context window and avoid diluting the signal with irrelevant information. However, this design was not evaluated itself in this study, and its effectiveness is not quantifiable, other than it permitted inference to be performed in more reasonable time and with less GPU memory, than if the full document was used as input.

As discussed in subsection 6.1.2, this thesis did not implement retrieval-augmented generation legal statutes, and did not ablate chain-of-thought reasoning, since it came built into the evaluated model rather than being separately elicited through prompting. Both of these techniques were part of the original H1b hypothesis and remain untested here. Also, explainability outputs were compared qualitatively but never validated against expert legal judgment at scale, so H2 remains open. Finally, this thesis did not conduct any fairness or disparate-impact analysis of either the heuristic labels or the model predictions, a prerequisite for any future real world deployment.

However, despite these limitations, I believe the thesis makes a meaningful contribution by establishing a baseline for Portuguese legal judgment prediction, demonstrating that LLMs can outperform fine-tuned BERT models under the tested conditions, and highlighting the challenges and considerations for future research in this domain and the broader legal AI ecosystem in Portugal.

6.6 Future Work

This thesis establishes a baseline across two case types and two modeling paradigms, but each component it touches is superficial enough to warrant a dedicated follow-up studies, breaking down the complex problems and limitations into smaller, more manageable piece and diving deeper into each, trying to achieve a more comprehensive understanding of them and testing them under more rigorous conditions.

For example, on the data side, the reliance on regex-based target and document sectioning extraction, could be improved for example, applying similar models to LegalBERTimbau, fine tuned on a larger validated samples, to extract and retrieve relevant information from the documents, in a more robust but specially scalable way, than the current rule based approach. Refining and expanding on the alogrithm proposed in section 4.5 could also be a direction for future work, to improve the quality of the summaries and reduce the loss of relevant information, which could improve not only the performance of the tasks assigned to the LLM, but also reduce the cost associated with its inference, by reducing the number of tokens processed and therefore the GPU memory allocated.

Particularly, studies like these could help legal data authorities like DGSi and CSM, by publishing more structured, richer in metadata, and more machine and non-expert human readable content at the source itself. This would benefit not just this line of research but the wider legal AI research and industry in Portugal, as it would allow for more robust and scalable data collection and processing pipelines, and reduce the need for manual validation by legal experts.

Regarding LLMs, a systematic multi-model comparison, by testing additional, similarly sized, open-source and proprietary LLMs on the same harness, would clarify how much of this thesis’s findings

are specific to DeepSeek-R1-8B versus general to instruction-tuned reasoning models. Furthermore, as mentioned briefly in section 6.4, the application of these models should be envisioned in a more agentic way, making them dynamically interact with the existing institutional workflows, rather than a static, single-purpose system.

To close the gap on explainability, future studies require researching XAI techniques that are theoretically well-grounded for this domain, applying them experimentally, and then, following the same logic used to build the judge annotation tool for this thesis, by packaging these techniques outputs into software with a legal expert friendly interface, so that domain experts can reliably assess a model’s decision making process at scale, therefore, allowing statistically meaningful conclusions rather than commentary anecdotes.

Finally, a fairness and responsibility audit of both the data and model outputs is a prerequisite that any future work aiming at real deployment would need to complete before even considering any real-world application, and should be a priority for any follow-up studies in this line of research.

Chapter 7

Conclusion

This dissertation asked whether modern language models can support judgment prediction in the Portuguese legal context, where no prior work had established that they could. It compared a fine-tuned LegalBERTimbau baseline against a locally-run reasoning LLM, DeepSeek-R1-8B, over two contrasting case types, Domestic Violence and Breach of Contract, and asked not only which approach predicts outcomes more accurately, but whether either can explain itself in a way a legal professional could follow and trust.

The findings were clear on some fronts and open on others. Fine-tuned LegalBERTimbau collapsed to majority-class prediction on both case types, while DeepSeek-R1-8B, using zero-shot reasoning over the smart summarization pipeline, correctly predicted minority classes and substantially outperformed it, answering RQ1 within the protocol tested here. However, few-shot prompting did not improve on zero-shot and in some cases actively hurt minority-class recall, cautioning against the assumption that more examples are always better.

In terms of explainability, RQ2, remains genuinely unresolved. Even though both models produced explanations, SHAP attributions for BERT and chain-of-thought reasoning for DeepSeek, neither was validated at scale against expert legal judgment, so legibility was demonstrated without demonstrating correctness.

Furthermore, RQ3, whether the human-centered DV domain would prove harder than the documentary BoC domain, was not settled either way. The confidence intervals overlapped, and with only 50 cases per task there was not enough statistical power to detect anything short of a large difference, so H3 is left as an open question rather than a resolved finding.

Interestingly, the judge validation exercise and the comments obtained from it, reinforced a broader point underlying all three questions, as even legal experts don't fully agree with each other on some cases, suggesting that judgment prediction is not a task with a single ground truth a model can simply converge on. This has implications for both model training and evaluation, as well as for the practical deployment of any system in a real-world judicial context, and underlined the complexity of such task.

Against the three objectives set out in section 1.2, the record is mixed. O1, establishing a research baseline for Legal Judgment Prediction in Portugal, was met, as this work leaves behind a documented corpus, a reproducible pipeline, two modelling paradigms evaluated on the same held-out sets, and a public code and data release. O2, experimenting with and evaluating DL and LLM techniques and recommending further directions, was also met, though bounded by the training protocol used here, which was not optimally tuned. O3 was only partly reached. XAI techniques were applied and explanations were produced for both models, but they were never validated by legal experts at scale, and the judge panel that did take part validated the label extraction rather than the explanations, so the trust-building

purpose behind O3 remains unfulfilled.

This work also has limitations, discussed at length in section 6.5. The corpus was small and needed aggressive cleaning, the labels were engineered heuristically and diverge from the terminology judges use in practice, the gold test sets were never fully validated by the judge panel, and neither the summarization pipeline nor retrieval-augmented generation nor a fairness audit was implemented or evaluated here.

The directions that follow, detailed in section 6.6, group into four. On the data side, better extraction models and a refined summarization pipeline would grow and clean the corpus, and could also make the case to DGSi and CSM for publishing richer decisions at the source. On the modelling side, running the same harness over several comparable models would show how specific these findings are to DeepSeek-R1-8B. On explainability, XAI outputs need to be packaged behind an expert-friendly interface so judges can assess them at scale rather than anecdotally. And before any real deployment, a fairness and responsibility audit is a prerequisite, alongside formal hypothesis testing of H3 on a larger gold test set.

To conclude this study, three key insights stand out.

First, reasoning-capable LLMs used zero-shot can outperform small fine-tuned encoders on Portuguese legal judgment prediction without any task-specific training data, which reframes where the practical effort of fine-tuning should be spent in future work.

Second, explainability and correctness are separate properties, since a model can produce a fluent, legible rationale for a wrong answer, so neither model evaluated here is ready for autonomous or direct decision-aiding. The realistic near-term applications are instead lower-risk, but deeply embedded in institutional workflows rather than a standalone tool.

Third, what this dissertation contributes is mostly a starting point. It leaves a corpus, an evaluation harness and a set of sharply defined open questions where there was previously nothing to build on in the Portuguese context, which is what a baseline study is for.

References

- Aletras, Nikolaos et al. (Oct. 24, 2016). “Predicting judicial decisions of the European Court of Human Rights: a Natural Language Processing perspective”. In: *PeerJ Computer Science* 2. Publisher: PeerJ Inc., e93. ISSN: 2376-5992. DOI: 10.7717/peerj-cs.93. URL: <https://peerj.com/articles/cs-93> (visited on 10/06/2025).
- Alfieri, Francesco et al. (2024). “Dynamic Demonstrations Selection for Few-Shot Legal Argument Mining”. In: *Proceedings of the CEUR Workshop Proceedings*. Vol. 4089. CEUR-WS.org. URL: <http://ceur-ws.org/Vol-4089/paper2.pdf>.
- Alva Principe, Renzo, Nicola Chiarini, and Marco Viviani (2025). “Long Document Classification in the Transformer Era: A Survey on Challenges, Advances, and Open Issues”. In: *WIREs Data Mining and Knowledge Discovery* 15.2, e70019. DOI: 10.1002/widm.70019. URL: <https://wires.onlinelibrary.wiley.com/doi/full/10.1002/widm.70019>.
- Anantathanavit, Rungpilin et al. (Feb. 2024). “Utilizing AI and Natural Language Processing to Predict Supreme Court Decisions in Thailand”. In: *2024 16th International Conference on Knowledge and Smart Technology (KST)*. 2024 16th International Conference on Knowledge and Smart Technology (KST). ISSN: 2473-764X, pp. 45–50. DOI: 10.1109/KST61284.2024.10499665. URL: <https://ieeexplore.ieee.org/abstract/document/10499665> (visited on 10/06/2025).
- Armour, John, Richard Parnham, and Mari Sako (2020). “Augmented Lawyering”. In: *SSRN Electronic Journal*. Publisher: Elsevier BV. ISSN: 1556-5068. DOI: 10.2139/ssrn.3688896. URL: <https://www.ssrn.com/abstract=3688896> (visited on 07/18/2025).
- Baldi, Pierre et al. (2000). “Assessing the accuracy of prediction algorithms for classification: an overview”. In: *Bioinformatics* 16.5, pp. 412–424. DOI: 10.1093/bioinformatics/16.5.412.
- Bellamy, Rachel K. E. et al. (2018). “AI Fairness 360: An extensible toolkit for detecting and mitigating algorithmic bias”. In: *arXiv preprint arXiv:1810.01943*. DOI: 10.48550/arXiv.1810.01943. arXiv: 1810.01943. URL: <https://arxiv.org/abs/1810.01943>.
- Bengio, Yoshua, Réjean Ducharme, and Pascal Vincent (2000). “A Neural Probabilistic Language Model”. In: *Advances in Neural Information Processing Systems*. Vol. 13. MIT Press. URL: https://papers.nips.cc/paper_files/paper/2000/hash/728f206c2a01bf572b5940d7d9a8fa4c-Abstract.html (visited on 07/19/2025).
- Bertalan, Vithor Gomes Ferreira and Evandro Eduardo Seron Ruiz (Mar. 2024). “Using attention methods to predict judicial outcomes”. In: *Artificial Intelligence and Law* 32.1, pp. 87–115. ISSN: 0924-8463, 1572-8382. DOI: 10.1007/s10506-022-09342-7. arXiv: 2207.08823[cs]. URL: <http://arxiv.org/abs/2207.08823> (visited on 10/06/2025).
- Cramer, Harald (1946). *Mathematical Methods of Statistics*. Princeton: Princeton University Press. ISBN: 0-691-08004-6.
- DeepSeek-AI (2025). *DeepSeek-RL: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning*. arXiv: 2501.12948 [cs.CL]. URL: <https://arxiv.org/abs/2501.12948>.

- DeepSeek-AI et al. (Jan. 22, 2025). *DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning*. DOI: 10.48550/arXiv.2501.12948. arXiv: 2501.12948[cs]. URL: <http://arxiv.org/abs/2501.12948> (visited on 07/27/2025).
- Gemini, Team et al. (May 9, 2025). *Gemini: A Family of Highly Capable Multimodal Models*. DOI: 10.48550/arXiv.2312.11805. arXiv: 2312.11805[cs]. URL: <http://arxiv.org/abs/2312.11805> (visited on 07/27/2025).
- Gu, Shangding (2026). *Long Context, Less Focus: A Scaling Gap in LLMs Revealed through Privacy and Personalization*. arXiv: 2602.15028 [cs.LG]. URL: <https://arxiv.org/abs/2602.15028>.
- Hannan, Md Abdul et al. (2026). “On the Difficulty of Selecting Few-Shot Examples for Effective LLM-based Vulnerability Detection”. In: *Proceedings of the Workshop on LLM Assisted Security and Trust Exploration (LAST-X 2026)*. DOI: 10.48550/arXiv.2510.27675. arXiv: 2510.27675 [cs.SE]. URL: <https://doi.org/10.48550/arXiv.2510.27675>.
- Homoki, Péter and Zsolt Zódi (June 27, 2024). “Large language models and their possible uses in law”. In: *Hungarian Journal of Legal Studies* 64.3. Publisher: Akadémiai Kiadó Zrt., pp. 435–455. ISSN: 2498-5473, 2560-1067. DOI: 10.1556/2052.2023.00475. URL: <https://akjournals.com/view/journals/2052/64/3/article-p435.xml> (visited on 07/18/2025).
- Jialin, Li (2026). “Exploring Bias Formation Mechanisms in Legal LLMs from a Cognitive Science Perspective”. In: *Big Data and Social Computing*. Ed. by Xiaofeng Meng et al. Singapore: Springer Nature, pp. 290–300. ISBN: 9789819508808. DOI: 10.1007/978-981-95-0880-8_24.
- Kalra, Dayal Singh and Maissam Barkeshli (2024). “Why Warmup the Learning Rate? Underlying Mechanisms and Improvements”. In: *Advances in Neural Information Processing Systems*. Vol. 37. Curran Associates, Inc. URL: <https://arxiv.org/abs/2406.09405>.
- Kmainasi, Mohamed Bayan, Ali Ezzat Shahroor, and Amani Al-Ghraibah (Feb. 28, 2025). *Can Large Language Models Predict the Outcome of Judicial Decisions?* DOI: 10.48550/arXiv.2501.09768. arXiv: 2501.09768[cs]. URL: <http://arxiv.org/abs/2501.09768> (visited on 10/06/2025).
- Lee, Wonyeol (2023). “Training with Mixed-Precision Floating-Point Assignments”. In: *Transactions on Machine Learning Research*. Published in TMLR. ISSN: 2835-8856. URL: <https://openreview.net/forum?id=t9z799BovW>.
- Liu, Shuang et al. (2026). *LLM Agents in Law: Taxonomy, Applications, and Challenges*. arXiv: 2601.06216 [cs.CL]. URL: <https://arxiv.org/abs/2601.06216>.
- Long, Shangbang et al. (Sept. 18, 2018). *Automatic Judgment Prediction via Legal Reading Comprehension*. DOI: 10.48550/arXiv.1809.06537. arXiv: 1809.06537[cs]. URL: <http://arxiv.org/abs/1809.06537> (visited on 10/06/2025).
- Lundberg, Scott M. and Su-In Lee (2017). “A Unified Approach to Interpreting Model Predictions”. In: *arXiv preprint arXiv:1705.07874*. DOI: 10.48550/arXiv.1705.07874. arXiv: 1705.07874.
- Matthews, Brian W. (1975). “Comparison of the predicted and observed secondary structure of T4 phage lysozyme”. In: *Biochimica et Biophysica Acta (BBA) - Protein Structure* 405.2, pp. 442–451. DOI: 10.1016/0005-2795(75)90109-9. URL: <https://www.sciencedirect.com/science/article/pii/0005279575901099>.
- Medvedeva, Masha, Michel Vols, and Martijn Wieling (2023). “Rethinking the field of automatic prediction of court decisions”. In: *Artificial Intelligence and Law* 31.1, pp. 195–212. DOI: 10.1007/s10506-021-09306-3. URL: <https://link.springer.com/article/10.1007/s10506-021-09306-3>.
- Mei, Lingrui et al. (2025). *A Survey of Context Engineering for Large Language Models*. arXiv: 2507.13334v2 [cs.CL]. URL: <https://arxiv.org/abs/2507.13334v2>.

References

- Melo, Rui, Pedro A. Santos, and João Dias (2023). “A Semantic Search System for the Supremo Tribunal de Justiça”. In: *Progress in Artificial Intelligence*. Ed. by Nuno Moniz et al. Cham: Springer Nature Switzerland, pp. 142–154. ISBN: 978-3-031-49011-8.
- Menezes-Neto, Elias Jacob de and Marco Bruno Miranda Clementino (July 28, 2022). “Using deep learning to predict outcomes of legal appeals better than human experts: A study with data from Brazilian federal courts”. In: *PLOS ONE* 17.7. Publisher: Public Library of Science, e0272287. ISSN: 1932-6203. DOI: 10.1371/journal.pone.0272287. URL: <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0272287> (visited on 10/06/2025).
- Meta AI (2025). *The Llama 4 herd: The beginning of a new era of natively multimodal AI innovation*. Accessed: 2025-08-03. Meta AI. URL: <https://ai.meta.com/blog/llama-4-multimodal-intelligence/>.
- Mikolov, Tomáš et al. (Sept. 26, 2010). “Recurrent neural network based language model”. In: *Interspeech 2010*. Interspeech 2010. ISCA: ISCA. DOI: 10.21437/interspeech.2010-343. URL: https://www.isca-archive.org/interspeech_2010/mikolov10_interspeech.html (visited on 07/19/2025).
- Min, Sewon et al. (2022). “Rethinking the Role of Demonstrations: What Makes In-Context Learning Work?” In: *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, pp. 11048–11064. DOI: 10.18653/v1/2022.emnlp-main.759. URL: <https://arxiv.org/abs/2202.12837>.
- Naveed, Humza et al. (Oct. 17, 2024). *A Comprehensive Overview of Large Language Models*. DOI: 10.48550/arXiv.2307.06435. arXiv: 2307.06435[cs]. URL: <http://arxiv.org/abs/2307.06435> (visited on 07/18/2025).
- Nigam, Shubham Kumar, Aniket Deroy, et al. (2024). “Rethinking Legal Judgement Prediction in a Realistic Scenario in the Era of Large Language Models”. In: *arXiv preprint arXiv:2410.10542*. URL: <https://arxiv.org/abs/2410.10542>.
- Nigam, Shubham Kumar, Balaramamahanthi Deepak Patnaik, et al. (Aug. 1, 2025). *NyayaRAG: Realistic Legal Judgment Prediction with RAG under the Indian Common Law System*. DOI: 10.48550/arXiv.2508.00709. arXiv: 2508.00709[cs]. URL: <http://arxiv.org/abs/2508.00709> (visited on 10/09/2025).
- OpenAI (Feb. 27, 2025). *OpenAI GPT-4.5 System Card*. Technical Report. Accessed: 2025-07-27. OpenAI. URL: <https://openai.com/index/gpt-4-5-system-card/>.
- Pedro, Ricardo (July 18, 2024). *DO USO DE IA GENERATIVA NOS TRIBUNAIS A UMA JUSTIÇA DEGENERATIVA: QUANDO A TECNOLOGIA ALUCINA* (From the Use of Generative Ai in Courts to Degenerative Justice: When Technology Hallucinates <p></p></p>. Rochester, NY. URL: <https://papers.ssrn.com/abstract=4904844> (visited on 08/17/2025).
- Pedroso, João and Andreia Santos (Aug. 2024). “Inteligência artificial e justiça criminal: Riscos e desafios”. In: *Revista APS* 35. DOI: 10.30553/sociologiaonline.2024.35.6.
- Peng, Xiao and Liang Chen (Oct. 15, 2024). *Athena: Retrieval-augmented Legal Judgment Prediction with Large Language Models*. DOI: 10.48550/arXiv.2410.11195. arXiv: 2410.11195[cs]. URL: <http://arxiv.org/abs/2410.11195> (visited on 10/09/2025).
- Pierce, Charles Sanders (1884). “The numerical measure of the success of predictions”. In: *Science* 4.93, pp. 453–454. DOI: 10.1126/science.ns-4.93.453.b. URL: <https://www.science.org/doi/10.1126/science.ns-4.93.453.b>.
- Rao, J. N. K. and C. F. Jeff Wu (1984). *Bootstrap Inference with Stratified Samples*. Technical Summary Report MRC-TSR-2629. DTIC Accession Number: ADA139313. Mathematics Research Cen-

- ter, University of Wisconsin-Madison. URL: <https://apps.dtic.mil/sti/tr/pdf/ADA139313.pdf>.
- Remolina, Nydia and David Socol De La Osa (2024). *AI at the Bench: Legal and Ethical Challenges of Informing - or Misinforming - Judicial Decision-Making Through Generative AI*. DOI: 10.2139/ssrn.4860853. URL: <https://www.ssrn.com/abstract=4860853> (visited on 10/17/2025).
- Rodrigues, Thyago M. et al. (June 2025). *Domain-Specific Stop-Word List for Brazilian Legal Texts*. Dataset. DOI: 10.21227/ea6g-ad33. URL: <https://ieee-dataport.org/documents/domain-specific-stop-word-list-brazilian-legal-texts>.
- Rosenfeld, R. (Aug. 2000). “Two decades of statistical language modeling: where do we go from here?” In: *Proceedings of the IEEE* 88.8, pp. 1270–1278. ISSN: 1558-2256. DOI: 10.1109/5.880083. URL: <https://ieeexplore.ieee.org/document/880083> (visited on 07/19/2025).
- Sasaki, Yutaka (2007). *The truth of the F-measure*. Tech. rep. 5. Technical Teaching Material. Manchester Interdisciplinary Biocentre, University of Manchester, pp. 1–5. URL: https://www.researchgate.net/publication/228548174_The_truth_of_the_F-measure.
- Schulhoff, Sander et al. (2024). *The Prompt Report: A Systematic Survey of Prompt Engineering Techniques*. arXiv: 2406.06608 [cs.CL]. URL: <https://arxiv.org/abs/2406.06608>.
- Shao, Peizhang et al. (July 10, 2025). *When Large Language Models Meet Law: Dual-Lens Taxonomy, Technical Advances, and Ethical Governance*. DOI: 10.48550/arXiv.2507.07748. arXiv: 2507.07748[cs]. URL: <http://arxiv.org/abs/2507.07748> (visited on 07/18/2025).
- Shukla, Anil (2025). “The Most Advanced AI Models of 2025 -Comparative Analysis of Gemini 2.5, Claude 4, LLaMA 4, GPT-4.5, DeepSeek V3.1, and Other Leading Models”. In: Publisher: Unpublished. DOI: 10.13140/RG.2.2.35858.36800. URL: <https://rgdoi.net/10.13140/RG.2.2.35858.36800> (visited on 07/27/2025).
- Siino, Marco et al. (2025a). “Exploring LLMs Applications in Law: A Literature Review on Current Legal NLP Approaches”. In: *IEEE Access* 13. Publisher: Institute of Electrical and Electronics Engineers (IEEE), pp. 18253–18276. ISSN: 2169-3536. DOI: 10.1109/access.2025.3533217. URL: <https://ieeexplore.ieee.org/document/10850911/> (visited on 07/18/2025).
- (2025b). “Exploring LLMs Applications in Law: A Literature Review on Current Legal NLP Approaches”. In: *IEEE Access* 13. Publisher: Institute of Electrical and Electronics Engineers (IEEE), pp. 18253–18276. ISSN: 2169-3536. DOI: 10.1109/access.2025.3533217. URL: <https://ieeexplore.ieee.org/document/10850911/> (visited on 07/18/2025).
- Song, Peiyang et al. (2025). *The Order Effect: Investigating Prompt Sensitivity to Input Order in LLMs*. arXiv: 2502.04134v2 [cs.CL]. URL: <https://arxiv.org/abs/2502.04134v2>.
- Souza, Fábio, Rodrigo Nogueira, and Roberto Lotufo (2020). “BERTimbau: pretrained BERT models for Brazilian Portuguese”. In: *9th Brazilian Conference on Intelligent Systems, BRACIS, Rio Grande do Sul, Brazil, October 20-23 (to appear)*.
- Sulea, Octavia-Maria et al. (Aug. 4, 2017). *Predicting the Law Area and Decisions of French Supreme Court Cases*. DOI: 10.48550/arXiv.1708.01681. arXiv: 1708.01681[cs]. URL: <http://arxiv.org/abs/1708.01681> (visited on 10/07/2025).
- Sun, Chi et al. (2019). “How to Fine-Tune BERT for Text Classification?” In: *Chinese Computational Linguistics: 18th China National Conference, CCL 2019*. Springer, pp. 194–206. DOI: 10.1007/978-3-030-32381-3_16. URL: <https://arxiv.org/abs/1905.05583>.
- Surden, Harry (2023). “CHATGPT, AI LARGE LANGUAGE MODELS, AND LAW”. In.

References

- Szegedy, Christian et al. (2016). “Rethinking the Inception Architecture for Computer Vision”. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 2818–2826. DOI: 10.1109/CVPR.2016.308. URL: <https://arxiv.org/abs/1512.00567>.
- Tang, Yongjian et al. (2025). *The Few-Shot Dilemma: Balancing Context Window and Demonstration Density in LLM Prompting*. arXiv: 2509.13196v1 [cs.CL]. URL: <https://arxiv.org/abs/2509.13196v1>.
- Turpin, Miles et al. (2023). “Language Models Don’t Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting”. In: *Advances in Neural Information Processing Systems*. Vol. 36. Curran Associates, Inc. URL: <https://arxiv.org/abs/2305.04388>.
- Van Rijsbergen, Cornelis Joost (1979). *Information Retrieval*. 2nd. London: Butterworth-Heinemann. URL: <http://www.dcs.gla.ac.uk/Hepple/Definition/Book/index.html>.
- Vaswani, Ashish et al. (2017). *Attention Is All You Need*. DOI: 10.48550/arXiv.1706.03762. arXiv: 1706.03762[cs]. URL: <http://arxiv.org/abs/1706.03762> (visited on 07/18/2025).
- Wang, Zichong et al. (Sept. 23, 2024). *History, Development, and Principles of Large Language Models- An Introductory Survey*. DOI: 10.48550/arXiv.2402.06853. arXiv: 2402.06853[cs]. URL: <http://arxiv.org/abs/2402.06853> (visited on 07/18/2025).
- Wengorovius, Sofia (2025). “TECNOLOGIA, PROTEÇÃO DE DADOS E INTELIGÊNCIA ARTIFICIAL – OS PASSOS EQUILIBRADOS PARA OS TRIBUNAIS”. In: *A INTELIGÊNCIA ARTIFICIAL E OS TRIBUNAIS EM PORTUGAL E NO BRASIL*. Ed. by Luiz Guilherme Marinoni and Ricardo Pedro. Chap. 9, pp. 198–220. URL: <https://csm.org.pt/wp-content/uploads/2025/01/Artigo-Sofia-W.-IA.pdf>.
- Woodcock, Frank (1976). “The Evaluation of Yes/No Forecasts for Scientific and Administrative Purposes”. In: *Monthly Weather Review* 104.10, pp. 1209–1214. DOI: 10.1175/1520-0493(1976)104<1209:TEOYFF>2.0.CO;2. URL: https://journals.ametsoc.org/view/journals/mwre/104/10/1520-0493_1976_104_1209_teoyff_2_0_co_2.xml.
- Youden, William John (1950). “Index for rating diagnostic tests”. In: *Cancer* 3.1, pp. 32–35. DOI: 10.1002/1097-0142(1950)3:1<32::aid-cnrcr2820030106>3.0.co;2-3.
- Yule, G. Udny (1912). “On the Methods of Measuring Association Between Two Attributes”. In: *Journal of the Royal Statistical Society* 75.6, pp. 579–652. DOI: 10.2307/2340126.
- Zhang, Jingzhao et al. (2019). “Why Gradient Clipping Accelerates Training: A Theoretical Justification for Adaptivity”. In: *Advances in Neural Information Processing Systems*. Vol. 32. Curran Associates, Inc. URL: <https://arxiv.org/abs/1905.11881>.
- Zhong, Haoxi et al. (2020). “How Does NLP Benefit Legal System: A Summary of Legal Artificial Intelligence”. In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics. Online: Association for Computational Linguistics. DOI: 10.18653/v1/2020.acl-main.466. URL: <https://www.aclweb.org/anthology/2020.acl-main.466> (visited on 07/18/2025).

Appendices

Github Repository

The full source code for the data scraping, preprocessing, modelling (BERT and LLM pipelines), evaluation, explainability, and judge-annotation study described throughout this thesis is publicly available at:

`https://github.com/Ecologykk/Master-Thesis---Msc-Data-Science`

The repository includes step-by-step documentation for reproducing every pipeline stage, an automated test suite, and a top-level README.md routing new readers through the codebase. It is released under the Apache License 2.0.

Data Files

The raw scraped court decisions, the processed and labelled datasets (including the temporal gold-test splits), the trained model checkpoints, and the full prediction outputs used throughout this thesis are archived on Zenodo with a permanent DOI:

<https://doi.org/10.5281/zenodo.22116675>

SHA-256 checksums for each archive are published alongside the Zenodo record so downloaders can verify integrity. A subset of these artefacts sufficient to check the headline results reported in this thesis without downloading the full archive — the 50-case gold-test sets, the model prediction outputs, and the consolidated evaluation results with confidence intervals — is committed directly under the `data/` folder of the Github repository referenced in Appendix 7.

As discussed in subsection 4.7.3, the raw participant responses collected during the judge-annotation study carry the judges' real identifiers and are not published, since they constitute personal data about identifiable legal professionals. An anonymised copy of the responses (judges relabelled J1–J5 per case type, with one self-identifying passage removed) is released in the GitHub repository, under `judge_labeling_app/reports/anonymised/`, alongside the source code used to collect them.

Class-Engineering Heuristics

This appendix provides the complete regular-expression patterns used to extract decision classes from case summaries, as referenced in the Data chapter.

.1 DV Binary Classification (KEPT vs. ALTERED)

```
def extract_decision_binary_from_summary_dv(summary_decision):
    """
    DV-only binary classification:
    - DECISÃO ALTERADA
    - DECISÃO MANTIDA
    """
    if not summary_decision or pd.isna(summary_decision):
        return None

    text = str(summary_decision).lower().strip()
    text_clean = re.sub(r"[^\w\s]", "_", text)

    # PRIORITY 0: negated alteration (must come first)
    negated_alteration_patterns = [
        r"\bnegado\s+provimento\b",
        r"\bn[ã]o\s+provido\b",
        r"\bdesprovido\b",
        r"\bnega[rd]\s+provimento\b",
        r"\brecurso\s+improcedente\b",
        r"\bapela[çc][ã]o\s+improcedente\b",
        r"\bimprocedente\b",
        r"\bimprocedência\b",
    ]

    for pattern in negated_alteration_patterns:
        if re.search(pattern, text_clean):
            return "DECISÃO MANTIDA"

    # PRIORITY 1: affirmative alteration (including partials)
    altered_patterns = [
        r"\bparcialmente\s+provido\b",
        r"\bprovido\s+parcialmente\b",
        r"\bprocedente\s+em\s+parte\b",
        r"\bparcial\b",
        r"\bparcialmente\b",
    ]
```

.2. BoC Ternary Classification (Favorable vs. Unfavorable vs. Partial)

```
    r"\bem\s+parte\b",
    r"\bprovido\b",
    r"\bprovimento\b",
    r"\bprocedente\b",
    r"\bconcedid[oa]\b",
    r"\balterad[oa]\b",
    r"\balterar\b",
    r"\brevogad[oa]\b",
    r"\brevoga[rd]\b",
    r"\breformad[oa]\b",
]

for pattern in altered_patterns:
    if re.search(pattern, text_clean):
        return "DECISÃO ALTERADA"

# PRIORITY 2: explicit maintenance
kept_patterns = [
    r"\bmantid[oa]\b",
    r"\bconfirmad[oa]\b",
    r"\bconfirma[rd]\b",
    r"\bconfirma[çc][aã]o\b",
]

for pattern in kept_patterns:
    if re.search(pattern, text_clean):
        return "DECISÃO MANTIDA"

return None
```

.2 BoC Ternary Classification (Favorable vs. Unfavorable vs. Partial)

```
def extract_decision_ternary_from_summary_boc(summary_decision):
    """
    Classificação ternária BoC:
    - FAVORÁVEL
    - DESFAVORÁVEL
    - PARCIAL

    Regras:
    0) Negação explícita tem prioridade absoluta
    1) Parcial é categoria própria
    2) Favorável só se for afirmativo e não negado
    3) Caso ambíguo → None
    """
    if not summary_decision or pd.isna(summary_decision):
        return None

    text = str(summary_decision).lower().strip()
    text_clean = re.sub(r"[^\w\s]", "", text)
```

```

# PRIORIDADE 0: negação explícita de provimento / procedência
negated_favorable_patterns = [
    r"\bn[aã]o\s+procede\b",
    r"\bn[aã]o\s+procedente\b",
    r"\bnega[rd]\s+provimento\b",
    r"\bnegado\s+provimento\b",
    r"\bn[aã]o\s+provido\b",
    r"\bdesprovido\b",
    r"\brecurso\s+improcedente\b",
    r"\bapela[çc][aã]o\s+improcedente\b",
    r"\bimprocedente\b",
    r"\bimprocedência\b",
]

for pattern in negated_favorable_patterns:
    if re.search(pattern, text_clean):
        return "DESFAVORÁVEL"

# PRIORIDADE 1: PARCIAL
partial_patterns = [
    r"\bparcial\b",
    r"\bparcialmente\b",
    r"\bem\s+parte\b",
    r"\bprocedente\s+em\s+parte\b",
    r"\bprovido\s+parcialmente\b",
    r"\bconcedid[oa]\s+parcialmente\b",
    r"\brevogad[oa]\s+parcialmente\b",
    r"\bconfirmad[oa]\s+em\s+parte\b",
]

for pattern in partial_patterns:
    if re.search(pattern, text_clean):
        return "PARCIAL"

# PRIORIDADE 2: FAVORÁVEL (afirmativo)
favorable_patterns = [
    r"\btotalmente\s+procedente\b",
    r"\bprocedente\b",
    r"\bprovido\b",
    r"\bprovimento\b",
    r"\bconcedid[oa]\b",
    r"\brevogad[oa]\b",
    r"\brevog[aou]?b",
]

for pattern in favorable_patterns:
    if re.search(pattern, text_clean):
        return "FAVORÁVEL"

# PRIORIDADE 3: DESFAVORÁVEL explícito

```

.2. BoC Ternary Classification (Favorable vs. Unfavorable vs. Partial)

```
unfavorable_patterns = [  
    r"\bmantid[oa]\b",  
    r"\bconfirmad[oa]\b",  
    r"\bconfirma[çc][aã]o\b",  
    r"\bnega[rd]\b",  
    r"\bnegad[oa]\b",  
]  
  
for pattern in unfavorable_patterns:  
    if re.search(pattern, text_clean):  
        return "DESFAVORÁVEL"  
  
return None
```

SHAP Explanation Examples

This appendix provides the visual SHAP outputs referenced in section 5.1 and subsection 5.1.3, complementing the sentence-level attributions computed for the fine-tuned LegalBERT_{Timbau} model on the same 5 DV and 5 BoC gold test cases used for judge validation.

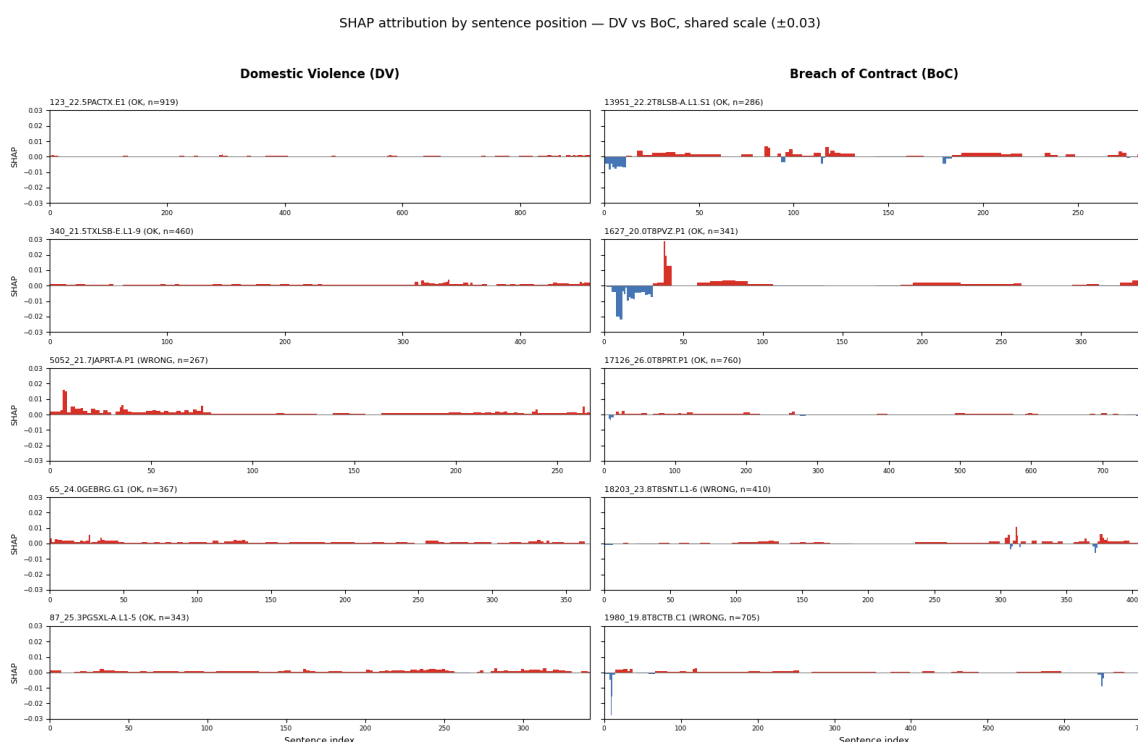


Figure 1: SHAP attribution by sentence position for all 5 DV and 5 BoC gold test cases, on a shared ± 0.03 y-axis so magnitudes are directly comparable across documents and case types. Red bars push toward the predicted class, blue bars push away from it. DV attributions are near-flat across all five documents. BoC shows sharper, mixed-sign bursts concentrated in the first few dozen sentences of some cases, which correspond to repeated monetary figures in the case report rather than legal reasoning, as discussed in section 5.1.

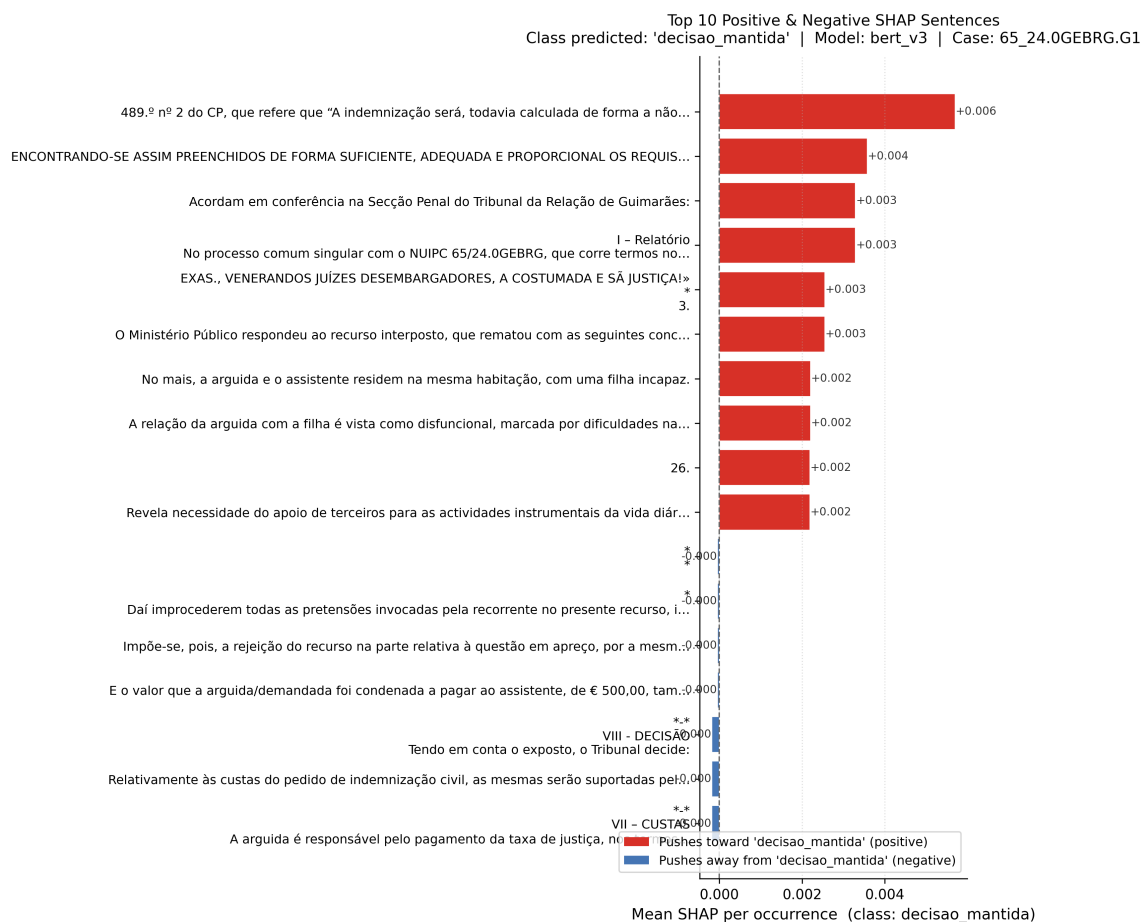


Figure 2: Top 10 positive and top 10 negative SHAP sentences (truncated) for DV case 65/24.0GEBRG.G1, predicted class KEPT. Nearly all contributing sentences are positive and of small, similar magnitude, consistent with the diffuse attribution pattern described in section 5.1.

Appendix . SHAP Explanation Examples

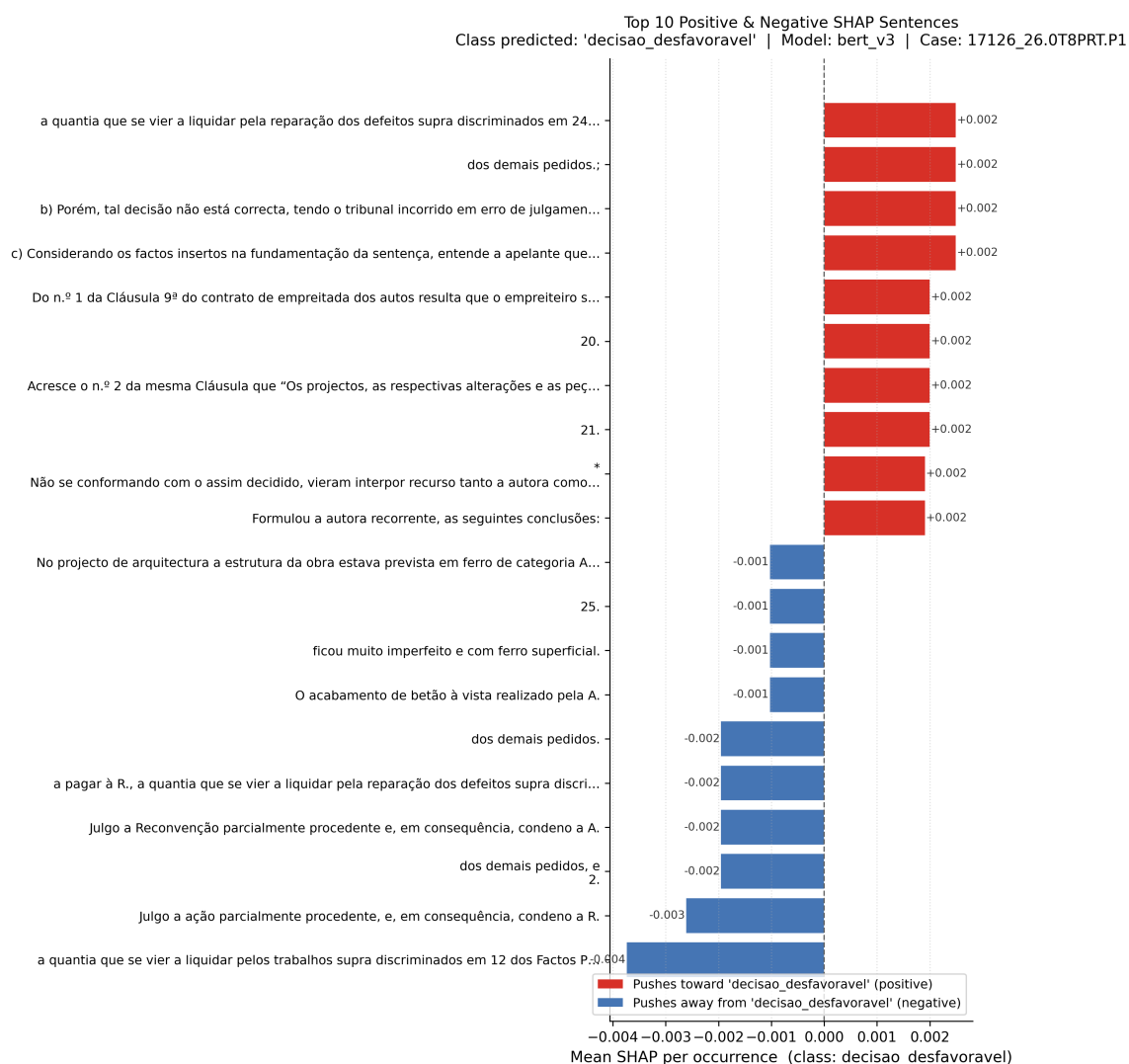


Figure 3: Top 10 positive and top 10 negative SHAP sentences (truncated) for BoC case 17126/26.0T8PRT.P1, predicted class DESFAVORAVEL. Unlike the DV example, both positive and negative sentences appear with comparable magnitude, several referencing monetary claim amounts from the case report.

Example DeepSeek Reasoning Trace

This appendix provides the full thinking log referenced in chapter 4, for case 236/23.6PBHRT.L1-9¹. DeepSeek-R1-0528-Qwen3-8B was asked to classify whether the appellate court kept or altered the lower court decision, and predicted a kept decision when the correct label was an altered decision. The trace below is reproduced verbatim from the model's <think> output.

Okay, let's tackle this case. The user provided a detailed legal document from a Portuguese court case involving domestic violence. The task is to classify whether the appellate court upheld or altered the lower court's decision.

First, I need to understand the context. The case is about domestic violence under Article 152 of the Penal Code. The defendant was initially acquitted by the lower court, and the prosecution appealed. The appellate court's decision is crucial here.

Looking at the text, the lower court found that the subjective elements of the crime weren't proven, leading to an acquittal. The prosecution's appeal argued that the lower court didn't properly assess the facts, specifically the defendant's knowledge of the victim's status and the presence of the child. The appellate court's reasoning is key.

The appellate decision states that the lower court's sentence was upheld. They mention that the appeal didn't meet the criteria for modification because the lower court's decision was well-founded. The appellate court also notes that the defendant's acquittal was correct based on the facts presented. There's a mention of the appeal being dismissed without any criticism of the lower court's decision.

The predicted_label should be "decisao_mantida" because the appellate court confirmed the lower court's acquittal. The prosecution didn't get a favorable change, so the decision was maintained.

¹<https://jurisprudencia.csm.org.pt/eccli/ECLI:PT:TRL:2025:236.23.6PBHRT.L1.9.58/>